A.F.O.N.S.O.
Asteroid Flyby and Observational iNsight Science Orbiter

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1 Mission Overview

1.1 Mission Summary (Logan Smith-Perkins)

The mission is designed to conduct a comprehensive survey of three distinct main-belt asteroids with diverse compositions: S-type (11 Parthenope), M-type (16 Psyche), and C-type (24 Themis). By targeting these specific bodies, the AFONSO mission (Asteroid Flyby and Observational iNsight Science Orbiter) aims to investigate the early history of the solar system, from the formation of planetesimal cores to the delivery of water and organic molecules to the inner planets.

The mission utilizes a direct architecture, departing Earth for a sequential rendezvous without the use of planetary gravity assists. Propulsion is provided by an all-electric Sims-Flanagan low-thrust system using a NEXT-C ion thruster with a specific impulse I_{sp} of 3100 s and a maximum thrust cap of 0.30 N. The flight system features a 2518 kg launch mass, utilizing a 7.0 km/s impulsive v_∞ launch provided by the by the Falcon Heavy launch vehicle. Following launch, the spacecraft executes a low-thrust trajectory with a total post-launch mission Δv of 13.0701 km/s.

Over the 16.81-year flight duration, the spacecraft will traverse heliocentric distances ranging from 1.015 to 3.247 AU, concluding with arrival at 24 Themis on May 20, 2046. To meet the scientific requirement of > 80 percent surface coverage at each target, the spacecraft will enter stable, low-altitude orbits optimized for the multispectral imager, IR spectrometer, and magnetometer. These proximity operations require robust thermal management to handle the low solar flux in the outer belt and autonomous navigation protocols to manage significant light-time delays.

1.2 Scientific Objectives (Hannah Ramsperger)

Asteroid missions offer an opportunity to study the earliest stages of our solar system and the processes that shaped the planets. Unlike planets, many asteroids have remained relatively unchanged for billions of years, preserving crucial information about the materials and conditions present during planetary formation. Thus, studying their composition, structure, and evolution can reveal details about how the inner planets, including Earth, developed over time [1]. Certain asteroids are of particular scientific interest because they may represent fragments of larger planetary bodies or even exposed planetary cores. Exploring such objects improves our understanding of collisions that characterized the solar system's early history. Asteroid research may also provide insight into the origins of life, as many scientists believe that impacts from asteroids and comets could have delivered water and organic molecules to the early Earth. By studying these primitive bodies, missions can help determine the role they may have played in creating the conditions necessary for life to emerge [2].

For these reasons, the AFONSO mission will visit three asteroids, 16 Psyche, 11 Parthenope, and 24 Themis, to collect data on the formation of the solar system. These asteroids exhibit distinct compositions and unique features, ensuring diverse data collection and a highly informative mission. Specifically, the mission aims to characterize the global physical prop-

erties, geologic history, composition, and magnetic field of these three bodies by mapping their mineralogy and composition via infrared spectrometry, determining their size, shape, mass, and surface history through visible light imaging, and analyzing their magnetic properties and solar wind interactions using a magnetometer. The detailed scientific objectives are stated in Table 1.

Psyche may contain large amounts of metal originating from the core of an early planetesimal. Observations suggest Psyche is composed of a mixture of rock and metal, with metal potentially making up a significant fraction of its volume [3]. Studying Psyche offers an opportunity to investigate metal core material similar to the interior of terrestrial planets like Earth.

S-type asteroid Parthenope is valuable as a primitive remnant from the early solar system. By studying its surface composition, mineralogy, shape, size, and structure, insight can be gained into the materials that existed during planet formation [4]. The topology analysis can also help improve the understanding of how collisions influenced the asteroid belt and reconstruct the history of the inner solar system.

M-type asteroid Themis is important because observations have revealed evidence of water ice and organic-rich material on its surface [5]. Since water and organic molecules are essential ingredients for life, studying Themis may help answer whether asteroids contributed to delivering these materials to Earth. It can also provide insight into how water ice can survive in the asteroid belt and how organic compounds evolved over time. Exploring this object could therefore improve our understanding of both solar system evolution and the origins of life. Furthermore, scientists suggest that Themis might be magnetized, and emphasize the value of magnetometry for investigating the interiors, and evolution of C-type asteroids [6].

Table 1: Science Objectives and Measurement Strategies

ID	Objective	Measurement Strategy
SO1	Map the topology, chemical composition, mineralogy, and temperature distribution of the carbonaceous asteroid Themis, M-type asteroid Psyche, and S-type Parthenope to characterize its geologic history.	IR spectrometry
SO2	Characterize the distribution and state of water ice and hydrated minerals on Themis.	IR spectrometry
SO3	Determine magnetic properties of the asteroids by measuring their magnetic field.	Magnetometer
SO4	Investigate how the asteroids' magnetic field interacts with the solar wind environment.	Magnetometer
SO5	Image ≥ 80 % of asteroids' surfaces to draw conclusions about their impact and tectonic history.	Visible Light Imaging
SO6	Determine size, shape and mass of Psyche, Themis and Parthenope.	Visible Light Imaging and IR spectrometry
SO7	Confirm the existence of organic molecules on Themis.	IR spectrometry

1.3 Functional Requirements (Hannah Ramsperger)

Given the mission objective of visiting the three asteroids Parthenope, Psyche and Themis and achieve the science objective in the given time of 16.81 years, following Level 1 functional requirements have been determined.

1. The flight system shall be capable of interplanetary transfer from Earth to the Parthenope, Psyche and Themis.
2. The flight system shall be capable of orbit insertion around Parthenope, Psyche and Themis, and orbit transfers around the asteroids.
3. The spacecraft shall maintain orbit around Parthenope, Psyche and Themis for a duration sufficient to achieve the science objectives.
4. The spacecraft shall be equipped with a visual light camera for optical navigation and high resolution imaging as well as with a magnetometer and infrared spectrometer to achieve the scientific objectives.
5. The power subsystem shall generate, regulate, store, and distribute electrical power throughout all mission phases, including cruise, science orbits and orbit transfers.

6. The spacecraft shall be capable of downlinking all of the collected science data by the end of the mission.
7. All spacecraft subsystems shall be capable of operating within the thermal environment from 1 AU to 3.5 AU encountered during cruise and asteroid encounters.
8. The avionics subsystem shall provide single-fault tolerant, reliable command, control, data handling, and onboard data storage capabilities to support all spacecraft and science operations throughout the mission.
9. The propulsion subsystem shall provide a total post-launch ΔV capability of at least 13.1 km/s using the electric propulsion system.
10. The spacecraft shall perform autonomous navigation to support rendezvous and orbit insertion at each asteroid.

1.4 Design Requirements

The following design requirements establish the minimum performance, environmental survivability, and operational criteria for each major spacecraft subsystem. These requirements ensure that the integrated vehicle can achieve the mission's functional goals and fulfill its scientific objectives. These requirements are the baseline against which the Vehicle Overview in Section 3 verifies each subsystem's design.

1.4.1 Payload (Hannah Ramsperger)

1. The payload suite should include a visible light camera, an infrared spectrometer and a magnetometer.
2. The instruments shall have flown on previous recent missions.
3. The Spectrometer shall provide spectral coverage from at least 0.2 μm to 4.0 μm.
4. The IR spectrometer shall provide spatially resolved spectral imaging to enable mineralogical mapping in geologic context.
5. The IR spectrometer shall be capable of acquiring spectral images from orbital altitudes between 20 km and 400 km, adjusting spatial resolution accordingly.
6. The camera shall accommodate a wide range of target albedos, from at least 0.06 to 0.4.
7. The magnetometer shall be able to operate during the whole mission.

1.4.2 Structure (Riley Foard)

1. The spacecraft shall be designed to survive the launch environments of the Falcon Heavy launch vehicle.
2. The spacecraft shall be designed to survive all conditions in the reference orbital environment for the duration of the mission.

3. The spacecraft's stowed diameter shall be less than 4.5 m.
4. The spacecraft's stowed height, including the antenna stow and fixed booms, shall be less than 10 m.
5. The spacecraft structure mounted on the Falcon payload adapter shall have a lateral mode of at least 10 Hz and an axial mode of at least 25 Hz with a factor of safety.

1.4.3 Thermal (Riley Foard)

1. The spacecraft shall maintain all subsystems within allowable temperature limits during all operational and non-operational mission phases, safe modes, eclipses, cruise/coast phases, and contingency conditions.
2. The thermal system shall reject excess internal and environmental heat during worst-case hot conditions.
3. The thermal system shall provide sufficient heater power or passive thermal isolation to maintain required temperatures during worst-case cold conditions.
4. The thermal control system shall meet thermal requirements within the allocated spacecraft power budget.
5. The thermal control system shall meet thermal requirements within allocated mass, volume, and interface constraints.
6. The thermal system shall provide stable thermal environments to support mission science objectives (imaging, spectroscopy, magnetic field measurements).
7. The thermal system should run on a local, closed-loop or passive control system architecture independent of continuous ground intervention.
8. The thermal control system shall provide redundancy or fault tolerance for critical thermal-control functions, including heaters, temperature sensors, and heat pipes.

1.4.4 Power (Miina Anvelt)

1. The power system consists of the solar arrays, the battery subunit, and the power control unit for taking power from the solar arrays and distributing it to the major subsystems of the spacecraft.
2. The 75 m^2 solar arrays and MA190-111 battery combined shall provide sufficient power to subsystems through post-launch low-thrust maneuvers and scientific proximity operations at each of the three chosen mission asteroids.
3. The power control unit consisting of the MA190-111 battery battery unit and associated voltage converters shall convert power from the solar arrays or battery into usable power for the subsystems of the spacecraft.
4. The power system shall convert any additional power from the solar arrays that is not required by the system into stored energy in the battery if battery levels are lower than

80 percent of full charge.

5. The power budget shall have a factor of safety between the input solar array power and the power consumption, and the power budget for the entire mission profile shall be closed.

1.4.5 Propulsion (Mahak Mathur)

1. The propulsion system shall be capable of meeting the mission velocity demands, including a **7.0 km/s launch** Δv provided by the launch vehicle and a **13.1 km/s post-launch** Δv provided by the spacecraft propulsion system, as demonstrated by validated low-thrust trajectory simulations.
2. The propulsion system shall be capable of providing up to **0.300 N (300 mN)** of thrust for trajectory and attitude control operations.
3. The propulsion system shall achieve high-specific-impulse operation, with a nominal $I_{sp} \approx 3100$ s, to ensure total xenon propellant consumption does not exceed the verified mass budget.
4. The attitude control system shall consist of **12 thrusters and 4 reaction wheels** to support both coarse and precise attitude control maneuvers.
5. The attitude control system shall be capable of maintaining thrust-vector alignment within 1° of the spacecraft center of mass, with a maximum slew rate of 0.1°/s during all burns.
6. The propulsion system shall support fully autonomous operation during all mission phases.

1.4.6 Avionics (Alexis Herfurth)

1. The onboard computer shall provide execution and coordination of all subsystem command and control functions, as well as payload data processing.
2. All avionics components shall be highly reliable, energy efficient, lightweight, and capable of surviving deep space environments, while operating within a single-fault tolerant architecture.
3. The avionics shall provide a minimum non-volatile storage capacity sufficient to store the data collected during the longest and most data-intensive asteroid observation orbit, supporting continuous operation throughout the orbit prior to transmission to the ground.

1.4.7 Telecommunications (Edward Ju)

The telecommunications subsystem handles all data exchange between the spacecraft and the ground throughout the mission, from the 2029 launch through the end of operations in 2047 (16.81 years to Themis arrival), including command uplink and engineering and science telemetry downlink. All links go through NASA's Deep Space Network at X-band.

The spacecraft carries one 2.0 m gimbaled parabolic high-gain antenna (HGA) mounted above the bus top deck, three low-gain antennas (LGAs), two redundant Small Deep Space Transponders (SDSTs), and two redundant 100 W traveling-wave tube amplifiers (TWTAs). The downlink uses QPSK with rate-1/6 turbo coding and the uplink uses BPSK with rate-1/2 convolutional coding. This design is based on the Psyche mission [7] and is sized for the worst case at the Themis conjunction (4.14 AU).

Requirements.

The telecom requirements come from the Dawn telecom specification [8] and JPL's standard design margins.

T-1 The science downlink shall sustain at least 120 kbps at the worst-case Earth-spacecraft range of 4.14 AU through a DSN 34 m beam-waveguide antenna.

T-2 The bit error rate of the downlink shall not exceed 10^{-3} at the worst-case range.

T-3 The command uplink shall sustain at least 2 kbps at the worst-case range, with bit error rate not exceeding 10^{-5}.

T-4 All active links shall maintain at least 3 dB of design margin.

The 120 kbps downlink and 2 kbps uplink are the minimum rates required at the worst case (Themis at 4.14 AU). At closer ranges the SDST runs faster.

1.5 Spacecraft Overview

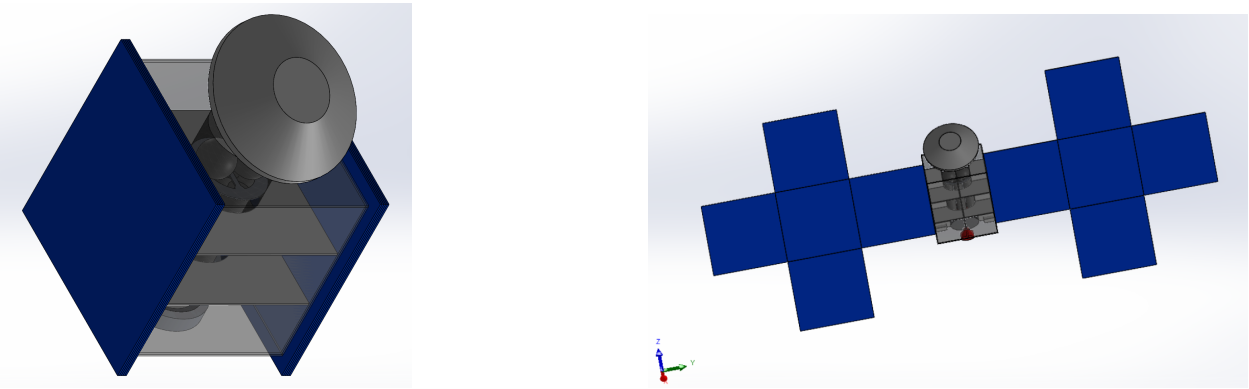


Figure 1: The spacecraft structure with the solar panels stowed (left) and the solar panels deployed (right).

Figure 1 is a CAD assembly showing the spacecraft bus with the solar panels attached to the bus in its stowed and deployed configurations. The spacecraft design is based on heritage from the Psyche and Dawn missions, both in its structural design and in selection of components and payload instruments, as detailed throughout this report.

1.6 Master Equipment List and Mass Budget (Tamar Levin)

The total launch mass estimate is **2480 kg**, with a total dry mass of **1515 kg**. The spacecraft mass budget is summarized in Table 2, with the detailed component-level breakdown shown in Table 3. The current dry mass estimate is 1377 kg before margin. A 10% dry mass margin is added to account for additional components that are not included in the preliminary design, such as deployment mechanisms for solar panels and booms, fasteners, brackets, mounting hardware, etc.

The xenon propellant mass was estimated using the rocket equation, $\Delta v = I_{sp} g_0 \ln\left(\frac{m_0}{m_f}\right)$.

with a requirement of $\Delta v = 13.1$ km/s and an average Specific Impulse of $I_{sp} = 3100$ s for the electric ion propulsion system. Using the current dry mass estimate the required xenon mass before margin is approximately 817 kg. This configuration maintains six xenon tanks, with 147.5 kg of xenon per tank. This is similar to the propellant loading per tank used on the Psyche mission [9].

The solar panel mass estimate is based on 75 m^2 of aluminum honeycomb panels, with the same material assumption as the main structure and a thickness of 2.5 cm. The actual mass of the solar panel system will be higher, since it also includes solar cells, deployment mechanisms, and additional electronics. This difference is currently accounted for by the dry mass margin.

Attempts were made to minimize the structure mass, as discussed in the Structures chapter.

Overall, the total launch mass estimate is 2450 kg, including the 10% dry mass margin, xenon propellant with 8% margin, and hydrazine propellant. This is comparable to the Psyche mission, which had a dry mass of 1648 kg and a launch mass of 2608 kg [10].

Table 2: Spacecraft mass budget summary.

Subsystem	**Mass (kg)**
Main structure	368
Fuel system (dry hardware)	138
Payload	23
Avionics	118
Telecommunications	25
Thermal	129
Propulsion system	105
Solar panels	473
Total dry mass	**1377**
Total dry mass + 10% margin	**1515**
Xenon propellant	885
Hydrazine propellant	80
Total fuel mass	**965**
Total mass	**2480**

Table 3: Detailed spacecraft component mass budget.

Subsystem	Component	Qty.	Unit kg	Total kg
Main structure	Aluminum honeycomb structural box	1	325.7	325.7
	Central cylinder	1	42.0	42.0
Fuel system	Xenon tanks, dry mass	6	11.0	66.0
	Xenon tank adapters	2	22.0	44.0
	Hydrazine tanks + Feed System	2	13.9	27.8
Payload	Dawn VIR instrument, including optics, electronics, and brackets	1	14.3	14.3
	Psyche magnetometer EUs and SUs	1	3.6	3.6
	Psyche multispectral imager, 2 heads and 1 DEA	1	5.1	5.1
Avionics	OBC, IMUs, star trackers, and sun sensors	1	9.9	9.9
	Honeywell HR16-100 reaction wheels	4	12.0	48.0
	Battery	1	60.0	60.0
Telecom	2.0 m parabolic HGA	1	13.3	13.3
	LGA antennas	3	0.5	1.5
	RF electronics and waveguide hardware	1	10.2	10.2
Thermal	Radiators	37.5	1.29	48
	Louvers	37.5	1.29	48
	Heaters	200	0.01	2.0
	Heat pipes	6	5	30
Propulsion	NEXT-C PPUs	2	36.0	72.0
	NEXT-C ion thrusters	2	14.0	28.0
	RCS thrusters	12	0.37	4.5
Solar panels	Solar panels	10	47.25	472.5
Fuel	Xenon	6	147.5	885.0
	Hydrazine	1	80.0	80.0
Fuel	**Subtotal**			**965.0**
Total dry mass				**1377**
Total dry mass + 10% margin				**1515**
Total mass				**2480**

2 Concept of Operations Overview

2.1 General Mission Overview (Logan Smith-Perkins)

The AFONSO mission is structured into three primary operational phases, each defined by a low-thrust interplanetary transfer followed by a dedicated science stay. The mission trajectory follows a direct architecture, departing Earth and proceeding through a sequence of target asteroids: Parthenope, Psyche, and Themis. Each phase consists of a cruise leg and a proximity operations period, with the spacecraft performing autonomous navigation to support rendezvous and orbit insertion at each body. The high-level timeline is as follows:

- Phase 1: Launch and cruise to 11 Parthenope, arriving on July 28, 2029, with a stay duration of approximately 6 months.
- Phase 2: Low-thrust transfer to 16 Psyche, arriving October 2040 for a 12-month science mission.
- Phase 3: Final transfer to 24 Themis, arriving in early 2046 with an 11.9 month operations period to conclude the baseline science mission.

The trajectory was developed using a refined optimization pipeline that integrates a Sims-Flanagan low-thrust model. This model discretizes each transfer leg into 15 segments, each with a three-component throttle vector to balance thrust efficiency with mass-update relations. The optimization criteria focused on minimizing total post-launch ΔV while adhering to a strict launch ΔV cap of 7.0 km/s provided by the Falcon Heavy launch vehicle. Key assumptions include a continuous thrust model with a maximum cap of 0.30 N and a constant engine I_{sp} of 3100.

long with trajectory optimization and scientific importance, the mission robustness was addressed in selecting the July 28, 2029 launch window. While the direct, no-flyby ordering was identified as the global optimal for the selected asteroid triplet, the low-thrust degrees of freedom provide inherent flexibility for launch delays. Preliminary analysis suggests that a 1-3 month launch delay can be compensated for by adjusting the throttle duty cycles during the Earth-to-Parthenope cruise, though this would result in a marginal increase in the 13.0701 km/s ΔV budget and potential reduction in science stay times.

2.2 Mission Timeline (Marissa Till)

The AFONSO mission launches on July 28, 2029, arriving at 11 Parthenope approximately 3 years and 9 months later. The spacecraft will remain at Parthenope for about 6 months before beginning a nearly 7-year transfer to 16 Psyche, arriving in October 2040. After spending roughly 1 year at Psyche, the mission will depart for 24 Themis on a transfer lasting about 4 years and 7 months, ultimately arriving at Themis in May 2046. The total mission duration from Earth launch to arrival at Themis is approximately 16.81 years.

Event	UTC Date	Elapsed (yr)
Earth launch	2029 JUL 28 18:56:49	0.00
Arrive 11 Parthenope	2033 APR 30 19:31:29	3.76
Depart Parthenope	2033 NOV 12 08:06:38	4.29
Arrive 16 Psyche	2040 OCT 11 03:01:16	11.20
Depart Psyche	2041 OCT 11 09:01:14	12.20
Arrive 24 Themis	2046 MAY 20 16:22:38	16.81

Table 4: Mission timeline

2.3 Trajectory Overview

2.3.1 Selected Architecture (Logan Smith-Perkins)

The AFONSO mission utilizes a direct, low-thrust interplanetary trajectory. The decision to utilize a direct (no-flyby) architecture was selected following a global optimization sweep conducted in Python, which compared direct transfers against various Mars Gravity Assist (MGA) trajectories. While gravity assists can often reduce fuel requirements, the specific orbital geometry of the 11 Parthenope, 16 Psyche, 24 Themis triplet, combined with the high performance of the NEXT-C ion engine, resulted in the direct architecture providing the most efficient mass-to-target ratio.

Additionally, this sequence is monotonic in heliocentric distance, moving from the inner to the outer main belt (approx. 2.45 AU to 3.14 AU), which aligns with the physical intuition of minimizing the energy required to traverse the asteroid belt. The optimization utilized a Sims-Flanagan model, discretizing each transfer into 15 segments to balance thrust efficiency with mass-update relations. By maintaining a strict launch ΔV cap of 7.0 km/s (consistent with Falcon Heavy performance), the direct architecture maximized delivered mass and simplified mission operations by removing the need for complex planetary encounter flyby geometries.

2.3.2 Optimization Framework (Donny Lu)

The post-launch trajectory was generated by a four-stage optimization pipeline implemented in Python and executed through the `run_ppt_lt_chain.py` runner on a Google Cloud Compute instance. The pipeline is layered so that the most expensive solver, the medium-fidelity Sims–Flanagan least-squares transcription, is invoked only on candidates that have already survived two cheaper screening passes. This structure allowed the entire PARTHENOPE–PSYCHE–THEMIS triplet to be optimized to convergence in a tractable number of solver evaluations while still spanning the global basin structure of the problem.

We emphasize at the outset that the integrated stage is a low-to-medium-fidelity model, not a high-fidelity one. The Sims–Flanagan transcription discretizes each leg into a finite number of segments, each carrying a single constant-thrust (impulse-plus-coast) control, and the heliocentric dynamics are propagated under a simplified two-body model with the Sun as the sole gravitating primary; flyby bodies enter only as point-mass, patched-conic

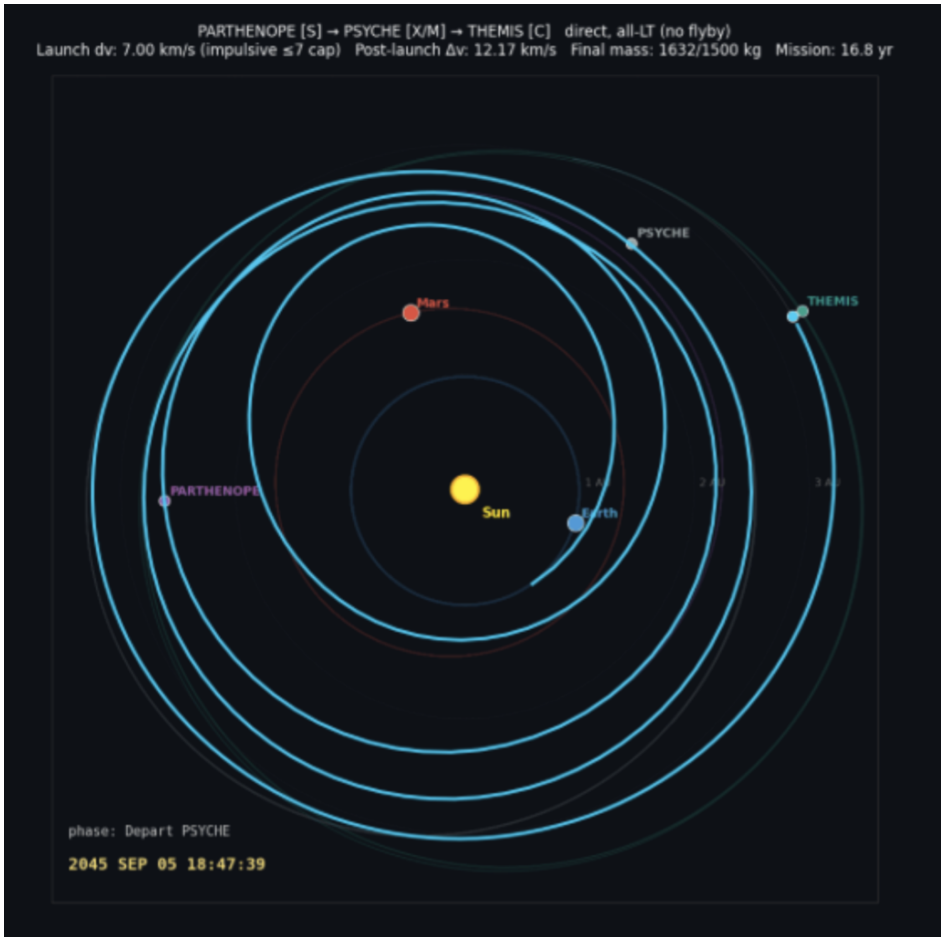


Figure 2: A visualization of the fully optimized trajectory.

perturbations evaluated from their SPICE states. The model will also omits higher-order gravitational perturbations, solar radiation pressure, and any continuously-varying thrust profile. The fidelity claims below are scoped accordingly: this pipeline is intended to resolve the global architecture and the leading ΔV ranking, not to serve as a flight-grade trajectory.

Stage 1: Global architecture sweep.

The optimizer first enumerated all permutations of the three target asteroids together with all admissible flyby architectures. Three architectures were considered (direct, a single Mars gravity assist, and a single Earth gravity assist) which with the $3! = 6$ visit orderings yields $3! \times 3 = 18$ candidate problem instances. Each instance was treated independently in the subsequent stages so that no architectural assumption was hard-coded into the global search. Including the Earth-flyby architectures was deliberate: because Earth supplies a substantially larger gravitational ΔV than Mars, a global-optimality argument requires that it be screened rather than excluded a priori. The cost of doing so is low precisely because the Stage 2 surrogate (below) is inexpensive, which is the intended use of a low-fidelity model, to explore the full design space before committing the expensive solver to a chosen direction. As reported under Results, the Earth-flyby orderings were carried through the surrogate and an exploratory global-optimizer pass but did not return a competitive post-launch ΔV, and were therefore not advanced to the integrated verification stage.

Stage 2: Fast surrogate search.

For each candidate, a low-cost surrogate objective was used to localize the global basin before the low-thrust solver was invoked. The surrogate is implemented in `compute_path_lt_chain_surrogate` and operates on a seven-element decision vector $\mathbf{x} \in \mathbb{R}^7$:

$$\mathbf{x} = \big(t_0,\ \tau_1,\ \tau_2,\ \tau_3,\ \sigma_1,\ \sigma_2,\ v_{\infty,\text{launch}} \big), \tag{1}$$

where t_0 is the launch epoch, τ_i is the time of flight for transfer leg i, σ_j is the stay duration at intermediate asteroid j, and $v_{\infty,\text{launch}}$ is the magnitude of the hyperbolic excess velocity supplied by the launch vehicle. The integrated low-thrust ΔV on each leg is approximated by scaling the corresponding ballistic Lambert solution by an empirical gravity-loss factor; this avoids the cost of a full Sims–Flanagan solve while still capturing the dominant low-thrust penalty.

Constraint handling distinguishes between simple variable bounds and functional inequalities. Each component of $\mathbf{x}$ carries a box bound that defines the searchable domain. As the reviewer correctly notes, these bounds are the optimizer's first check and reject out-of-domain candidates before any objective evaluation, so they cost essentially nothing; they are therefore enforced natively rather than penalized. The smooth soft penalties are reserved only for the functional inequalities that cannot be written as a bound on any single variable, the per-leg thrust-magnitude feasibility, which is a nonlinear function of the whole vector, and the total mission-duration ceiling, which is a coupled (linear) combination of all τ_i and σ_j. A further practical constraint is that the L-BFGS-B polish step (Stage 3) admits box bounds only, with no facility for general linear or nonlinear constraints; routing the duration

and thrust constraints through a smooth exterior penalty lets the refiner be guided back into the feasible region from the "outside" rather than stalling on a hard boundary or requiring a switch to a slower constrained solver.

Stage 3: Multimodal differential-evolution sweep.

The surrogate-localized candidates from Stage 2 are then handed to a multimodal differential-evolution (DE) optimizer. The trajectory cost surface for an interplanetary low-thrust transfer is multi-basin in two distinct ways: the Lambert solver itself admits multiple revolution branches, and stochastic local minima exist in the time-of-flight and stay-duration directions. Both are handled explicitly. Five Lambert multi-revolution combinations $(m_1, m_2, m_3) \in \{0, 1\}^3$ were enumerated as parallel optimization branches, and six independent random seeds were swept within each branch to escape stochastic basins. The DE solver was configured with a population size of 22, a dithered mutation parameter $F \in [0.5, 1.3]$, and a maximum of 400 generations. Each DE run was followed by an L-BFGS-B local polish step to drive the residual gradient toward zero.

Restricting the full multi-seed, multi-revolution DE sweep to the competitive direct and Mars-flyby architectures amounted to approximately $5 \times 6 \times 12 \approx 360$ DE invocations for the full triplet, which on the 12-vCPU instance completed in roughly 20 minutes of wall-clock time. The Earth-flyby architectures were additionally subjected to an exploratory global-optimizer pass but, as noted in Stage 1, did not yield a competitive basin and so did not consume the full sweep budget.

Stage 4: Integrated verification.

Only the surviving best candidate from Stage 3 is passed to the medium-fidelity Sims–Flanagan low-thrust solver. Each transfer leg is discretized into $N = 15$ segments, each carrying a three-component throttle vector $\mathbf{u}_k \in \mathbb{R}^3$ subject to $\|\mathbf{u}_k\| \leq 1$. The half-coast / impulsive-kick / half-coast scheme is solved as a constrained least-squares problem in `scipy.optimize.least_squares` that simultaneously enforces asteroid-rendezvous endpoint matching (position and velocity), the Tsiolkovsky mass-update relation at each segment, and the per-leg thrust ceiling. The constant-thrust-per-segment transcription and the two-body heliocentric dynamics noted above place this stage firmly in the low-to-medium-fidelity class; it is a verification of the surrogate-localized optimum, not a flight-grade integration. Because the stage is invoked only on the single winning candidate, its cost is amortized across the ~ 360 cheaper DE evaluations rather than incurred per candidate. For flyby architectures, the same stage also runs a geometric audit: the periapsis altitude, the required turn angle, and the natural v_∞-conserving turn at the safe periapsis are recomputed from the SPICE state of the flyby body and compared against the constraint floor. For the direct PPT architecture selected, this audit is unnecessary and was not performed. With surrogate-localized starting guesses, this final refinement stage converged in approximately 120 seconds, compared with the roughly 20 minutes spent on the global DE sweep which illustrats the efficiency of reserving the expensive solver for a single pre-localized candidate.

Results

Of the eighteen candidate architectures enumerated in Stage 1, the six Earth-flyby orderings were eliminated at the global-optimizer screening stage for failing to improve on the direct ΔV. The mechanism is the same one that disfavors the Mars assist: the low-thrust degree of freedom already supplies the required heliocentric energy change cheaply, so the phasing detour and the geometric constraints of a planetary encounter (the v_∞-matching tolerance and the turn-angle bound at the safe periapsis altitude) over-constrain the transfer more than the slingshot recovers. Among the remaining twelve instances, all six direct orderings produced a lower converged post-launch ΔV than any of the six Mars-gravity-assist orderings. The ballistic flyby constraint $\big||\mathbf{v}_{\infty,\text{in}}| - |\mathbf{v}_{\infty,\text{out}}|\big| \leq 0.05\,\text{km/s}$, together with the geometric turn-angle bound at the 200 km Mars altitude floor, was found to over-constrain the trajectory more than the slingshot recovers in heliocentric energy. Under the all-electric model, the low-thrust degree of freedom recovers the ballistic gain without the geometric penalty.

The selected solution is therefore the direct, no-flyby ordering PARTHENOPE → PSYCHE → THEMIS, with a converged post-launch integrated ΔV of 13.0701 km/s, a final delivered mass of 1638.1 kg, and a total flight time of 16.81 yr. The full per-leg verification of this solution is reported in Table 5, with all 146 independent checks passing.

While this solution represents a single optimized point, future analysis would benefit from two extensions. First, the Earth-encounter exploration considered only single direct flybys; resonant and multi-revolution Earth encounters, which exchange increased mission duration for additional ΔV, were not enumerated and could in principle reopen the flyby trade. Second, generating a full Pareto frontier of ΔV versus time-of-flight would make the duration–propellant trade explicit rather than reporting a single design point; the present solution was prioritized for its balance of a <17-year duration and high mass-to-target margin.

2.3.3 Delta V Breakdown (Marissa Till)

Table 5 summarizes the mission ΔV budget used for verification. The impulsive Earth launch contributes a v_∞ injection of 7.00 km/s, while the low-thrust transfer sequence contributes a total post-launch integrated ΔV of 13.0701 km/s. All three low-thrust trajectory legs satisfy the verification objectives and reproduce the expected transfer performance.

The detailed per-leg audit in Table 6 confirms consistency in time-of-flight, propellant usage, integrated ΔV, thrust statistics, and endpoint convergence metrics. Leg 1 (Earth → Parthenope) is the highest-propellant segment and operates continuously at full duty cycle, while Leg 2 (Parthenope → Psyche) is the lowest-thrust cruise segment with a 67% duty cycle. Leg 3 (Psyche → Themis) restores continuous-thrust operation and delivers the second-largest integrated ΔV. Position closure errors remain below $1 \cdot 10^{-7}$ AU for all legs, indicating satisfactory Sims-Flanagan endpoint convergence throughout the trajectory.

Table 5: ΔV breakdown verification summary

Burn	ΔV **(km/s)**	**Type**
Earth launch (v_∞)	7.0000	Impulsive (Falcon Heavy)
Earth $\rightarrow$ Parthenope	6.1471	Integrated LT
Parthenope $\rightarrow$ Psyche	2.4758	Integrated LT
Psyche $\rightarrow$ Themis	4.4472	Integrated LT
Total post-launch	**13.0701**	LT Sum

Table 6: Detailed Per-Leg Trajectory Audit

Metric	**Leg 1**	**Leg 2**	**Leg 3**	**Total**
TOF (years)	3.76	6.91	4.61	15.28
Propellant Burnt (kg)	461.0	160.9	258.0	879.9
Peak Thrust Used (mN)	241.4	48.5	99.9	–
Duty Cycle (%)	100%	67%	100%	–
Pos. Error (AU)	4.46×10^{-7}	4.98×10^{-8}	1.51×10^{-7}	–
Vel. Error (km/s)	1.87×10^{-5}	1.95×10^{-6}	5.79×10^{-6}	–

2.3.4 Mass Progression (Marissa Till)

The mass evolution across the mission timeline is consistent with the integrated Δv and the NEXT-C engine's specific impulse of 3100 s. The total propellant starts at 1100 kg, and the total amount consumed during the interplanetary cruise phases is 879.90 kg. The spacecraft's mass begins at a launch-vehicle-constrained 2518.0 kg and reduces to a final arrival mass at 24 Themis of 1638.1 kg.

The propulsion system is designed with a robust fuel reserve to ensure mission success in the event of launch delays or unexpected trajectory corrections. After the completion of the primary interplanetary legs, approximately 219.98 kg of Xenon propellant remains in the tanks. This represents a fuel margin of 25.0% relative to the nominal cruise propellant used, or 8.7% of the total initial spacecraft mass.

This surplus provides a significant safety margin for proximity operations, station-keeping at the three target asteroids, and potential extended mission phases. Furthermore, because the maximum thrust commanded by the optimizer remains below the engine's hard cap of 0.30 N, the spacecraft retains a thrust margin that allows for increased throttle to compensate for any temporal deviations in the mission timeline.

2.3.5 Throttle/Thrust Profiles (Logan Smith-Perkins)

The low-thrust trajectory was verified using a 15-segment Sims-Flanagan integration per leg to generate integrated throttle and thrust profiles. As shown in the per-leg audit, the mission utilizes the NEXT-C ion thruster at a tuned I_{sp} of 3100 s. The maximum thrust commanded by the optimizer across all mission phases is 270.3 mN (during Leg 3), which

Figure 3: Thrust and mass throughout the mission.

remains below the engine's tuned maximum capability of 300 mN, providing a thrust margin for trajectory contingencies.

Leg-Specific Performance Summary:

- Leg 1 (Earth → Parthenope): This leg requires a continuous 100% duty cycle. It features a peak thrust of 241.4 mN and a mean thrust of 119.2 mN, corresponding to a maximum throttle factor of 0.8045.
- Leg 2 (Parthenope → Psyche): Characterized as a cruise phase, this leg operates at a reduced 66.7% duty cycle. The peak thrust is significantly lower at 48.5 mN, with a mean thrust of 22.5 mN, and a maximum throttle of only 0.1616.
- Leg 3 (Psyche → Themis): The final leg restores a 100% duty cycle to reach the outer belt. It reaches the highest peak thrust of the mission at 270.3 mN (Throttle: 0.901) with a mean thrust of 54.3 mN.

The throttle profiles indicate that the spacecraft spends the majority of the mission at mid-to-low throttle settings, which preserves engine life and ensures that the power subsystem is not over-taxed at high heliocentric distances where solar flux is reduced. Position and velocity closure errors at the leg endpoints are negligible, confirming that the integrated thrust vectors accurately achieve the required rendezvous conditions at each target asteroid.

2.4 Mission Schedule of Operations (Hannah Ramsperger)

The mission schedule of operations shown in 7 outlines the detailed timeline from launch on July 28, 2029, to mission end on May 20, 2047, indicating key events, mission phases, and science objectives. The schedule includes interplanetary cruise phases, orbital insertion and departure dates, altitude-specific observation orbits (A, B, C), and transfer periods. The transfer periods have been estimated based on the Psyche mission's 23-day transfer time between its 190 km altitude orbit and its 303 km altitude orbit [10]. Since the transfer times have not been optimized or confirmed by code, the timeline at each asteroid remains an educated estimate and requires further work.

Table 7: Mission Schedule of Operations

Date	**Event**	**Mission Phase**	**Science/Functional Objective**
7/28/2029	Launch Earth		
7/28/29–4/30/33	Cruise from Earth to Parthenope	Solar System Traversal	
8/3/2029	Turn on magnetometer		Measure magnetic field throughout mission for noise calibration
10/1/2029	Camera Calibration		
11/1/2029	Spectrometer Calibration		
1/20/33–4/30/33	40 OpNav Images		Day-to-day orbit determination for accurate orbit insertion
4/30/2033	Arrival Parthenope		
4/30/33–7/16/33	Orbit A (77 days) Mean Altitude: 307 km	Coverage Orbit	Size and shape characterization, imaging coverage, topology, geologic and temperature mapping, composition investigation
7/16/33–8/14/33	29-day Transfer	Orbit Transfer	
8/14/33–11/12/33	Orbit B (90 days) Mean Altitude: 78 km	Coverage Orbit	Magnetometry, high-resolution imaging, composition investigation, refine shape model

Continued on next page

Table 7 – continued from previous page

Date	Event	Mission Phase	Science/Functional Objective
11/12/2033	Departure Parthenope		
11/12/33–10/11/40	Cruise from Parthenope to Psyche	Solar System Traversal	
7/3/40–10/11/40	40 OpNav Images		Day-to-day orbit determination for accurate orbit insertion
10/11/2040	Arrival Psyche		
10/11/40–1/21/41	Orbit A (102 days) Mean Altitude: 309 km	Coverage Orbit	Size and shape characterization, imaging coverage, topology, geologic and temperature mapping, composition investigation
1/21/41–2/13/41	23-day Transfer	Orbit Transfer	
2/13/41–5/24/41	Orbit B (100 days) Mean Altitude: 192 km	Coverage Orbit	Improved topography, magnetometry, composition analysis, refine shape model
5/24/41–6/3/41	10-day Transfer	Orbit Transfer	
6/3/41–10/11/41	Orbit C (130 days) Mean Altitude: 91 km	Stability Orbit	High-resolution imaging, composition investigation, magnetometry
10/11/2041	Depart Psyche		
10/11/41–5/20/46	Cruise from Psyche to Themis	Solar System Traversal	
2/9/46–5/20/46	40 OpNav Images		Day-to-day orbit determination for accurate orbit insertion
5/20/2046	Arrival Themis		
5/20/46–8/18/46	Orbit A (90 days) Mean Altitude: 152 km	Coverage Orbit	Size and shape characterization, imaging coverage, topology, geologic and temperature mapping, composition investigation
8/18/46–9/10/46	23-day Transfer	Orbit Transfer	

Continued on next page

Table 7 – continued from previous page

Date	Event	Mission Phase	Science/Functional Objective
9/10/46–12/22/46	Orbit B (103 days) Mean Altitude: 78 km	Coverage Orbit	High-resolution imaging, composition investigation, refine shape model
12/22/46–12/27/46	5-day Transfer	Orbit Transfer	
12/27/46–5/16/47	Orbit C (140 days) Mean Altitude: 23 km	Stability Orbit	Magnetometry, high-resolution imaging, composition investigation
5/16/47	End of mission		

2.5 Asteroid Orbit Design and Optimization (Rajat Gupta)

2.5.1 Overview and Key Goal

Once the interplanetary cruise trajectory delivers the spacecraft to the vicinity of each target asteroid, the mission transitions to its science phase: a sustained orbital observation campaign around each body. This design and optimization is performed using **Tudatpy** combined with **PyGMO**. The primary science objective driving orbit design is the requirement to image at least **80% of each asteroid's surface** (Science Objective SO5), while simultaneously satisfying the altitude constraints imposed by the onboard instruments and spacecraft safety.

Because the three targets 24 Themis (C-type), 16 Psyche (M/X-type), and 11 Parthenope (S-type) differ substantially in size, shape, rotation period, and gravitational environment, a single orbit design cannot serve all three bodies. Instead, a dedicated orbit optimization is performed for each asteroid. Multiple orbits at different altitudes are required at every target because the visible-light camera, the VIR spectrometer, and the magnetometer each have distinct field-of-view widths and resolution requirements that map to different optimal operational altitudes.

The overarching design goals for the orbital phase are:

1. **Coverage**: image $> 80\%$ of the asteroid surface to satisfy SO5.
2. **Stability**: maintain a controlled altitude band above the surface throughout the observation period to protect the spacecraft from collision and to keep the instruments within their operational resolution window.
3. **Feasibility**: the orbit must remain within the altitude corridor [10 km, 550 km] for its full planned duration without active propulsion.

2.5.2 Asteroid Environment Model

Accurately propagating a spacecraft orbit around a small, irregular body requires a high-fidelity dynamical model that captures the dominant forces acting on the spacecraft. Four modelling components are assembled for each asteroid.

Physical Model: Shape and Rotation

The asteroid shape is represented as a **polyhedral mesh** sourced from the DAMIT (Database of Asteroid Models from Inversion Techniques) database. Some orbits are designed to have a really close proximity with the asteroid its suitable to get the actual altitude instead of an approximation using simple spherical shape. The polyhedral representation captures the true surface geometry of the body and is used both for computing the true altitude above the surface (as opposed to the distance from the centre of mass) and for evaluating occultation of solar radiation pressure. It should be noted that gravity model and shape model are considered as independent in tudatpy.

The rotation state is modelled as a simple constant-spin model parameterised by:

- the pole declination δ_p and right ascension α_p, which define the spin-axis orientation in the inertial frame.
- the meridian angle at the reference epoch (set to zero).
- the sidereal rotation rate $\dot{\theta} = 2\pi/T_{\text{rot}}$, where T_{rot} is the body's rotation period.

To find the most accurately available values of pole declination and pole right ascension, we find sourced the value of pole ecliptic latitude and longitude from the DAMIT (Database of Asteroid Models from Inversion Techniques) database, referenced in the inertial ecliptic frame (ECLIPJ2000), and then used a simple geometric conversion given below to get the required value.
The declination is derived using the sine of the latitude:

$$\sin(\delta) = \sin(\beta)\cos(\epsilon) + \cos(\beta)\sin(\epsilon)\sin(\lambda)$$

The right ascension is derived using the ratio of the components, typically solved using the `atan2` function for numerical stability:

$$\tan(\alpha) = \frac{\cos(\beta)\sin(\lambda)\cos(\epsilon) - \sin(\beta)\sin(\epsilon)}{\cos(\beta)\cos(\lambda)}$$

where,

- λ : Ecliptic Longitude
- β : Ecliptic Latitude
- α : Right Ascension
- δ : Declination

- ϵ : Obliquity of the Ecliptic ($\approx 23.439°$)

For 24 Themis, the adopted parameters are $\delta_p = 59.8°$, $\alpha_p = 125.6°$, and $T_{\text{rot}} = 8.374187\,\text{h}$. *The reader can have note that most of the values are presented for 24 Themis to avoid confusion as its the main example we discuss, but the same in depth analysis is performed for each asteroid.*

Dynamical Model: Spherical Harmonics Gravity Field

The gravitational field of each asteroid is modeled with a **spherical harmonic expansion to degree and order 10** (SH10×10). The normalized cosine and sine coefficients $\bar{C}_{nm}$ and $\bar{S}_{nm}$ are computed from the polyhedral shape model, which evaluates the exact gravitational potential of a constant-density polyhedron at any exterior point and then fits the spherical harmonic series. This approach is consistent with the irregular, non-spherical mass distribution of the asteroid. More details about how its derived is given in subsection 2.5.3

For 24 Themis the adopted gravitational parameter is $\mu = 3.8 \times 10^8\,\text{m}^3\,\text{s}^{-2}$ and the SH reference radius is $R_{\text{ref}} = 99\,\text{km}$.

The SH gravity model is preferred over a simple point-mass model because orbital resonances and secular drifts driven by the low-order harmonic terms (J_2, C_{22}, etc.) are the dominant perturbation at the close orbital altitudes required for science operations, and ignoring them would produce significant errors in the predicted ground track, coverage and stability of the orbit.

Perturbations

In addition to the asteroid's own gravity field, the following external perturbations are included in the equations of motion:

- **Third-body point-mass gravity**: contributions from the Sun, Earth, Mars, Jupiter, and Saturn are included. Jupiter and Saturn are dominant perturbers in the asteroid belt due to their large masses and relatively close proximity.
- **Spherical harmonic self-gravity**: the SH10×10 model is applied as described above.
- **Solar radiation pressure (SRP)**: modelled using the cannonball approximation with a reference cross-sectional area of $A = 75 + 2.4 \times 2.2 = 80.28\,\text{m}^2$ (solar panels plus bus), a radiation pressure coefficient $C_R = 1.2$, and a mass of the spacecraft. Shadowing of the spacecraft by the asteroid body (occultation) is computed using the polyhedral shape model.

Propagation and Integrator

All orbits are propagated using the **Cowell formulation** (full Cartesian translational dynamics) implemented in TudatPy (TU Delft Astrodynamics Toolbox in Python). The numerical integrator is a **Runge–Kutta–Fehlberg 7(8) variable-step-size** scheme (RKF7(8)) with:

- minimum step size: 10^{-6} s;
- maximum step size: 1 day;
- relative and absolute error tolerances: 10^{-8}.

These tight tolerances ensure that the orbit remains accurate over the multi-month observation windows. Propagation is carried out in the ECLIPJ2000 inertial frame and terminates if the true altitude above the polyhedral surface falls below $h_{\min} = 10\,\text{km}$ (collision risk) or exceeds $h_{\max} = 550\,\text{km}$ (instrument resolution limit / orbit escape).

Table 8 summarises the complete environment model specification.

Table 8: Asteroid environment model specification.

Component	**Model**	**Reference**
Shape	Polyhedral mesh	DAMIT Database
Rotation	Constant-spin (simple)	DAMIT Database
Reference frame	Inertial ecliptic (ECLIPJ2000)	Spice kernels
Gravity field	Spherical harmonics, 10×10	ESA polyhedral-gravity-model
Third-body perturbers	Sun, Earth, Mars, Jupiter, Saturn	DE440 ephemeris
SRP	Cannonball + polyhedral occultation	–
Propagator	Cowell (translational)	TudatPy
Integrator	RKF7(8) variable step	TudatPy
Tolerances	10^{-8} (rel. & abs.)	–
Altitude floor	$h > 10\,\text{km}$	Collision avoidance
Altitude ceiling	$h < 550\,\text{km}$	Instrument resolution

2.5.3 Spherical Harmonics and Polyhedral Models (Mohanad Ahmed)

Unlike large planets, asteroids are typically irregularly shaped. As such their gravity field is not spherically symmetric. Farther away from the asteroid, the asteroid appears as a point mass and we can approximate the field as spherical. However, close to the asteroid particularly for the orbits we are considering, this is not true. The typical approach towards this is to expand the gravitational *potenital* field in spherical harmonic expansion and then take the gradient. The spherical harmonic expansion is given by:

$$U(r,\lambda,\theta) = \frac{GM}{r}\left[1 + \sum_{n=1}^{\infty}\left(\frac{a}{r}\right)^n \sum_{m=0}^{n} P_{nm}(\cos\theta)\left(C_{nm}\cos(m\lambda) + S_{nm}\sin(m\lambda)\right)\right].$$

where (r,λ,θ) are the radial distance, longitude and colatitude of the point at which the field is evaluated, a is a reference radius, P_{nm} are the associated Legendre polynomials and C_{nm}, S_{nm} are the coefficients of the expansion, sometimes called the cosine and sine coefficients.

In a series of papers, Robert Werner, showed how to compute these coefficients more efficiently than direct simulation [11, 12]. The idea starts from assuming that the polyhedron

has faces that can be decomposed into a triangular mesh. Each face (a triangle) with the origin form a *simplex*. The formula for the gravity of each simplex can be computed analytically, followed by summing over all simplices. Due to space considerations we will leave it to the interested reader to see [12] particularly Equations (2), (11), (12) and the implementation section.

The data representing the simplices for the three asteroids Parthenope (11), Psyche (16) and Themis (24) was obtained from the Database of Asteroid Models from Inversion Techniques (DAMIT) website: `https://damit.cuni.cz/`. For details on the DAMIT database see [13]. To confirm the correctness of the implementation, we verified against the oblate spheroid model of earth and obtained a C_{20} value of -6.26×10^{-4}, which is not very far from the value for earth -4×10^{-4}. Further validation by cross-checking against results from `polyhedral_gavity_model` library: `https://github.com/esa/polyhedral-gravity-model`.

Since the `TudatPy` library used in designing the orbits supports spherical harmonic models, the computed values were used directly in the program. Note that to compute the gravity force field one must compute the gradient:

$$\vec{F} = -\nabla U \tag{2}$$

Although this is doable in spherical coordinates, it suffers from singularities around the poles of the asteroid. The typical method is to transform the potential into Cartesian coordinates and compute the gradient in Cartesian. To do this efficiently, there are algorithms described in [14]. A more modern presentation of this can be found in [15].

A final note, is that the DAMIT 3D model of an asteroid is given in the local (Fixed body) coordinate of asteroid. To carry out computations in the inertial frame one must be able to transform back and forth between these two coordinate systems. Thus one must know the: spin axis of the asteroid, its angular speed and phase of rotation at a certain epoch. All this information can be obtainde from the DAMIT database as: (λ, β) Ecliptic Longitude and Latitude of the spin axis, P the peroid of revolution, initial phase ϕ_0 at initial epoch t_0 expressed in Julian date. The equation now to move from local frame to inertial is given by:

$$\vec{r}_{ecl} = R_z(\lambda)\, R_y(90^\circ - \beta)\, R_z\left(\phi_0 + \frac{2\pi}{P}(t - t_0)\right) \vec{r}_{ast}$$

Note that t is the current time. The values we used for each asteroid are shown in Table 9 and the 3D mesh model is shown Fig. 4.

2.5.4 Orbit Optimization Framework (Rajat Gupta)

Design Variables

The optimizer searches for an optimal set of four initial orbital elements at the target mission epoch:

$$\mathbf{x} = \begin{bmatrix} a & e & i & \Omega \end{bmatrix}^T \tag{3}$$

Table 9: Spin-state parameters from DAMIT and mass, with earth for comparison

Parameter	Parthenope	Psyche	Themis	Earth
λ (deg)	315	36	146	270.0
β (deg)	18	-10	73	66.56
P (h)	13.72205	4.195948	8.374187	23.93447
ϕ_0 (deg)	0	0	0	0
t_0 (JD)	2435475	2457357.4	2438904	—
Mass (kg)	5.45×10^{18}	2.29×10^{19}	2.39×10^{19}	5.9722×10^{24}

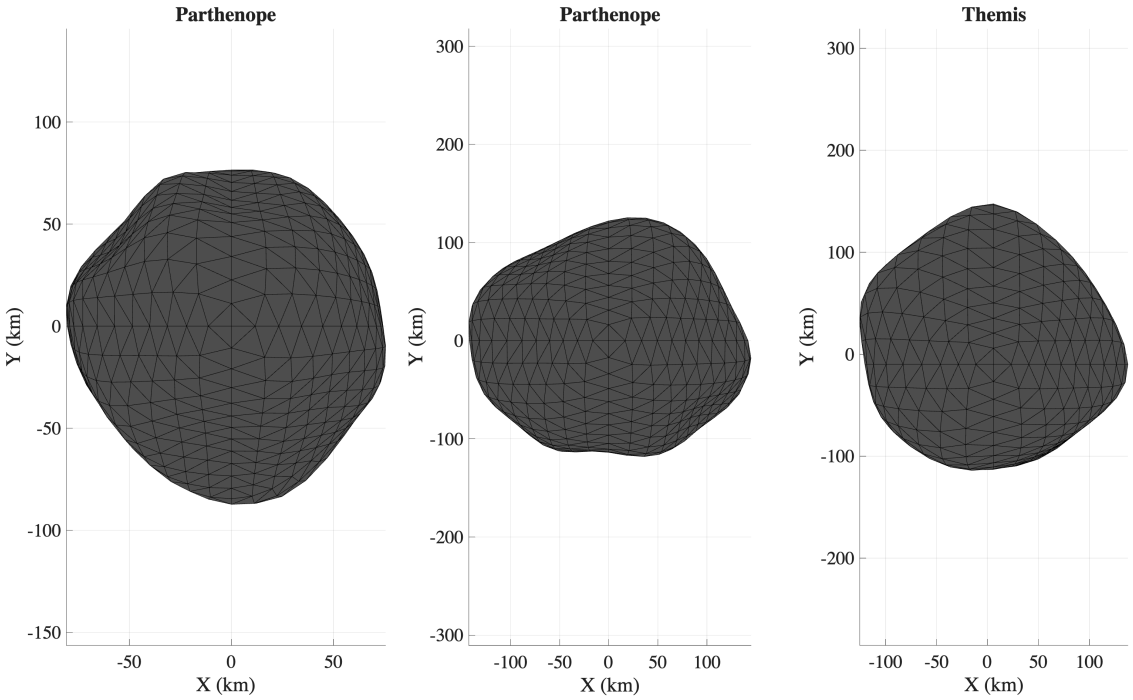


Figure 4: 3D Model of Asteroids from DAMIT

where a is the semi-major axis, e the eccentricity, i the inclination, and Ω the right ascension of the ascending node (RAAN). The argument of periapsis ω and the true anomaly ν at epoch are held fixed at design values since the multi-revolution nature of the orbit renders the coverage insensitive to these parameters over the full mission window. This is a tradeoff made between more accurate orbital element and the simulation time. Adding two more design variable will require more population size and number of evolution which increases the simulation time significantly.

The design variable bounds for the Themis orbit optimization are:

$$a \in [R+h_{req}-10,000, R+h_{req}+10,000]\,\text{m}, \quad e \in [0,\ 0.15], \quad i \in [0^\circ,\ 180^\circ], \quad \Omega \in [0^\circ,\ 360^\circ]. \tag{4}$$

The $\pm 10\,\text{km}$ window on a about a target altitude allows the optimizer to find the precise semi-major axis that balances resonance avoidance with surface proximity.

Multi-Objective Optimization: MOEA/D via PyGMO

Because coverage and orbital stability are competing objectives—a high-inclination retrograde orbit covers the poles but drifts aggressively in altitude due to third-body secular effects—the problem is cast as a **two-objective minimization** problem and solved with the **NSGA-II** (Non-dominated Sorting Genetic Algorithm II) implementation in **PyGMO** (Python Parallel Global Multi-Objective Optimizer).

PyGMO is interfaced with TudatPy through a User Defined Problem (UDP) class. At each function evaluation the optimizer provides a candidate $\mathbf{x}$, the UDP converts it to a Cartesian initial state, runs the full high-fidelity propagation, and returns the two objective values.

Optimizer settings are:

- Fixed random seed: 112987 (for reproducibility);
- Population size: 48 individuals (increased from the baseline 100 to improve Pareto front density at the cost of computational time);
- Number of generations: 60.

Fitness Function Formulation and Rationale (Rajat Gupta)

Objective 01 *[1 - Surface Coverage] (minimize).*

The first objective maximizes the fraction of the asteroid surface that is *observed* during the propagation window. The surface is tessellated into a $5^\circ \times 5^\circ$ latitude-longitude grid of $N_{\text{lat}} \times N_{\text{lon}} = 36 \times 72 = 2{,}592$ cells. Rather than counting cells by number (which would over-weight equatorial cells that subtend larger solid angles), each cell is weighted by its true spherical surface area:

$$A_{ij} = R^2 \left[\sin(\varphi_{i+1}) - \sin(\varphi_i)\right] \Delta\lambda \tag{5}$$

where φ_i are latitude bin edges and $\Delta\lambda = 2\pi/N_{\text{lon}}$ is the longitude bin width. A cell is marked as covered if the spacecraft ground-track position (extracted from the body-fixed spherical coordinates saved as dependent variables) falls within that cell at any propagation step. Because the RKF7(8) integrator uses variable step sizes, the ground-track time series is **uniformly resampled at 60-second intervals** by linear interpolation before coverage is evaluated; this eliminates a systematic bias that would otherwise over-count near-apoapsis passes (slow motion, many raw data points) and under-count equatorial crossings (fast motion, sparse raw data points).

The covered area fraction is then:

$$f_{\text{cov}} = \frac{A_{\text{covered}}}{A_{\text{total}}} = \frac{\sum_{(i,j)\,\in\,\text{visited}} A_{ij}}{\sum_{i,j} A_{ij}} \tag{6}$$

A **survival fraction** f_{surv} is introduced to penalize orbits that crash or escape before completing the planned observation duration T_{obs}:

$$f_{\text{surv}} = \min\left(\frac{t_{\text{final}} - t_0}{T_{\text{obs}}},\ 1\right) \tag{7}$$

The effective coverage is $f_{\text{cov}} \cdot f_{\text{surv}}$, and the first objective to be *minimized* is:

$$\boxed{F_1 = 1 - f_{\text{cov}} \cdot f_{\text{surv}}} \tag{8}$$

Why this choice? A coverage function based purely on geometric footprint counting ignores the latitudinal area distortion of a Cartesian latitude–longitude grid, causing polar regions to appear more visited than they are. The spherical-area weighting in Eq. (5) corrects this. The survival penalty in Eq. (7) prevents the optimizer from exploiting a loophole: an orbit that crashes after 5 days but happens to scan the entire visible hemisphere before impact would score 100% coverage on a naive metric. Multiplying by f_{surv} forces the optimizer to find orbits that are *both* highly covering *and* stable enough to survive the full observation window. A more complicated coverage function can be made if we further multiply it by $Mean(N_{pass})$ which is the mean of number of passes over all the cells, but the results obtained were having lower coverage than desired by the avionics team so this option is discarded.

Objective 02 — *Orbital Stability (minimize).*

The second objective quantifies altitude variability. An orbit that spends part of its time above $h_{\max}$ degrades instrument resolution, while one that dips below $h_{\min}$ risks a surface impact. Altitude variation is measured as the **coefficient of variation** of the true altitude above the polyhedral surface:

$$\boxed{F_2 = \frac{\sigma(h)}{\bar{h} \cdot f_{\text{surv}}}} \tag{9}$$

where $\sigma(h)$ is the standard deviation of the altitude time series and $\bar{h}$ is its mean. The same survival-fraction denominator is applied: an orbit that exits the corridor in the first hour has contributed almost no altitude data, so its $\sigma(h)/\bar{h}$ would be artificially low (stable) without the penalty.

Why altitude and not semi-major axis? An earlier iteration of the objective used $\sigma(a)/\bar{a}$ (relative SMA variation) as the stability metric. However, for an irregular body like Themis, two orbits with identical semi-major axes can have very different altitude profiles: one may hover close to a protruding ridge while the other stays above a depression. Altitude above the polyhedral surface directly encodes the safety margin relevant to collision avoidance and instrument resolution, making it the physically meaningful quantity to stabilize.

Combined objective and Pareto trade-off.

The optimizer minimizes $\mathbf{F} = [F_1,\, F_2]^T$ simultaneously. Because the two objectives conflict (raising inclination toward 90° - 180° improves polar coverage but excites secular inclination oscillations driven by third-body perturbations, increasing altitude variance), no single orbit is optimal in both senses. The NSGA-II algorithm therefore converges to a **Pareto front**: a set of non-dominated solutions from which a mission operator must make a trade-off decision. Two canonical solutions are extracted from the final Pareto front:

- **Best-Coverage (BC) orbit**: the Pareto member with the smallest F_1 value, maximizing surface coverage regardless of altitude stability.
- **Knee-Point (KP) orbit**: the Pareto member closest to the ideal point ($F_1 = 0, F_2 = 0$) in the normalized objective space, i.e., the solution offering the best simultaneous improvement in both objectives the classic "best trade-off" orbit.

2.5.5 Detailed Example: Orbit Optimization around 24 Themis (Rajat Gupta)

24 Themis is the final and scientifically most critical destination of the A.F.O.N.S.O. mission, hosting evidence of water ice and organic molecules on its surface. The spacecraft arrives at Themis on **20 May 2046** and departs on **12 June 2047**, providing approximately 362 of observation time divided across three orbital phases.

Themis Physical Parameters

The polyhedral shape model consists of 402 vertices and 800 triangular facets. After re-centering the mesh to the body's centre of mass (the raw DAMIT mesh had a 6.694 km centroid offset, which was subtracted), the derived radii are:

The large difference between minimum and maximum radius ($\sim 67\,\text{km}$ peak-to-trough) underscores the need for a high-fidelity gravity model: a point-mass assumption would fail to capture the significant non-uniform gravitational pull of this elongated body.

Table 10: 24 Themis derived radii from polyhedral shape model.

Quantity	**Value (km)**
Mean radius	115.893
Minimum radius	85.714
Maximum radius	152.716
SH reference radius R_{ref}	99.000

Optimization Setup for the Themis B Orbit

The Themis orbital phase is divided into three sub-phases (Table 11). The most science-intensive sub-phase is **Orbit B** (103 days, best-coverage orbit at $\sim 78\,\text{km}$ mean altitude), which is selected for the detailed optimization presented here. This orbit directly addresses SO2 (water ice distribution), SO5 (80% surface imaging), SO6 (size and shape determination), and SO7 (organic molecule detection).

Table 11: Themis orbital phase timeline. The detailed mission schedule can be found in section 2.4

Date range	**Phase**	**Duration**	**Mean alt.**	**Orbit type**
20 May 2046	Arrival	–	–	–
20 May - 18 Aug 2046	Orbit A	90 days	152 km	Best coverage
18 Aug - 10 Sep 2046	Transfer	23 days	–	Orbit transfer
10 Sep - 22 Dec 2046	Orbit B	103 days	78 km	Best coverage
22 Dec - 27 Dec 2046	Transfer	5 days	–	Orbit transfer
27 Dec 2046 - 12 Jun 2047	Orbit C	140 days	23 km	Knee-point

The design variable bounds for the Orbit B optimization are: $a \in [175\,000,\ 185\,000]\,\text{m}$, $e \in [0,\ 0.15]$, $i \in [0°,\ 180°]$, $\Omega \in [0°,\ 360°]$. The mission epoch is set to **10 September 2046, 00:40:56 UTC** and the propagation duration is $T_{\text{obs}} = 103$ days.

Convergence of the Optimizer

The NSGA-II optimizer was run for 60 generations with a population of 48 individuals. Figure 5 (reproduced from the optimization output) shows that by generation 28 the mean Pareto-front coverage reaches $\sim$ 84.4% settled around (85-88%), and the best coverage achieved is **98.57%** by the final generation.The Pareto front stabilizes after approximately generation 14, indicating the search has found the global basin of attraction.

Pareto Front Analysis

Figure 6 and Figure 7 shows the Pareto fronts for the initial (random) and final (optimized, generation 60) populations respectively. Each point in the scatter plot represents one candidate orbit, the green staircase curve is the Pareto-optimal boundary. The color of each point encodes one of the four design variables.

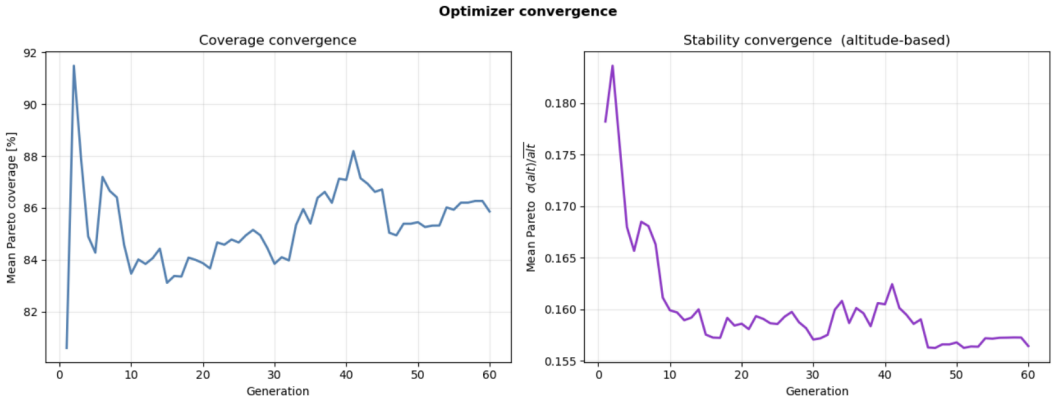


Figure 5: Convergence of the NSGA-II optimizer for the Themis Orbit B case. Left: mean Pareto-front coverage [%] vs. generation. Right: mean Pareto-front altitude stability $\sigma(h)/\bar{h}$ vs. generation. Both objectives improve monotonically and stabilize by generation 30.

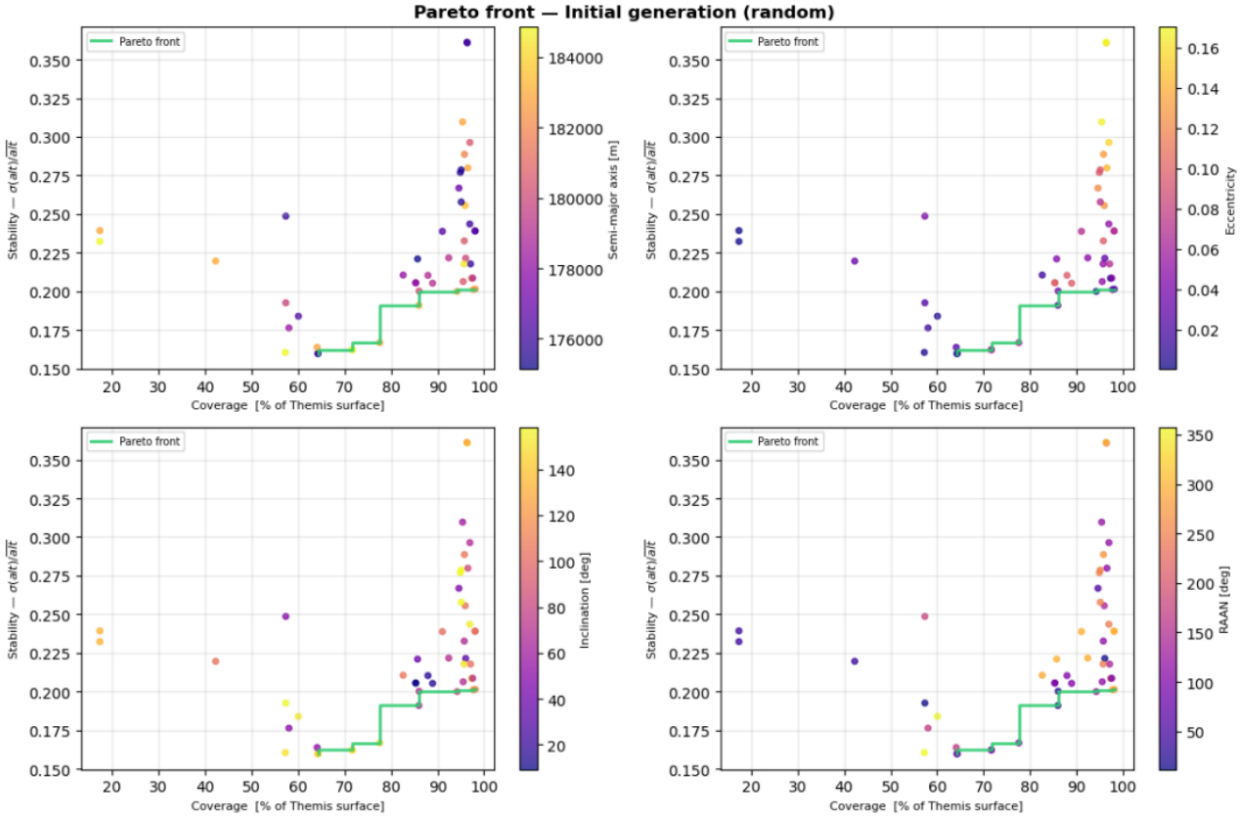


Figure 6: Pareto fronts for the Themis Orbit B optimization. Each subplot colors the population by a different design variable. The initial generation spans the full feasible space at random.

Three key observations from the Pareto analysis:

1. **Semi-major axis** converges tightly to $\sim$ 184.7–185.0 km in the final generation, indicating that the optimal orbit altitude is well-constrained by the combined requirements

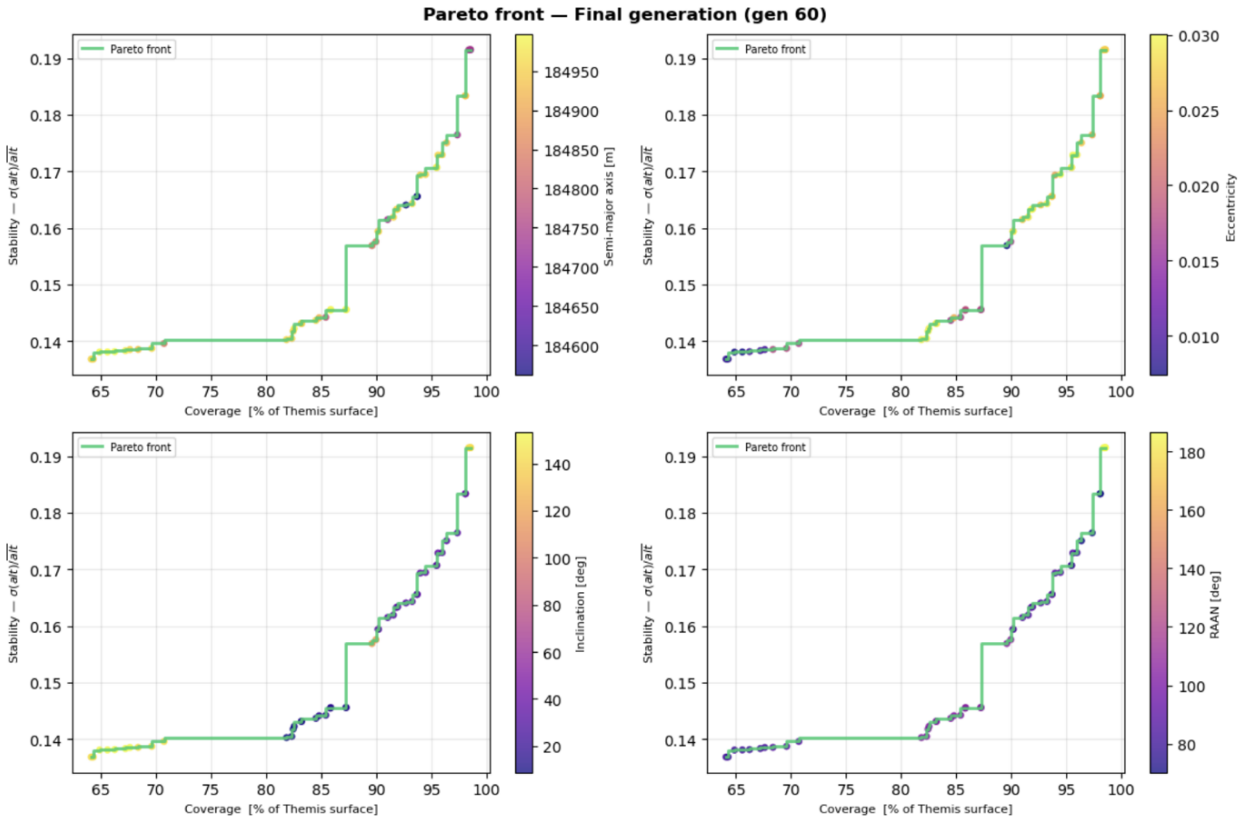


Figure 7: Pareto fronts for the Themis Orbit B optimization. Each subplot colours the population by a different design variable. The final generation is tightly packed near the Pareto boundary, indicating convergence..

of surface proximity and gravity resonance avoidance.

2. **Eccentricity** is consistently low ($e < 0.03$) for high-coverage, high-stability solutions, confirming that nearly circular orbits are optimal around Themis at this altitude.

3. **Inclination** bifurcates: the highest-coverage orbits favor $i \approx 140°$–$143°$ (retrograde near-polar), while the most stable orbits prefer $i \approx 8°$–$16°$ (low-inclination prograde). This bifurcation is the fundamental tension captured by the two-objective formulation.

Selected Orbits and Key Results

Two orbits are extracted from the final Pareto front and characterized in detail. Table 12 lists their orbital elements and performance metrics.

Both orbits survive the full 103-day observation window without crashing or escaping. The BC (Best Coverage) orbit achieves **100% surface coverage** (when evaluated on the uniformly resampled 60-second ground track) at the expense of higher altitude variability ($\sigma_h = 14.9\,\text{km}$). The KP (Knee Point) orbit covers 89% of the surface but maintains a tighter altitude band ($\sigma_h = 11.7\,\text{km}$), making it better suited for operations requiring a predictable instrument-to-surface distance, such as the magnetometer measurements (SO3,

Table 12: Orbital parameters and performance metrics for the two selected Themis Orbit B solutions.

Parameter	**Knee-Point (KP) Orbit**	**Best-Coverage (BC) Orbit**
Semi-major axis a [km]	184.990	184.739
Eccentricity e [-]	0.01776	0.02758
Inclination i [deg]	15.99	142.62
RAAN Ω [deg]	105.52	186.65
Coverage (optimizer) [%]	87.26	98.57
Coverage (resampled 60-s) [%]	89.18	100.00
$\sigma(h)/\bar{h}$ [-]	0.1456	0.1915
Altitude min [km]	38.8	33.3
Altitude mean [km]	80.6	77.9
Altitude max [km]	104.3	107.8
Altitude std dev [km]	11.736	14.917
Survival [days / planned]	103 / 103	103 / 103

SO4).

Operational Decision

For Themis Orbit B, the **Best-Coverage orbit is adopted as the baseline** by the avionics subgroup for imaging and spectrometry operations (SO1, SO2, SO5, SO7). Its retrograde inclination of 142.6° ensures that the ground track sweeps both polar regions within a few orbit periods, achieving 100% coverage in 103 days. The Knee-Point orbit is retained as the design for any extended low-altitude phase (Orbit C, mean altitude $\sim 23\,\mathrm{km}$) where altitude stability is paramount for the close-approach magnetometer and spectrometry passes.

Trajectory Visualization

Figure 8 shows the 3D trajectory of the Best-Coverage orbit over the 103-day observation window, rendered with the polyhedral Themis mesh. The dense, quasi-uniformly distributed ground track is the geometric consequence of the high inclination combined with the fast RAAN drift ($\dot{\Omega} \approx +1.28^\circ\,\mathrm{day}^{-1}$, completing a full precession in ~ 280 days) driven by the ten-degree gravity harmonics and third-body perturbations.

Surface Coverage Maps

Figure 9 shows the $5^\circ \times 5^\circ$ surface coverage grids for both selected orbits after resampling.

Altitude Stability Analysis

Figure 11 shows the altitude above the polyhedral surface as a function of time for both orbits.

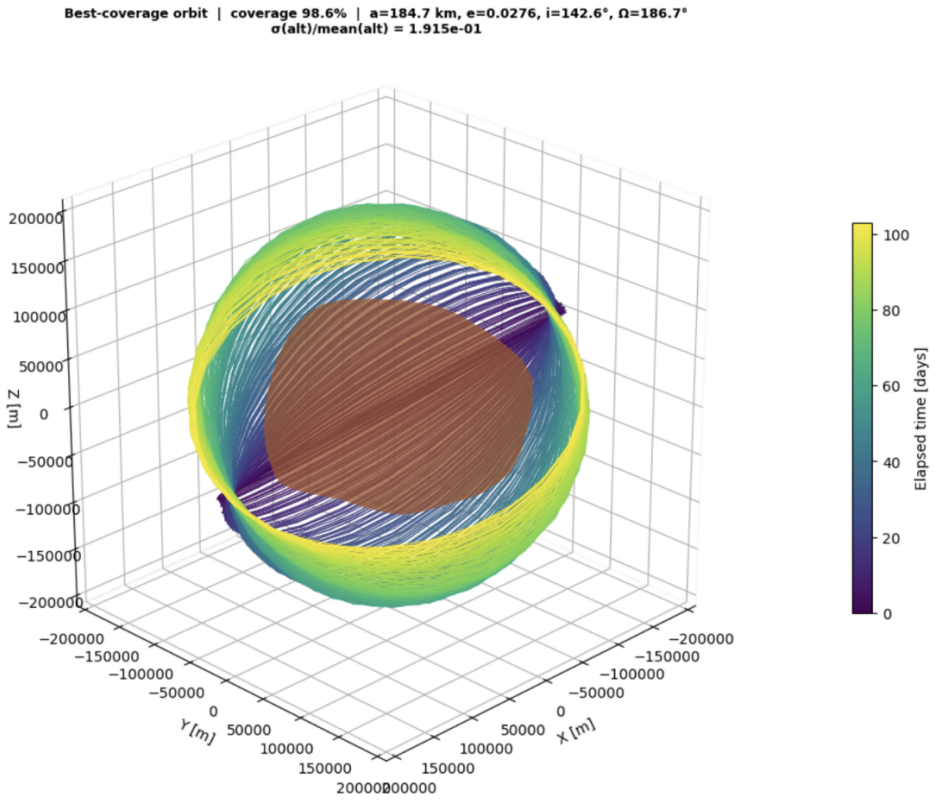
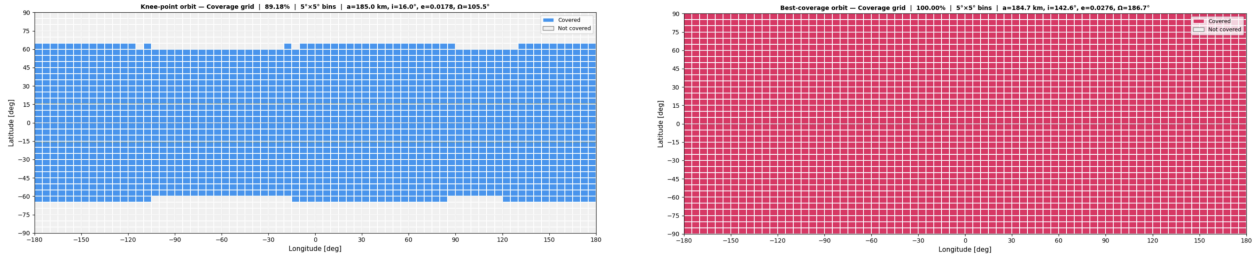


Figure 8: 3D trajectory of the Themis Best-Coverage orbit (Orbit B, 103 days). Orbital elements: $a = 184.74\,\text{km}$, $e = 0.028$, $i = 142.6°$, $\Omega = 186.7°$. The trajectory is coloured from purple (day 0) to yellow (day 103). The polyhedral Themis body is rendered in brown.



(a) Knee-Point orbit (89.2% coverage)

(b) Best-Coverage orbit (100.0% coverage)

Figure 9: Themis surface coverage grids ($5° \times 5°$ cells) for the Knee-Point orbit (left) and Best-Coverage orbit (right). Blue cells are covered; light grey cells are uncovered. The KP orbit leaves a small uncovered band near the poles consistent with its low inclination of 16°. The BC orbit achieves complete global coverage owing to its near-polar retrograde inclination of 143°.

Key altitude statistics for the Best-Coverage orbit:

- Minimum altitude: 33.3 km (no collision risk; floor at 10 km).
- Mean altitude: 77.9 km.

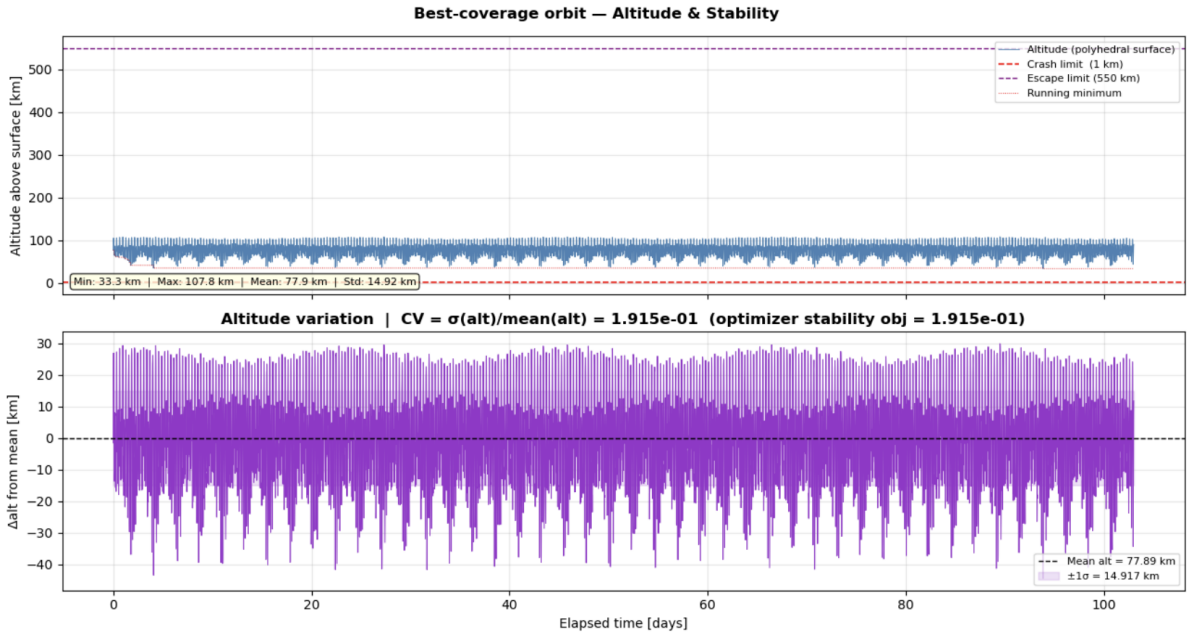


Figure 10: Altitude above Themis surface for the Best-Coverage orbit over 103 days. The oscillation at the orbital period reflects the true topographic variation of the irregular body. The minimum altitude of 33.3 km remains well above the 10 km collision-avoidance threshold throughout the observation window.

- Maximum altitude: 107.8 km (well below the 550 km ceiling).
- Standard deviation: 14.9 km ($\approx 19\%$ of mean).

The altitude oscillates at approximately the orbital period (~ 7.5 h), driven by the eccentricity and by the irregular mass distribution of Themis. There is no secular drift in the altitude envelope over the 103-day window, confirming that the orbit remains bounded for the full planned observation duration.

Image and Spectrometer Acquisition Coverage Verification

In addition to the geometric coverage optimization, the actual science data acquisition coverage was verified by simulating the instrument footprints along the orbital ground track. With an image acquisition rate of 2 minutes per image and a spectrometer acquisition rate of 2 minutes per spectral cube. The same plots are generated for all the asteroids orbit in collaboration with the avionics subgroup to decide the suitable acquisition rate for imager and spectrometer.

- **Imager** (1200×1600 px, footprint width $\approx$ alt$\times$IFOV): achieves **100% area coverage** of Themis surface during Orbit B.
- **VIR Spectrometer** (slit width = alt $\times\, 64$ m/km, slit length = $v \times 15$ s): also achieves **100% area coverage** of Themis surface during Orbit B.

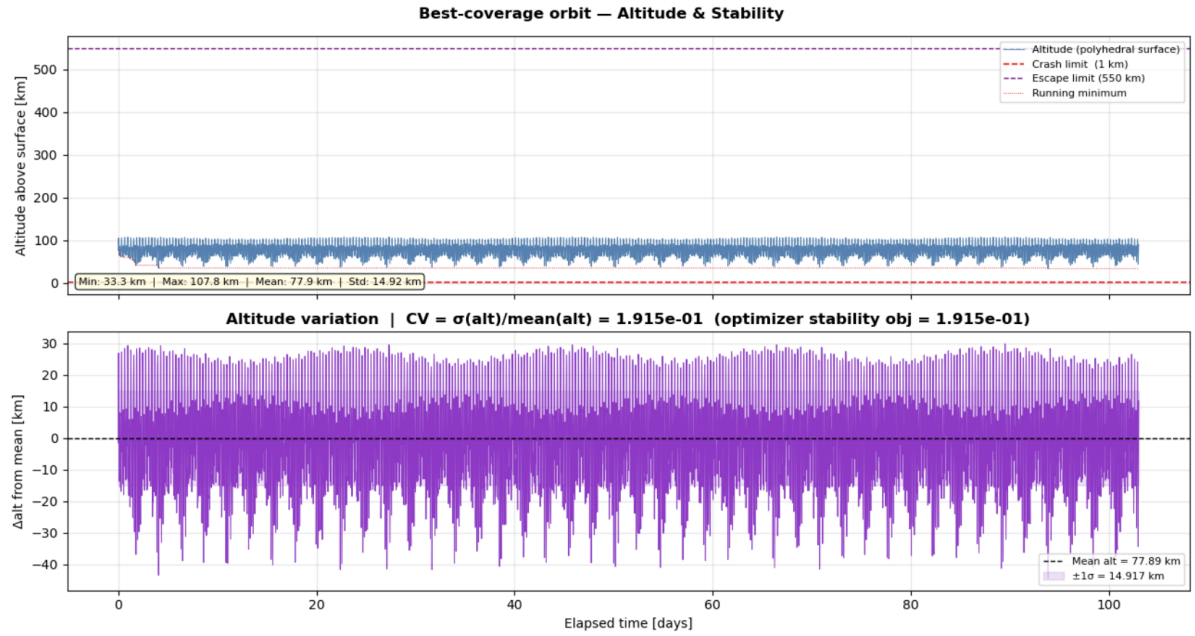


Figure 11: Altitude above Themis surface for the Best-Coverage orbit over 103 days. The oscillation at the orbital period reflects the true topographic variation of the irregular body. The minimum altitude of 33.3 km remains well above the 10 km collision-avoidance threshold throughout the observation window.

Both instruments satisfy the $> 80\%$ coverage requirement of Science Objective SO5 with substantial margin when operating in the Best-Coverage orbit.

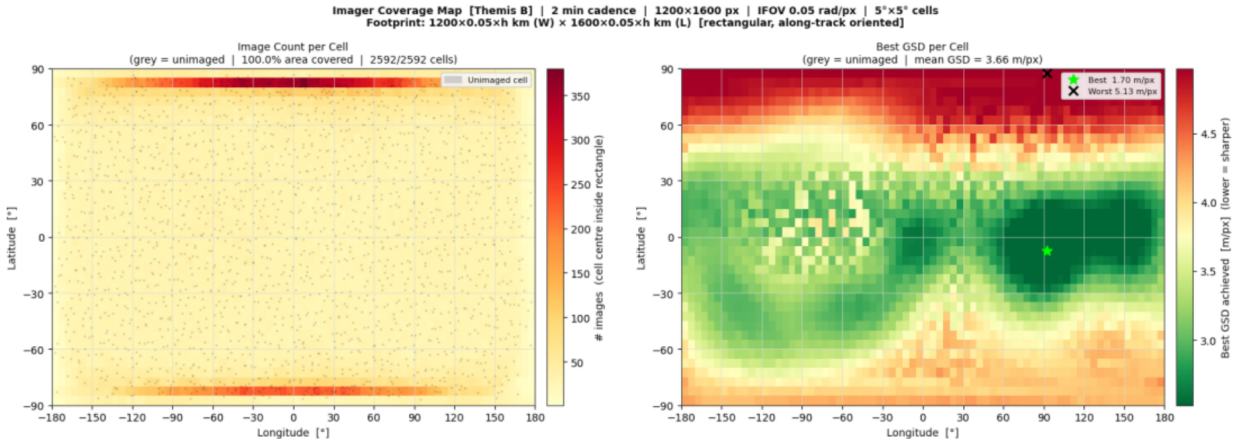


Figure 12: Imager Coverage Map for Themis B

2.5.6 Summary of Orbit Results for All Three Asteroids

Detailed multi-phase orbit optimization was performed for all three target asteroids following the same framework described above. Table 13 provides a consolidated summary of the

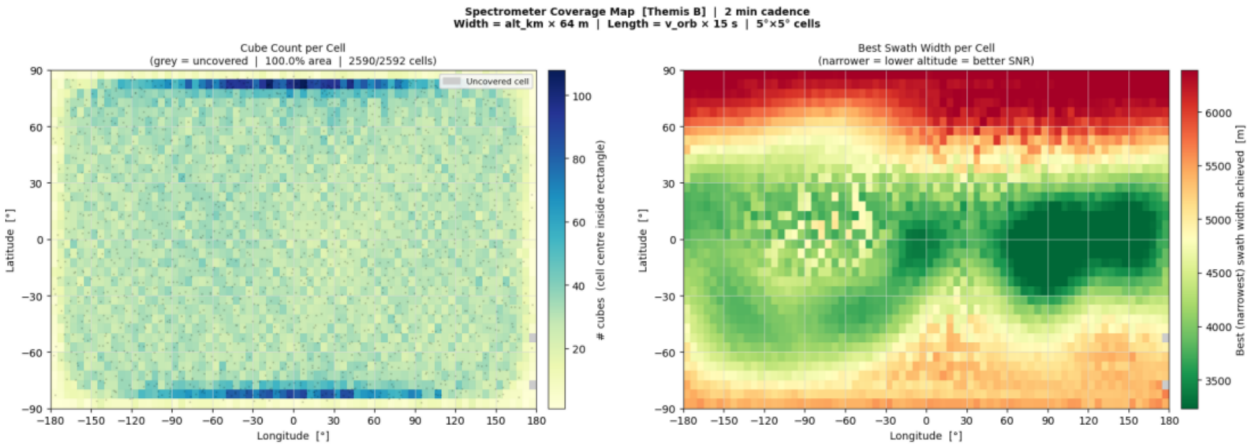


Figure 13: Spectrometer Coverage Map for Themis B.

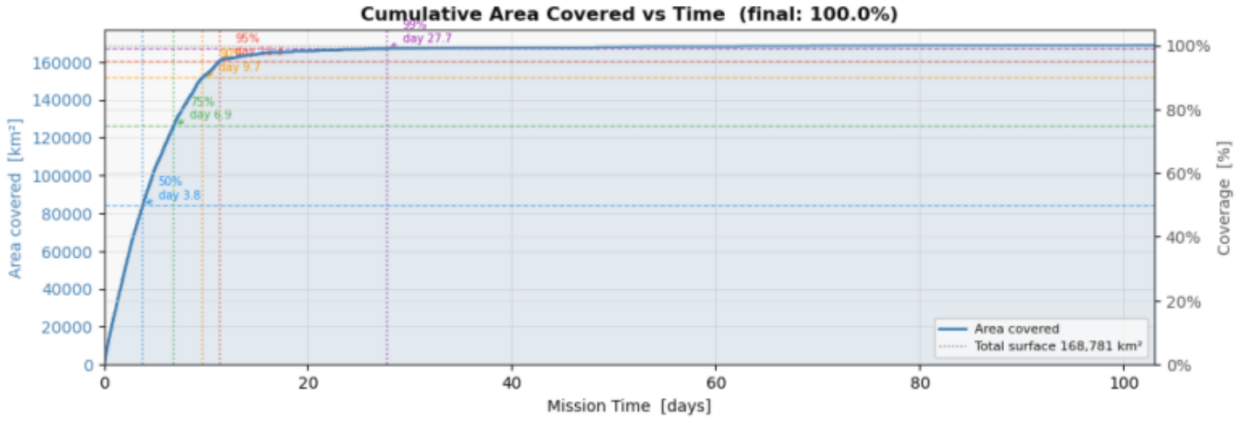


Figure 14: Area Covered vs Time for Themis B

selected operational orbits, and Figure 15 shows representative 3D trajectory plots for each body.

2.5.7 Future Work

The current framework optimizes each science orbit independently, with inter-phase transfers designed separately as fixed manoeuvre windows. Two extensions would improve overall mission optimality.

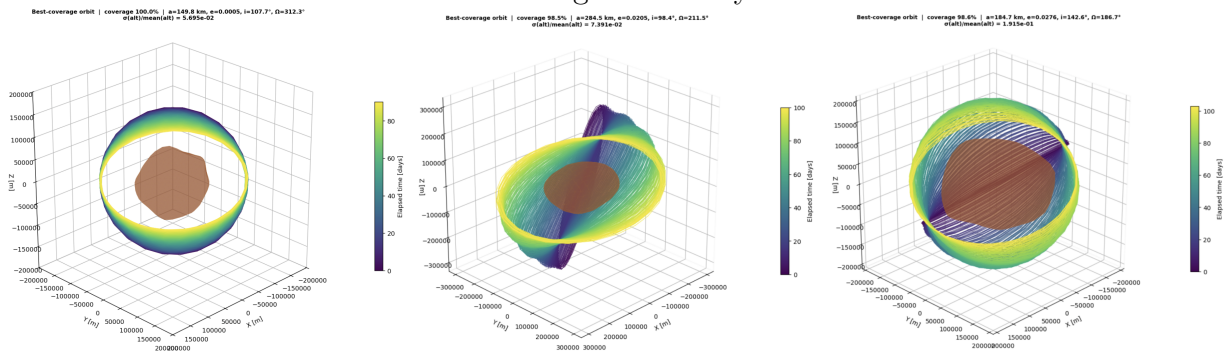
First, the orbit-to-orbit transfer arcs within each asteroid encounter could be optimized using the same Sims–Flanagan low-thrust transcription already employed for the interplanetary cruise, minimizing propellant consumption during altitude descent manoeuvres. Since the transfer epoch also determines the initial RAAN of the subsequent science orbit directly affecting its coverage performance jointly optimizing the transfer duration and target orbital elements would close this currently neglected coupling.

Second, a unified end-to-end optimization could treat all science orbits and connecting trans-

Table 13: Summary of optimized operational orbits for all three A.F.O.N.S.O. target asteroids. "KP" = Knee-Point orbit; "BC" = Best-Coverage orbit. All altitudes are mean altitudes above the polyhedral surface. * corresponds to the coverage obtained after resampling

Asteroid	**Orbit**	**Type**	**Duration**	**Alt. [km]**	a **[km]**	e	i **[deg]**	**Cov. [%]**
11 Parthenope	A	BC	77 days	307	384.5	0.0185	38.3	100.0
	B	BC	90 days	78	149.8	0.0005	107.7	100.0
16 Psyche	A	BC	102 days	302	406.4	0.00964	147.57	98.40
	B	BC	100 days	182	284.451	0.02046	98.36	100
	C	KP	130 days	91	185.0	0.001	89.0	90.6
24 Themis	A	BC	90 days	152	257.9	0.0193	133.6	100*(91.6)
	B	BC	103 days	78	184.7	0.0276	142.6	100*(98.6)
	C	KP	140 days	23	139.9	0.0045	154.1	29.5

Three subfigures side by side



(a) Parthenope Orbit B (78 km) (b) Psyche Orbit B (192 km) (c) Themis Orbit B (78 km)

Figure 15: Best-coverage orbit 3D trajectories for the primary science orbit at each asteroid (trajectory coloured by elapsed time, polyhedral body mesh shown in brown). All three bodies exhibit the characteristic dense ground-track weave of a near-polar retrograde orbit, which ensures global surface coverage within the available observation window.

fer arcs within a single asteroid encounter as one combined optimal control problem. This would allow the optimizer to discover solutions invisible to the sequential approach, such as a marginally sub-optimal first orbit whose RAAN at handover dramatically reduces the cost of reaching the second orbit. Given that PyGMO, TudatPy, and the Sims-Flanagan solver are already in place, this unified formulation is a natural next step for future mission design iterations, but it should be kept in mind that this can be very computational heavy.

2.5.8 Conclusion

The orbit optimization framework developed for the A.F.O.N.S.O. mission successfully identifies science orbits for all three main-belt asteroids that satisfy the 80% surface coverage requirement. The key design choices are:

1. A **multi-fidelity asteroid environment model** combining a polyhedral shape representation with a SH10×10 gravity field and full third-body perturbations ensures that predicted orbit lifetimes and coverage maps are physically reliable.
2. A **two-objective NSGA-II optimization** balances surface coverage against altitude stability, yielding a Pareto front from which two canonical orbits are extracted for operational use: the Best-Coverage orbit for imaging and spectroscopy phases, and the Knee-Point orbit for close-approach magnetometry phases requiring a predictable instrument altitude.
3. The **area-weighted spherical coverage metric** with survival-fraction penalization provides a physically correct and manipulation-resistant objective function that guides the optimizer toward orbits that are both scientifically productive and operationally safe.
4. All selected orbits survive their full planned observation durations without collision or escape, and all achieve coverage well in excess of the 80% science requirement.

2.6 Spacecraft Attitude for Cruise and Proximity (Mohanad)

As briefly described in §1.5 and further described in §3.2, the spacecraft has its main thrust along the +Z axis, the solar panels along the Y-axis with their normal along the +X direction but can be rotated anywhere in the XZ plane. The main payload cameras along the +X axis. The high gain antenna is along +Z axis. During the cruise stage other than brief communication windows, the main requirement is to align the +Z axis with the desired thrust direction and keep the solar cells pointed as much as possible with the sun. In fact most of the trajectory the spacecraft is not thrusting along its direction of travel, see Fig. 16.

During the asteroid proximity part of the mission there are competing requirements. Collecting science data requires pointing the payload (+X) towards the asteroid. Communicating this data required pointing the antenna towards Earth, in case of no occlusion. We also want to point the solar cells towards the Sun. We use a very simple decision procedure, upon arrival at the asteroid we begin science collection for a certain window. Beyond this initial window, communication takes priority. If we are not occluded we point towards earth, otherwise we collect science data. The communication windows and science collection can be seen in Fig. 68.

In brief the attitude planning task simply consists of deciding two variables the orientation of the spacecraft $R_{SC}(t)$, a rotation matrix, and the angle of the solar panels $\theta_{SP}(t)$, at different points in time, t. With some computations, it can be seen, that due to the extra degree θ_{SP} in addition to the 3 degrees of freedom in R_{SC}, the solar panels requirement

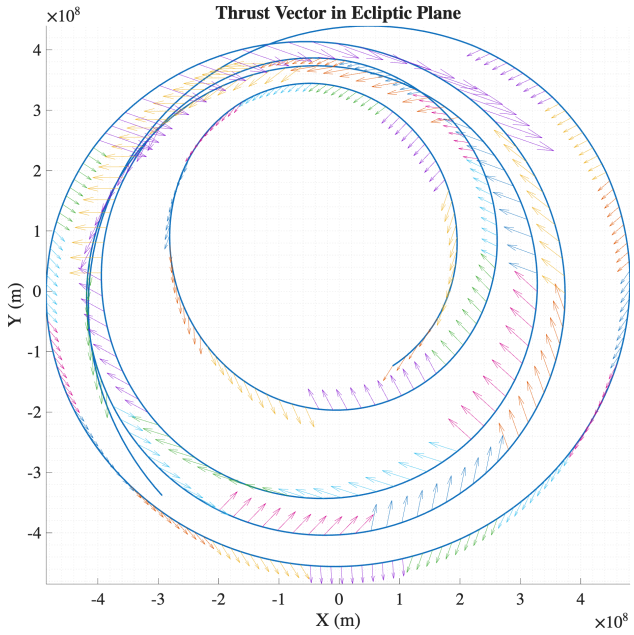


Figure 16: Thrust Direction during cruise phases of the mission

does not contradict thrusting during cruise phase, and does not contradict both science and communication during proximity operations. With simple vector calculations the matrix $R(t)$ and the angle $\theta_{SP}(t)$ can be found.

The spacecraft is equipped with Star Trackers and Solar sensors, in addition to inertial measurement units to well estimate its attitude. The controller that implements this attitude adjustment has not been undertaken in any depth in this study. A slew manuevers and desaturation study was conducted to asses how much fuel is needed. This can be found in Table 38 and surrounding text .

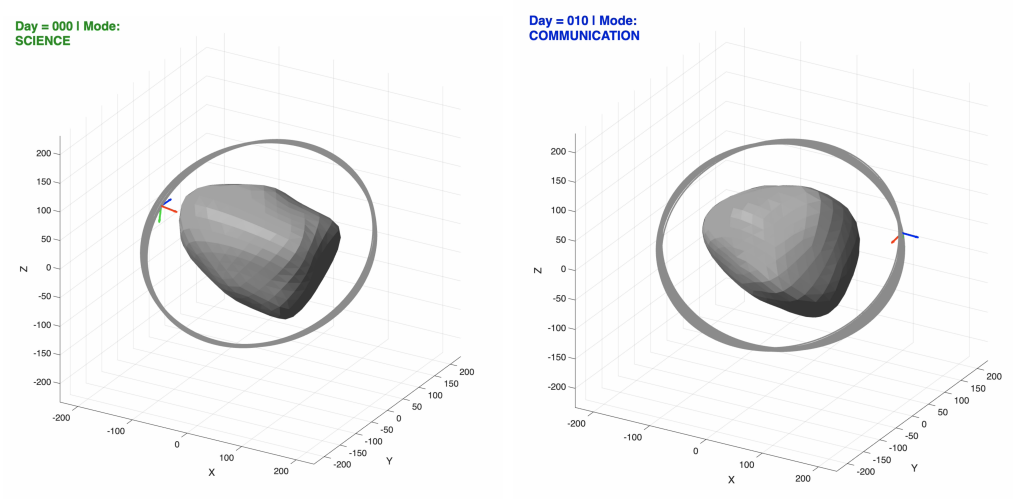


Figure 17: Atittudes during Themis B Orbit for Science collection and Communication. Red: X-axis Green: Y-axis Blue: Z-axis

3 Vehicle Overview

3.1 Payload (Hannah Ramsperger)

3.1.1 Visible Light Camera

The Multispectral Imager will be the high-resolution imaging instrument used to achieve scientific objectives as well as navigation requirements of the mission to the three asteroids. The Imager is inspired by the Multispectral Imager used for the Psyche mission (Figure 18). The Multispectral Imager system also shares significant heritage with imaging instruments flown on the Mars Climate Orbiter, the Mars Science Laboratory and Mars 2020 rovers, and Juno. This lineage provides confidence in the instrument's radiation tolerance, thermal stability, and long-duration performance.

The instrument consists of a pair of block-redundant cameras and associated electronics. The two identical camera heads are connected to a separate Digital Electronics Assembly (DEA) that interfaces with the spacecraft avionics and provides power, commanding, data processing, and onboard image storage. Each head has a mass of 1.77 kg, and the DEA has a mass of 1.58 kg. The system has a maximum power consumption of 8.7 W. The minimum operating temperature is -10°C and the maximum operating temperature is 25°C [16].

The instrument directly supports the mission's primary science goals by acquiring the data necessary to fully characterize the shape, topography, and geology of each asteroid. Additionally, the imager supports optical navigation and provides calibration data for pointing accuracy, jitter, detector stability, and geometric distortion. Each camera has a 1600 by 1200 pixel charge-coupled device detector, a nine-position filter wheel assembly, a compact catadioptric f/2.9 telescope with a fixed focal length of 148 mm, and a sunshade to minimize stray and scattered light. The camera has nine filters: a broadband polychromatic channel centered at 540 nm with a 250 nm bandwidth (clear filter), a neutral density filter to protect from direct solar illumination, and seven narrowband filters spanning wavelengths from 439 nm to 1015 nm [16]. The clear filter will be used during the mission for high-resolution and topographic imaging as well as optical navigation. The narrowband filters overlap with the wavelength range of the Visual and Infrared Spectrometer, and therefore serve as a redundant feature. They can be commanded to cover some parts of the composition analysis if the spectrometer encounters issues as elaborated in 3.1.2 IR Spectrometer.

Each camera has a field of view of 4.6 degrees by 3.4 degrees and an instantaneous field of view of 50 microradians per pixel, enabling imaging of the asteroid at scales ranging from approximately 35 meters per pixel from 700 km altitude to approximately 4 meters per pixel at 75 km altitude [16]. The backup camera, Head B, is toed-out by 3.7 degrees to potentially enable greater surface area coverage per unit time if both imagers were operated simultaneously. However, Head A is planned to perform imaging while Head B serves as a backup. The expected image resolution and surface coverage are specified in 16.

Regarding the science data management, each uncompressed image requires 2.88 MB of storage, while companded images reduce this to 1.92 MB. The instrument is equipped with 8 GB of flash storage, capable of holding approximately 4000 images [16]. The sampling rate

in each orbit and the total data size are provided in 3.1.6 Data Collection.

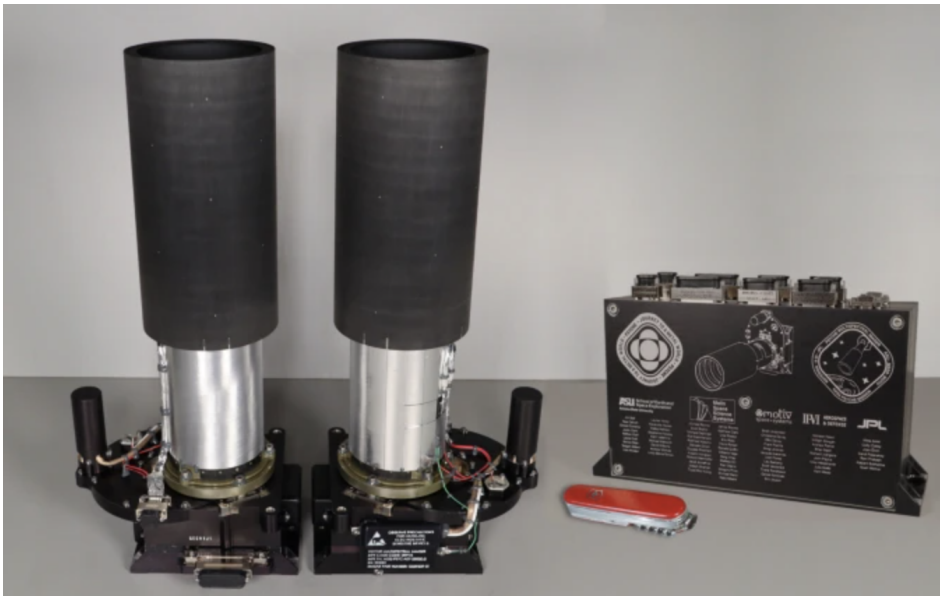


Figure 18: Multispectral Imager camera heads (left) and flight Digital Electronics Assmembly (right) before integration. The 90 mm long pocket knife is pictured for scale. [16]

3.1.2 IR Spectrometer

For mineralogical and chemical composition analysis of the three asteroid targets, the mission will employ a Visual and Infrared Spectrometer (VIR), adapted from the flight-proven design operated on the Dawn mission at both Vesta and Ceres. The VIR instrument is a Visual and Infrared Imaging Spectrometer contained within a single optical head. Spectral images are formed on two matrix detectors to cover the wavelength ranges from 0.25 µm to 1 µm and from 1 µm to 5 µm. The instrument produces spatially resolved spectral images, enabling the determination of the mineral composition of each asteroid's surface and the placement of that composition within its geologic context. The VIR instrument shares significant heritage with spectrometers flown on the Rosetta and Venus Express missions, as well as on Dawn. The spectrometer consists of three primary modules: the optical system with a mass of 5.0 kg, the proximity electronics with a mass of 3.0 kg and power consumption of 5 W, and the cryocooler including its driving electronics with a mass of 1.3 kg and power consumption of 12.6 W. An additional mechanical and thermal mounting structure of 5.0 kg accommodates the spectrometer subsystems [17]. This broad wavelength coverage is essential for detecting the diverse mineralogies expected across three different asteroid types. Wavelengths from 0.2 μm to 0.55 μm detect metals such as iron and nickel as well as sulfides, characteristic of M-type asteroids. The 0.55 μm to 0.7 μm range captures iron charge transfer in iron oxides and sulfides, and the 0.7 µm to 2.5 µm region is diagnostic of olivine, pyroxene, and phyllosilicates, indicative of S-type asteroids. The 1 μm to 3 μm range reveals hydrated minerals and water ice, typical of C-type asteroids such as Themis. Finally, the 3 μm to 5 μm region detects organics and carbonates which are also expected on Themis.

The VIR is a single high-reliability unit with no hardware backup. In case of VIR failure or issues, the imager's filter range of 439–1015 nm serves as a backup for partial science return. Failures can include mechanical failure such as the degradation or contamination of the optical elements, thermal stress cracking or delamination of optical coatings since spectrometers are highly sensitive to thermal cycling, or detector readout failure due to radiation damage to the readout integrated circuit. However, severe radiation is not expected beyond nominal deep-space levels. It is also possible that only one of the two detector arrays for visible and infrared fails, leading to partial failure of the instrument. The thermal and structural systems mitigate these risks by maintaining the detector within its operating temperature range through passive thermal control, reducing thermal cycling stress on optical coatings and electronics. In the event of VIR failure, the imager's filter range of 439–1015 nm still provides partial compositional coverage. The silicates on Themis can be detected within the 0.7–1.0 µm overlap region, however the hydrated minerals, water ice, and organics, which require coverage beyond 1 µm, cannot be recovered. Since Parthenope is an S-type asteroid, its science objectives are still fully covered by the imager alone, as olivine and pyroxene features fall within the 0.7–1.0 µm window. The rocky composition of Psyche can similarly be identified, however its metallic character can only be partially assessed, as the 0.2–0.438 µm range diagnostic of iron and nickel is not covered by the imager. Based on which wavelength bands are lost and which science objectives they serve, full VIR failure is estimated to preserve 65% of the total science return, as Parthenope's objectives are fully met while Themis hydration and Psyche metal science are significantly curtailed. In the case of infrared detector failure alone, coverage of the visible range is retained, preserving Parthenope's full objectives and partial Psyche characterization, yielding an estimated 70% science return. In the case of visible detector failure, the infrared channel continues to provide hydrated mineral, water ice, and organic detection on Themis, with only the short-wavelength metal identification on Psyche is partially lost, preserving an estimated 95% of the desired science return. Given the successful heritage of spectrometer use on the Dawn mission, which also flew a single VIR instrument, combined with the risk mitigation provided by the thermal and structural systems and the demonstrated science return floor in case of failure, a single VIR on board is satisfactory.

The VIR instrument is designed to accommodate a wide variety of albedos, which is necessary given the range of expected target reflectivities. Themis has an albedo of 0.067, where as metallic Psyche's is 0.4. Dawn's mission confirmed this ability by acquiring data for Ceres with an albedo of 0.073 and for Vesta with a maximum albedo of 0.76. Another requirement that is fulfilled as shown by previous missions, is the ability to gather data at different altitudes. The VIR instrument on Dawn successfully measured composition data at orbital altitudes of 385 km at Ceres and 210 km at Vesta, while Rosetta operated at altitudes between 10 km and 450 km and Venus Express between 131 km and 135 km.

Due to the VIR instrument's wavelength limit of 5 µm, thorough thermography is not feasible because the peak thermal emission of these asteroid surfaces, determined from Wien's displacement law, $\lambda_{peak} = b/T$ where b = 2898 µm K, lies approximately between 10-20 µm for the expected surface temperatures. Consequently, the thermal emission peak falls outside the VIR spectral range. Nevertheless, surface temperature measurements remain achievable during daytime observations when surface temperatures are above 170 K, which is sufficiently

high for the short-wavelength tail of the blackbody spectrum to produce detectable radiance within the VIR range below 5 µm. Although the instrument cannot directly observe the peak thermal emission, surface temperature distributions can still be derived from measured thermal radiance within the short-wavelength tail of the blackbody spectrum [18].

For the mission, the VIR instrument will achieve a spatial resolution of approximately 100 meters per pixel from a nominal orbital altitude of 370 km [17], which is sufficient to resolve major geologic units and map compositional variations across each asteroid's surface. The instrument will operate in a nadir-pointed configuration during nominal mapping orbits, acquiring hyperspectral cubes that will be stored in the 6 GB internal memory and data storage, and later processed to produce global and regional mineral maps [19]. The VIR acquires hyperspectral data cubes with a frame size of 432 spectral × 256 spatial pixels and an instantaneous field of view of 250 × 250 µrad [20]. Each cube consists of 15 slit frames collected over 15 s at a sampling rate of 1 Hz [17]. The 1 Hz frame rate was selected as a trade-off between spatial coverage, SNR, and data volume. While higher frame rates reduce the pixel gap between consecutive slit frames, they also increase data rates and decrease SNR. Similar sampling rates have been employed by spectrometers operating at lower altitudes for science measurements [21]. With a bit depth of 16 bits per pixel [22] and a selected compression factor of 5 [23], the combined visual and infrared channels generate approximately 0.42 MB of uncompressed data per frame, corresponding to a compressed data volume of about 1.3 MB per hyperspectral cube.

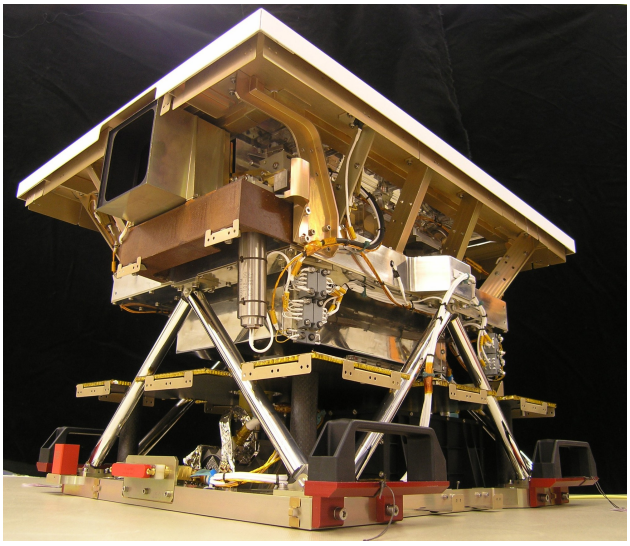


Figure 19: Visual and Infrared Spectrometer (VIR) [24].

Dawn's VIR vs OSIRIS REx' OVIRS Spectrometer Tradeoff

The mission targets three asteroid classes with significantly different surface albedos and operational constraints on orbital altitude and stay time. Therefore, the spectrometer must be capable of acquiring high-quality spectral data across varying orbital distances while remaining effective for both low- and high-albedo surfaces. OSIRIS-REx Visible and Infrared Spectrometer (OVIRS) provides very high-resolution spot spectroscopy and was specifically

designed to achieve high SNR for low-albedo targets such as 101955 Bennu. It covers a wavelength range of 0.4–4.3 µm and is highly effective for targeted compositional analysis [25]. However, achieving its full performance typically requires low-altitude operations of 5 km and long observation periods to map the composition, which are not compatible with the operational constraints of this mission. In contrast, Visible and Infrared Spectrometer (VIR) from Dawn mission offers greater operational flexibility for a multi-target asteroid mission. VIR has demonstrated successful performance over a wide range of orbital altitudes, from approximately 10 to 385 km, whereas OVIRS was optimized primarily for very low-altitude observations around Bennu. This makes VIR better suited for missions requiring measurements from variable orbital distances. VIR also accommodates a broad range of surface albedos (0.067–0.76), making it applicable to both dark and relatively bright asteroid surfaces. In addition, VIR provides broader wavelength coverage with 0.25–5 µm compared to OVIRS' 0.4–4.3 µm, extending further into the ultraviolet and infrared regions. This extended range improves the detection of diagnostic absorption features associated with hydrated minerals, carbonates, and silicates [17]. A major advantage of VIR is its ability to perform spatially resolved imaging spectroscopy, enabling the generation of global compositional maps. OVIRS, by comparison, performs point spectroscopy and does not provide the same imaging capability. For a mission focused on comparative analysis of multiple asteroid bodies, global mapping is a significant scientific benefit. Furthermore, VIR benefits from extensive flight heritage through its use on the Dawn, Rosetta, and Venus Express missions, reducing technical and operational risk. Consequently, for a mission requiring orbital flexibility, compositional mapping, and compatibility with diverse asteroid environments, VIR represents the more suitable instrument choice.

3.1.3 Magnetometer

The primary science objective of the magnetometer investigation is to measure the magnetic field of Themis, Psyche and Parthenope. The choice of magnetometer is the dual-sensor triaxial fluxgate magnetometer with direct heritage from the Psyche mission. The instrument provides a measurement range of +/- 80'000 nT, and samples at up to 50 Hz for science operations, but the sampling frequency can be adjusted to 10 Hz, 1 Hz, or 0 Hz, the latter effectively powering off the sampling function although the instrument remains powered. [26]

The instrument features full redundancy through two sensor units and two electronics units. The primary string consists of Sensor Unit A paired with Electronics Unit A, while Sensor Unit B paired with Electronics Unit B provides a redundant backup. The two sensor also allow for gradiometry to compensate for spacecraft-generated magnetic fields through different placements on the boom [27]. Cross-strapping capability allows either sensor unit to interface with either electronics unit.

To ensure that the science data are not contaminated by spacecraft-generated fields, the magnetometer sensor units are mounted on a boom. The major magnetic subsystems are placed at distal locations on the spacecraft. Furthermore, the power and grounding design for the magnetometer uses separate power return and grounding paths to minimize magnetic interference from other subsystems. The magnetometer calibration, which starts shortly after launch, will make sure that the B-field generated by the spacecraft is properly

characterized such that the noise can be accounted for when evaluating the B-field measurements in the science orbits. Also, magnetic field measurements are affected when the ion thrusters are active due to electromagnetic interference. Although the noise may be partially removed through calibration, this process cannot fully recover the data quality during thrusting periods compared to during coasting phases.

The dominant noise source for this measurement are solar wind fluctuations, which vary with heliocentric distance and solar activity. Because solar wind noise cannot be eliminated by instrument design alone, the mission design incorporates three specific operational strategies to achieve the required signal-to-noise ratio. First, dense spatial sampling is performed during orbital phases. Second, multiple altitude passes are conducted, with high-altitude passes characterizing the background solar wind field and low-altitude passes enhancing the signal from the asteroid. Third, repeated passes over the same terrain are executed to stack data and improve the signal-to-noise ratio through averaging of uncorrelated noise.

To meet these objectives, the magnetometer must resolve magnetic fields with sufficient SNR to detect weak signals. The minimum SNR ratio to detect signals is approximately 5. However, for magnetic field characterization, a SNR of at least 20 is required [28]. The magnetometer samples at 50 Hz for science operations, and all data are transmitted to the ground. One science data packet contains 50 samples, which corresponds to one second of data at the nominal 50 Hz sampling rate. Each uncompressed packet has a size of 500 bytes. This yields an uncompressed science data transmission rate of 500 bytes/s, which is equivalent to 4 kbps. Each compressed packet is approximately 200 bytes, reducing the data rate to 1.6 kbps. Data are transmitted to the ground approximately once per second, meaning one packet per second. Telecommands can be sent to the magnetometer, but they are not required except for the initial power-on sequence [26]. After that sequence, the magnetometer operates autonomously for the remainder of the mission.

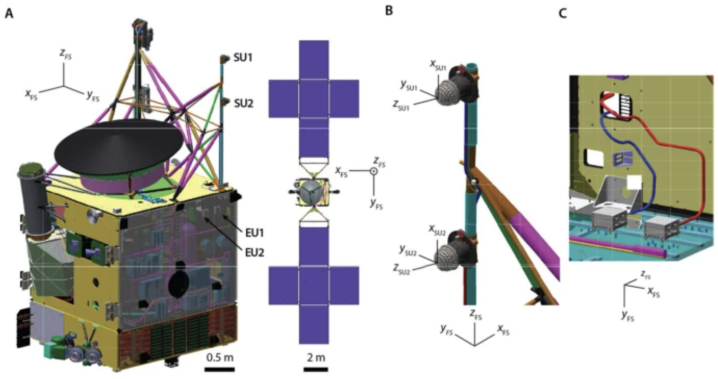


Figure 20: (A) Psyche spacecraft and location of Magnetometer Sensor Units and Electronics Units. (B) Close-up of Sensor Units, (C) Close-up of Electronics Units [26].

3.1.4 Duty Cycles of Science Instruments

The duty cycle shows the operating times of each scientific instrument as well as the power consumption during certain periods. As shown in Figure 21, the magnetometer will be turned

on six days after launch and will operate constantly during the mission due to the importance of measuring the magnetic field of the environment which will determine the noise of the B-field during science measurement around the asteroids. The spectrometer will be turned on three months after launch to calibrate it and afterwards put into standby until arriving at Parthenope. The camera will be turned on two months after launch for calibration and the put into Standby until 100 days before arriving at Parthenope, where it will take 40 optical navigation (OpNav) images in this period of time supporting the telecommands. The same OpNav procedure is performed in the cruise period from Parthenope to Psyche and from Psyche to Themis. The exact OpNav window dates are specified in Table 7.

In general, the VIR will be operating in the orbits around the asteroids during a data collection window and will be in Standby during cruise and telecommunication windows. The imager will be turned on while the spacecraft is in the orbits around the asteroid during data collection windows and will be in standby during telecommunication window. The magnetometer will collect data during telecommunication windows as well. The duty cycles and maximum power consumptions during the asteroid stay times are show in Figures 22,23, 24.

	Event	Spectrometer	Camera	Magnetometer	Power
7/28/2029	Launch Earth	0 W	0 W	0 W	0 W
7/28-8/3		0 W	0 W	0 W	0 W
8/3/2029	Turn on Magnetometer	0 W	0 W	4.4 W	4.4 W
8/3-10/1		0 W	0 W	4.4 W	4.4 W
10/1/2029	Calibration	0 W	8.7 W	4.4 W	13.1 W
10/10/2029	Put in Standby	0 W	8.7 W	4.4 W	13.1 W
10/10/29-11/1/29		0 W	5.6 W	4.4 W	10 W
11/1/2029	Calibration	17.6 W	5.6 W	4.4 W	27.6 W
10/10/29-4/30/33	Camera On/Off for OpNav	5 W	5.6 W	4.4 W	15 W

Figure 21: Duty cycle of the science instruments showing peak power consumption in certain periods during the mission phase between launch and the cruise to Parthenope. During the final 100 days of the cruise phase, the camera operates at 8.7 W only long enough to acquire a total of 40 images and will be on standby with 5.6 W consumption otherwise.

3.1.5 Science Orbits Requirements and Choices

To achieve the scientific objectives specified in Table 1, the orbits must satisfy requirements for stay time, stability, coverage, and altitude to gather useful science data. The optimized trajectory design limits the stay time at the first two asteroid encounters. The stay time at Parthenope is 196 days, and at Psyche, one year. The stay time at Themis is not constrained

Date	Event	Spectrometer	Camera	Magnetometer	Power
4/30/2033	Arrival Parthenope	17.6 W	8.7 W	4.4 W	30.7 W
4/30/33-7/16/33	Orbit A	17.6 W	8.7 W	4.4 W	30.7 W
7/16/33-8/14/33	Orbit Transfer	5 W	5.6 W	4.4 W	15 W
8/14/33-11/12/33	Orbit B	17.6 W	8.7 W	4.4 W	30.7 W
11/12/2033	Departure Parthenope	5 W	5.6 W	4.4 W	15 W
11/12/33-10/11/40	Camera On/Off for OpNav	5 W	5.6 W	4.4 W	15 W

Figure 22: Duty cycle of the science instruments showing peak power consumption in certain periods during the mission phase between arrival at Parthenope and cruise to Psyche. During the final 100 days of the cruise phase, the camera operates at 8.7 W only long enough to acquire a total of 40 images and will be on standby with 5.6 W consumption otherwise.

Date	Event	Spectrometer	Camera	Magnetometer	Power
10/11/2040	Arrival Psyche	17.6 W	8.7 W	4.4 W	30.7 W
10/11/40-1/21/41	Orbit A	17.6 W	8.7 W	4.4 W	30.7 W
1/21/41-2/13/41	Orbit Transfer	5 W	5.6 W	4.4 W	15 W
2/13/41-5/24/41	Orbit B	17.6 W	8.7 W	4.4 W	30.7 W
5/24/41-6/3/41	Orbit Transfer	5 W	5.6 W	4.4 W	15 W
6/3/41-10/11/41	Altitude: 80 km	17.6 W	8.7 W	4.4 W	30.7 W
10/11/2041	Depart Psyche	5 W	5.6 W	4.4 W	15 W
10/11/41-5/20/46	Camera On/Off for OpNav	5 W	5.6 W	4.4 W	15 W

Figure 23: Duty cycle of the science instruments showing peak power consumption in certain periods during the mission phase between arrival at Psyche and cruise to Themis. During the final 100 days of the cruise phase, the camera operates at 8.7 W only long enough to acquire a total of 40 images and will be on standby with 5.6 W consumption otherwise.

by the solar system traversal trajectory and was instead optimized for science requirements. Specifically, three science orbits were demanded: a low-altitude orbit with a stay time of 140 days for magnetometry, and two higher-altitude orbits with a minimum stay time of 80 days each to achieve the coverage objective.

The stay time constraints at Parthenope and Psyche limit the number of science orbits. Given Psyche missions stay times in each science orbit, an evaluation was made of how much time in each orbit is required. Therefore, for a six month stay at Parthenope, two

Date	Event	Spectrometer	Camera	Magnetometer	Power
5/20/2046	Arrival Themis	17.6 W	8.7 W	4.4 W	30.7 W
5/20/46-8/18/46	Orbit A	17.6 W	8.7 W	4.4 W	30.7 W
8/18/46-9/10/46	Orbit Transfer	5 W	5.6 W	4.4 W	15 W
9/10/46-12/22/46	Orbit B	17.6 W	8.7 W	4.4 W	30.7 W
12/22/46-12/27/46	Orbit Transfer	5 W	5.6 W	4.4 W	15 W
12/27/46-5/16/47	Orbit C	17.6 W	8.7 W	4.4 W	30.7 W
5/16/47	End of mission	0 W	0 W	4.4 W	4.4 W

Figure 24: Duty cycle of the science instruments showing peak power consumption in certain periods during the mission phase between arrival at Themis and the end of the mission. During the final 100 days of the cruise phase, the camera operates at 8.7 W only long enough to acquire a total of 40 images and will be on standby with 5.6 W consumption otherwise.

science orbits have been chosen where the higher altitude orbit was selected to be shorter. For Psyche, the longer stay time allows for three science orbits. Again, the low altitude orbit was selected to have the longest stay time to ensure good coverage as well, while the two higher altitude orbits still have around 100 days of stay time.

Multiple orbits at different altitudes are necessary because each instrument has its own requirements for performing valuable measurements. Furthermore, during data collection is the spacecraft required to be nadir-pointed. For all three asteroids, Orbit A provides sufficient imaging resolution to meet the topology and geologic mapping requirements, as well as the coverage percentage (SO5). In addition, the higher-altitude orbit is needed to determine the size and shape and to confirm the spin rate of the asteroids (SO6). The VIR spectrometer will also begin its data collection there for composition analysis, as the Dawn and Rosetta missions proved that measurements at this altitude are feasible.

Psyche is expected to yield an interesting magnetometer investigation. Since the magnetic field can be detected at altitudes below 400 km, and in fact will already be detectable at 300 km, the mission does not transfer directly into a low-altitude orbit below 100 km. Instead, an intermediate-altitude Orbit B ($\geq$ 170 km) has been selected, because a thorough, complete magnetic field investigation requires measurements at several altitudes. This orbit will also be used to refine topology mapping and obtain additional coverage images. For Parthenope and Themis, this intermediate orbit was skipped. In the case of Parthenope, the stay time only allows for two science orbits, and a low-altitude orbit of around 75 km is required for magnetometry, precise spectrometer measurements, and high-resolution imaging. For Themis, the intermediate orbit was also omitted, instead, Orbit B serves for high-resolution imaging and composition analysis, given that Themis is expected to contain water-ice and

organic material, requiring precise spectrometry to fulfill SO2 and SO7.

The scientific instrument that has the most demands on the orbit selection is the magnetometer due rapid magnetic field strength decay over distance, resulting in weak SNR in higher science orbits as shown in Figure 25. This shows that at Psyche a B-field can already be characterized at an altitude of 400 km. For Parthenope, detectability depends strongly on the as-yet-unknown magnetic field. If Parthenope lacks a global magnetic field, the magnetometer will not measure a signal. If it possesses remanent magnetism, data collection can be performed below 100 km, following the faster magnetic field decay model. At Themis, uncertainty remains regarding B-field detectability due to the simplified model and the unknown field strength. Nevertheless, to have any chance of receiving a signal, an altitude of 25 km is required. Consequently, at Psyche, a thorough magnetic field investigation is feasible, with the magnetometer expected to collect valuable data in all three orbits. For Parthenope and Themis, the lowest-altitude orbit may still enable magnetic field characterization sufficient to achieve SO3 and SO4.

Given the magnetometer constraints and the coverage objectives, the mean altitude requirements were specified as shown in Table 14, where the allowed mean altitude deviation determined whether a given trajectory was acceptable.

Table 14: Orbit altitude requirements to achieve scientific objectives.

Asteroid	**Orbit A**	**Orbit B**	**Orbit C**
Parthenope	300 km $\pm$ 25 km	75 km $\pm$ 10 km	–
Psyche	300 km $\pm$ 25 km	$\geq$ 170 km	75 km $\pm$ 10 km
Themis	170 km $\pm$ 25 km	75 km $\pm$ 10 km	$<$ 25 km

The operations subsystem provided two possible trajectories for each science orbit: one prioritizing altitude stability, the other prioritizing coverage. In general, the magnetometer investigation requires a stable orbit to achieve measurements at similar altitudes across different positions, as well as multiple passes and dense sampling. However, the stability orbit's lower inclination did not allow imaging coverage at latitudes higher than 60° for any asteroid. Furthermore, coverage is also constrained by the field of view. Therefore, the knee orbit was chosen only for the lowest-altitude orbits, where altitude was optimized for magnetometry. In the case of Parthenope, the coverage orbit exhibited an altitude fluctuation profile similar to that of the stability orbit, so the coverage orbit was selected even for the lower-altitude orbit.

The resulting mean altitudes and stay times in each orbit are summarized in 7, and achieve the science orbits requirements. The imaging coverage is also achieved as shown in 16.

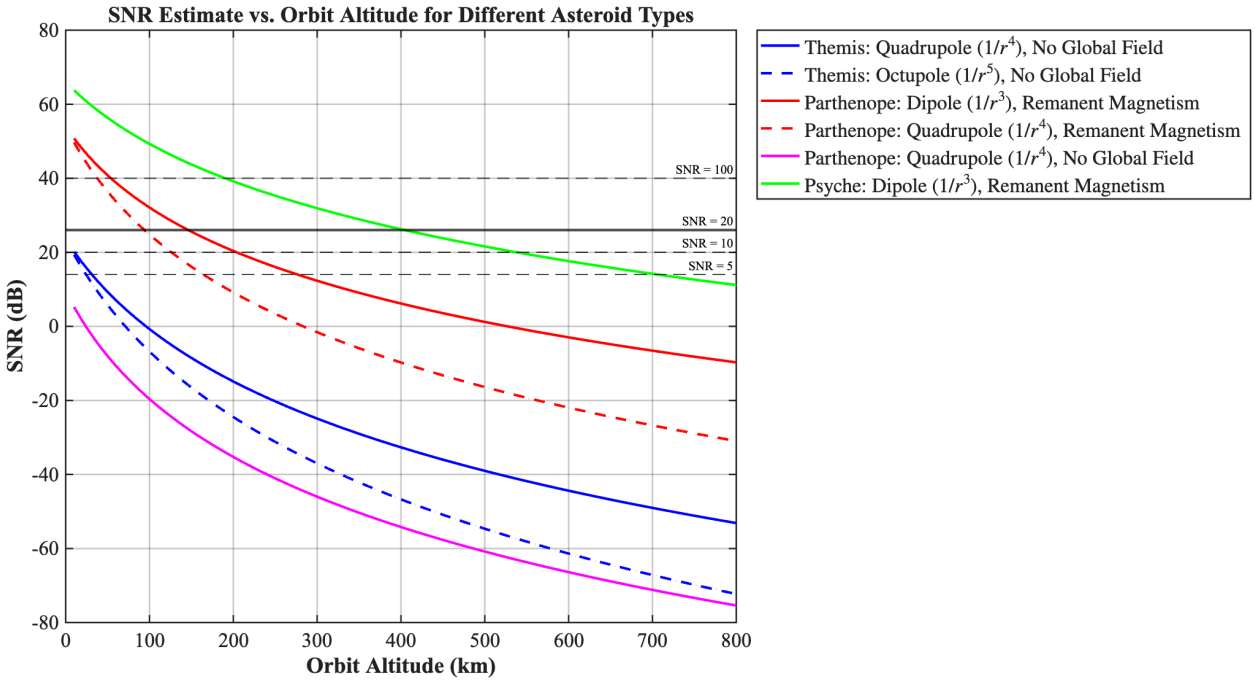


Figure 25: Plot showing the estimated SNR with different models at specific altitudes around the asteroids to determine the lowest orbit altitude at each asteroid. SNR levels of 5, 10, 20, and 100 (i.e., 14, 20, 26, and 40 dB) are marked as reference lines for detectability judgment. Several magnetometer specifications and magnetic field assumptions have been made in the model. The magnetometer has a noise density of 39 $pT/Hz^{-1/2}$ over a 0.9 Hz bandwidth, giving an instrument noise floor around 37 pT. A 1 nT interplanetary magnetic field background was applied for all the asteroids from the solar wind, making the IMF the dominant noise source [26]. Rather than assuming a pure dipole ($1/r^3$) for every body, the model assumed different and multiple multipole orders depending on the asteroid, such as dipole ($1/r^3$) [29], quadrupole ($1/r^4$), and octupole ($1/r^5$), reflecting uncertainty about whether a global organized field or only localized remanence exists [30]. The surface field of Psyche was estimated to be 2 µT given the 0.2 nT to 38 µT range from literature [31]. Parthenope's magnetic field was estimated based on two scenarios: either 500 nT with remanent magnetism or 3 nT for no global field informed for example by the NEAR Shoemaker mission to an S-type asteroid [32]. Themis 15 nT optimistic upper bound was loosely informed by Ryugu flyby data for primitive C-type bodies. [33]

3.1.6 Data Collection

The data collection at the given orbits with the defined stay times was a tradeoff between coverage, sampling rate, data size, and downlinking. Initial sampling rate estimates have been provided to operations, who confirmed the surface coverage ability of that sampling rate. A time versus coverage table was provided, which was used to determine the main science data collection window. Afterwards, the rates and data sizes, as well as the chosen first data collection window, were provided to the telecommunications subsystem. In case of a too large data size, the imager and spectrometer cadence were adjusted. The final

data acquisition rates of the spectrometer and Imager are shown in Table 15. The Imager and VIR cadence ensure satisfactory imaging coverage, spectrometer data collection, and a feasible data size for the telecommunications system and downlinking windows.

Table 15: Data acquisition rates for each asteroid orbit of the mission.

Asteroid Orbit	**Mean Altitude**	**Imager Acquisition Rate**	**VIR Acquisition Rate**	**Data Rate**
Parthenope A	309 km	1 image every 5 minutes	1 cube every 2 minutes	0.01743 MB/s
Parthenope B	78 km	1 image every 2 minutes	1 cube every 2 minutes	0.02703 MB/s
Psyche A	301 km	1 image every 5 minutes	1 cube every 2 minutes	0.01743 MB/s
Psyche B	192 km	1 image every 3 minutes	1 cube every 2 minutes	0.0217 MB/s
Psyche C	91 km	1 image every 2 minutes	1 cube every 2 minutes	0.02703 MB/s
Themis A	152 km	1 image every 5 minutes	1 cube every 2 minutes	0.01743 MB/s
Themis B	78 km	1 image every 2 minutes	1 cube every 2 minutes	0.02703 MB/s
Themis C	23 km	1 image every 3 minutes	1 cube every 2 minutes	0.0217 MB/s

The resulting imaging coverage percentage as well as computed imaging resolution for the mean altitude are given in Table 16. The altitude range shows that the resolution will vary in each orbit and also indicates that data processing after downlinking is required to account for measurements taken at different altitudes within the same science orbit. Note that SO5 is itself satisfied in each science orbit except for Themis C, which is a low-altitude stability orbit meant for magnetic field characterization. The coverage range indicates the lower bound, which is the percentage covered during the first data collection window, and the upper bound, which is the maximum coverage possible, the latter might be achieved in the remaining data collection windows that have been optimized for telecommunication. The scheduled downlink windows and data collection windows are shown in Figure 68. During downlink windows, the imager and spectrometer do not acquire data. However, the magnetometer constantly collects data at 50 Hz. Therefore, dense sampling with multiple passes at different altitudes is performed, which is necessary for magnetometry. The main data collection window, the first, long data collection window, has been determined such that during that period all science objectives are fulfilled and the compressed data can be downlinked. The remaining shorter data collection windows provide extra measurements; however, the coverage of these measurements has not been studied separately. Optimizing data collection windows and downlink windows and including them in the coverage code remains future work.

Given the imager and spectrometer cadence and the duration of the data collection windows,

Table 16: Multispectral Imager resolution, coverage and stability in each science orbit.

Orbit	**Altitude**	**Imager Resolution**	**Coverage**	**Altitude Range**
Parthenope A	309 km	15.5 m/pixel	90-94.3%	286-325 km
Parthenope B	78 km	3.9 m/pixel	80-81.4%	290-323 km
Psyche A	301 km	15.1 m/pixel	90-95.8%	266-324 km
Psyche B	192 km	9.6 m/pixel	95-97%	135-249 km
Psyche C	91 km	4.6 m/pixel	88-90.6%	52-129 km
Themis A	152 km	7.6 m/pixel	90-98.8%	107-188 km
Themis B	78 km	3.9 m/pixel	90-100%	33-108 km
Themis C	23 km	1.2 m/pixel	42-43.5%	1-48 km

the total number of images captured and spectrometer hyperspectral cubes were computed. Given the estimated orbit transfer times, these are subject to change based on actual times and stay time hours. The images and cubes collected in the first data collection window are given in Table 17, and those collected for the remaining windows are shown in Table 18. Overall, the first window is more data-heavy than the remaining windows together. The total number of cubes and images taken at each asteroid and during the mission is shown in Table 19. A total of approximately 189,570 images and 277,530 spectral cubes are taken during the whole mission. Finally, the total science data size was determined. The magnetometer will return 112 GB of data throughout the mission. The multispectral imager will downlink 364 GB of compressed science data, and the spectrometer will downlink 361 GB. In total, 837 GB of compressed science data will be collected during the mission.

Table 17: Images and Spectrometer data taken during the first data collection window at each orbit.

Orbit	Window	Duration	Imager Images	VIR Cubes
Parthenope A	4/30/33-6/2/33	33 days	$\sim$ 10,100	$\sim$ 25,200
Parthenope B	8/14/33-9/22/33	39 days	$\sim$ 28,800	$\sim$ 28,800
Psyche A	10/11/40-11/9/40	29 days	$\sim$ 8,600	$\sim$ 21,600
Psyche B	2/13/41-3/24/41	39 days	$\sim$ 19,200	$\sim$ 28,800
Psyche C	6/3/41-6/22/41	19 days	$\sim$ 14,400	$\sim$ 14,400
Themis A	5/20/46-7/13/46	54 days	$\sim$ 15,800	$\sim$ 39,600
Themis B	9/10/46-9/19/46	9 days	$\sim$ 7,200	$\sim$ 7,200
Themis C	12/27/46-1/31/47	34 days	$\sim$ 16,800	$\sim$ 25,200

Table 18: Images and Spectrometer data taken during the data collection window after the first telecommunication window at each orbit.

Orbit	Duration	Imager Images	VIR Cubes
Parthenope A	27.9 days	$\sim$ 8,000	$\sim$ 20,100
Parthenope B	33.8 days	$\sim$ 24,300	$\sim$ 24,300
Psyche A	0.6 days	$\sim$ 170	$\sim$ 430
Psyche B	5.1 days	$\sim$ 2,400	$\sim$ 3,700
Psyche C	18.1 days	$\sim$ 13,000	$\sim$ 13,000
Themis A	0.0 days	0	0
Themis B	16.9 days	$\sim$ 12,200	$\sim$ 12,200
Themis C	18.0 days	$\sim$ 8,600	$\sim$ 13,000

Table 19: Images and Spectrometer data summary.

	Total Imager Images	Total VIR Cubes
Parthenope	$\sim$ 71,200	$\sim$ 98,400
Psyche	$\sim$ 57,770	$\sim$ 81,930
Themis	$\sim$ 60,600	$\sim$ 97,200
1st Data Collection Window	$\sim$ 120,900	$\sim$ 190,800
Remaining Data Collection Windows	$\sim$ 68,670	$\sim$ 86,730
Mission	$\sim$ 189,570	$\sim$ 277,530

3.2 Structure

3.2.1 Primary Structure Design (Riley Foard)

The primary structure of the spacecraft is based on the Maxar 1300 series bus used by the Psyche mission ([7]), and this bus is a rectangular panel-box construction supported by an internal cylindrical tube made of graphite composite, which achieves a strong primary launch load path through a core and includes flat exterior panels for mounting hardware and radiators. The spacecraft will be launched on the Falcon Heavy rocket, which includes a fairing with a 5.2 m diameter and a 13.2 m overall height ([34]). However, the spacecraft must fit inside a keep-in volume that is smaller than the space inside the fairing to account for payload dynamic deflections relative to the fairing. Thus, the maximum stowed diameter of the spacecraft should be no more than 4.5 m (leaving at least 0.7 m of gross diameter margin) before accounting for fairing wall thickness and keep-in constraints. The maximum stowed height of the spacecraft (including the fixed booms and the antenna stow) should be no more than 10 m, which is significantly less than the 13.2 m overall height. This height constraint reserves several meters for the payload adapter and separation system.

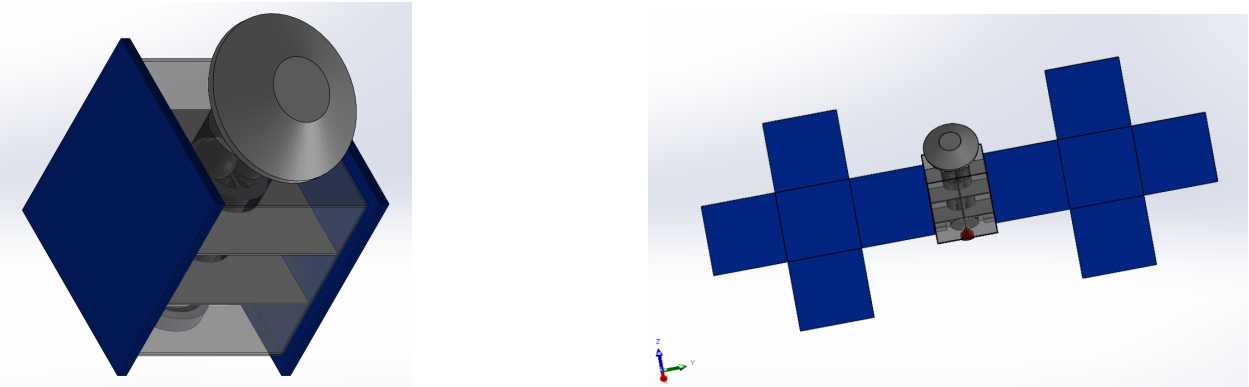


Figure 26: The spacecraft structure with the solar panels stowed (left) and the solar panels deployed (right).

Figure 26 is a CAD assembly showing the spacecraft bus with the solar panels attached to the bus in its stowed and deployed configurations. The bus envelope's dimensions are based on heritage from the Psyche mission. The bus is bound by a rectangular envelope measuring approximately 2.2 m × 2.4 m × 2.4 m, with the height limited to 2.4 m to allow room for the 2 m-long magnetometer booms that will be installed on the +Z panel. This envelope provides the starting geometric constraint for the spacecraft bus and defines the maximum external dimensions used for the initial CAD model. The structure is organized around a central thrust tube, multiple horizontal decks, corner posts, side honeycomb panels, and internal shear bulkheads to create the main load-carrying framework of the spacecraft.

The bottom thrust deck forms the lower structural interface of the spacecraft bus. It connects the spacecraft to the launch vehicle adapter and carries launch loads from the adapter into the spacecraft primary structure. For the initial CAD model, the bottom deck is modeled as a rectangular honeycomb panel with dimensions 2.2 m x 2.4 m x 0.0254 m. The panel thickness of 0.0254 m corresponds to 1 inch, matching the stock honeycomb panels selected

from McMaster-Carr. The stock panels were imported into SolidWorks and their dimensions were edited to match the required spacecraft bus geometry.

At the center of the bus is a central thrust tube (dimensions are 1.0 m ID, 0.004 m thickness, 2.4 m height), which serves as the main compressive load path for launch loads and propulsion loads. The thrust tube is centered within the rectangular bus and extends vertically through the full bus height. The thrust tube should structurally connect the launch adapter ring, bottom thrust deck, internal decks, and top deck. In this configuration, launch loads are transferred from the launch vehicle adapter into the bottom thrust deck and then into the central tube, which carries the primary axial compressive loads upward through the spacecraft. The exact thrust tube dimensions will be refined later using buckling calculations, modal frequency requirements, and mass optimization.

The top deck uses the same general structural concept as the bottom deck but is slightly inset from the side panels (2.15 m x 2.35 m x 0.0254 m) to prevent interference with the side panel attachments and to make room for corner rails or edge closeout features. The top deck can support externally or internally mounted hardware such as antennas, star trackers, avionics boxes, payload brackets, and instrument mounts. Like the bottom deck, it is treated as a honeycomb sandwich panel in the initial structural layout.

Three internal decks are included to divide the spacecraft bus into equipment bays and provide mounting surfaces for spacecraft subsystems. The lower deck is intended for tanks and propulsion supports. The middle deck is intended for avionics, batteries, and power electronics. The upper deck is intended for payload, telecom, and thermal hardware. For the initial CAD model, the internal decks are assumed to have the same size as the top deck. Each internal deck includes a circular cutout for the central thrust tube that is about 0.05 m larger than the central cylinder's outer diameter to account for clearance.

The exterior of the bus is closed using side honeycomb panel tiles. These panels attach to the vertical corner posts and horizontal rails, forming the side walls of the spacecraft bus. The design also includes four internal vertical shear bulkheads help the bus resist lateral shear loads, bending loads, and torsional loads during launch and spacecraft maneuvers. In the initial CAD model, they can be placed in a cross-like arrangement radiating from the central thrust tube toward the four side walls.

The average density of the aluminum honeycomb sandwich panels was estimated so that the panels could be included in the preliminary spacecraft mass budget and later assigned representative material properties in structural simulations. The selected honeycomb panel configuration uses a 5052 aluminum honeycomb core (material density $\rho_{face} = 2680$ kg/m^3) with face sheets. The honeycomb core has a cell size of 1/4 in and a cell wall foil thickness of 0.003 in, corresponding to a nominal core density of $\rho_{core} = 96.1$ kg/m^3 (6 lb/ft^3) ([35]). The core thickness is $t_{core} = 0.0254$ m, and each face sheet has thickness $t_{face} = 8.13 \times 10^{-4}$ m.

The effective density of the honeycomb panel is defined as the total panel mass divided by the total panel volume:

$$\rho_{eff} = \frac{m_{total}}{V_{total}} \tag{10}$$

For a panel with area A, the total mass is the sum of the core mass and the two face-sheet masses:

$$m_{total} = \rho_{core} A t_{core} + 2\rho_{face} A t_{face} \tag{11}$$

The total volume of the full panel is

$$V_{total} = A(t_{core} + 2t_{face}). \tag{12}$$

Therefore, the effective density is

$$\rho_{eff} = \frac{\rho_{core} t_{core} + 2\rho_{face} t_{face}}{t_{core} + 2t_{face}}, \tag{13}$$

resulting in $\rho_{eff} = 252$ kg/m^3. This effective density is much lower than the density of solid aluminum because most of the panel thickness is occupied by the low-density honeycomb core. For preliminary mass budgeting, each honeycomb panel can therefore be modeled using an effective density of approximately 252 kg/m^3 and a total thickness of approximately 27.03 mm.

A preliminary mass optimization was performed to select a starting wall thickness for the spacecraft central cylindrical tube before building the full FEA modal model. The central tube was modeled as a thin-walled cylindrical shell made from an equivalent isotropic graphite composite material with $E = 70$ GPa, $\nu = 0.30$, inner diameter $D_i = 1.0$ m, and height $h = 2.4$ m. The design variable was the tube wall thickness t, and the objective was to minimize the tube mass, $m_{tube} = \rho(2\pi Rht)$, subject to selected lower and upper bounds on manufacturable thickness.

The loading conditions were based on Falcon Heavy quasi-static load cases for payloads above 1800 kg ([34]). Each load case was defined by an axial acceleration factor $a_{z,i}$ and lateral acceleration factor $a_{lat,i}$. For each case, the axial and lateral loads were computed as

$$P_i = m_{sc} a_{z,i} g_0, \;\; F_{lat,i} = m_{sc} a_{lat,i} g_0,$$

where m_{sc} is the spacecraft wet mass and $g_0 = 9.80665$ m/s^2. The lateral force was also used to estimate the bending moment at the tube base, $M_i = F_{lat,i} z_{cg}$, where z_{cg} is the spacecraft center-of-gravity height above the tube base. The tube cross-section was approximated using thin-wall cylinder relations:

$$A = 2\pi Rt, \;\; I = \pi R^3 t, \;\; J = 2\pi R^3 t.$$

The primary structural constraint was cylindrical shell buckling under combined axial compression and bending. The critical buckling stress was modeled as

$$\sigma_{cr} = \gamma \frac{Et}{R\sqrt{3\,(1-\nu^2)}}, \tag{14}$$

where γ is the cylindrical shell knockdown factor. A buckling factor of safety was then applied as

$$\sigma_{allow} = \frac{\sigma_{cr}}{FS_{buckling}}. \tag{15}$$

For each load case, the maximum compressive stress was calculated as

$$\sigma_{max,i} = \frac{\max\left(P_i,\ 0\right)}{A} + \frac{|M_i|\, R}{I},$$

and the buckling constraint was enforced as $\sigma_{max,i} \leq \sigma_{allow}$ for every Falcon Heavy load case.

Additional stiffness and dynamic constraints were included to ensure that the optimized tube would be suitable as a starting point for modal analysis. The lateral deflection constraint was

$$\delta_i = \frac{|F_{lat,i}|\, h^3}{3EI} \leq \delta_{max},$$

with $\delta_{max} = 0.005$ m. The first lateral bending frequency was estimated using a cantilever beam approximation,

$$f_{lat} = \frac{1}{2\pi} 1.875^2 \sqrt{\frac{EI}{\left(m_{sc} + m_{tube}\right) h^3}}, \tag{16}$$

and constrained by the Falcon-recommended primary lateral frequency requirement, $f_{lat} \geq 10$ Hz. The axial frequency was estimated using

$$f_{axial} = \frac{1}{2\pi} \sqrt{\frac{EA}{h\left(m_{sc} + m_{tube}\right)}} \tag{17}$$

and constrained by $f_{axial} \geq 25$ Hz.

The optimization initially produced a minimum feasible tube thickness of approximately 2.173 mm, but this corresponded to a buckling safety factor of approximately 1. To increase the design margins, a 4 mm wall thickness was selected for the initial structural model, resulting in the tube mass of approximately 49.703 kg, a worst-case cylindrical buckling safety factor of 3.388, a lateral frequency safety factor of 3.64, and an axial frequency safety factor of 2.73. Therefore, the selected 4 mm thickness provides a more conservative starting design with all major safety factors above 2 while keeping the central tube mass reasonably low.

3.2.2 Component Placement (Tamar Levin)

The major spacecraft components were placed based on propulsion, structural, thermal, and communication requirements. Here, coordinate origin is taken at the center of the spacecraft. The levels of the spacecraft are defined, from bottom to top: bottom plate, bottom deck, upper deck, upper plate. For reference and coordinate system, see figures 27–29.

- **Fuel tanks:** The xenon fuel tank subsystem is designed to be axisymmetric, with one tank placed in the middle and five tanks surrounding it. It is located inside the inner cylinder, at the geometric center of the structure. The center of the inner tank is designed to coincide with the spacecraft center of mass. The tanks are mounted between two adapters that connect the subsystem to the inner cylinder. The hydrazine fuel tanks are also placed inside the inner cylinder and are mounted on the top adapter of the xenon fuel tank subsystem.

- **Ion thrusters:** The ion thrusters are mounted on the bottom plate, with one facing the $+X$ direction and the other facing the $-X$ direction. The thrusters are tilted so that the thrust vector passes through the center of mass. In practice, some offset is expected, so small torques may still be generated and must be countered by the attitude control system. The two-thruster configuration is used for redundancy, with one thruster designed to operate at a time.
- **ACS thrusters:** Based on propulsion team requirements, the ACS thrusters are placed at four spacecraft corners: $(1.2, 1.1, 1.2)$, $(-1.2, 1.1, -1.2)$, $(-1.2, -1.1, 1.2)$, and $(1.2, -1.1, -1.2)$. They are arranged as four groups, with three thrusters in each cluster.
- **Reaction wheels:** The four reaction wheels are placed in a pyramid configuration, as close as practically possible to the spacecraft center of mass. This reduces torques caused by reaction wheel offsets and reduces the effect of jitter. The wheels are mounted on the outside of the central cylinder, on the bottom deck.
- **Payload:** The payload cameras are placed on the upper plate in the $+Z$ direction such that the cameras face the $+X$ direction. The payload magnetometer is placed on a 2.1 m long fixed boom mounted on the top plate corner at $(-1.2, +1.1)$, extending in the $+Z$ direction.
- **HGA:** The High Gain Antenna is located at the center of the upper plate and is generally pointed in the $+Z$ direction. A gimballed configuration was proposed to provide additional flexibility in communication windows. Based on the current placement of the magnetometer boom and the HGA, the gimbal would allow an azimuth range of approximately $-135°$ to $45°$, with an elevation range up to $90°$. This is the maximum range available, so that the boom does not interfere with the antenna line of sight. The gimbal mast height can be determined based on communication requirements and will define the minimum achievable elevation angle. A mast of 1 m will allow minimum elevation of $0°$. A negative elevation would be impossible since the antenna is places in the center of the upper plate. At the time of this report, the gimballed antenna option had not been implemented in the design.
- **Thermal control components:** The heaters and louvers are placed on the $\pm Y$ faces, covering approximately 3 m^2 of the 5.3 m^2 area on each side.
- **Avionics and battery:** The battery is placed on the upper deck per thermal team requirements. The avionics box is placed on the bottom deck per thermal team requirements.
- **Solar panels:** There are 10 solar panels, five on each side of the spacecraft. Each panel has an area of 7.5 m^2, giving a total area of 75 m^2. The dimensions of each panel are 2.8 m in height, along the Z direction, and 2.7 m in width. When stowed, the solar panels are flush against the $\pm Y$ structural panels. They deploy in the $\pm Y$ direction, such that they face the $\pm X$ direction when deployed. The solar panels have one degree of freedom: rotation about the Y axis. The solar panel mounting and deployment mechanism remains for future work.

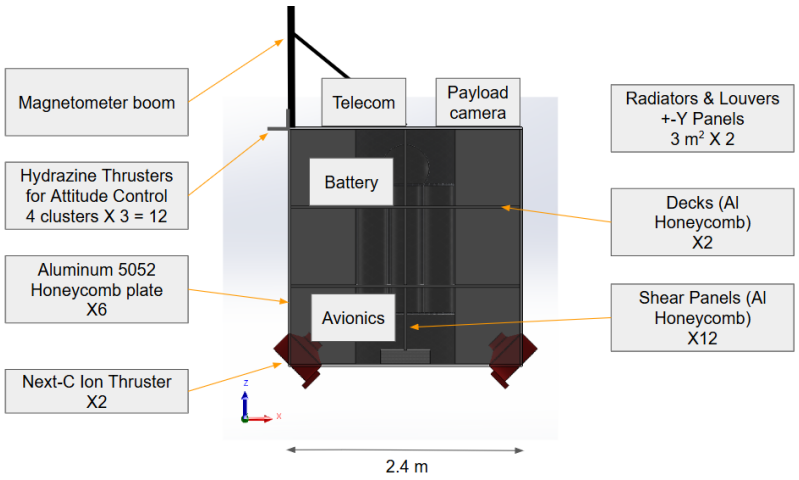


Figure 27: Component Placement - Primary Structure

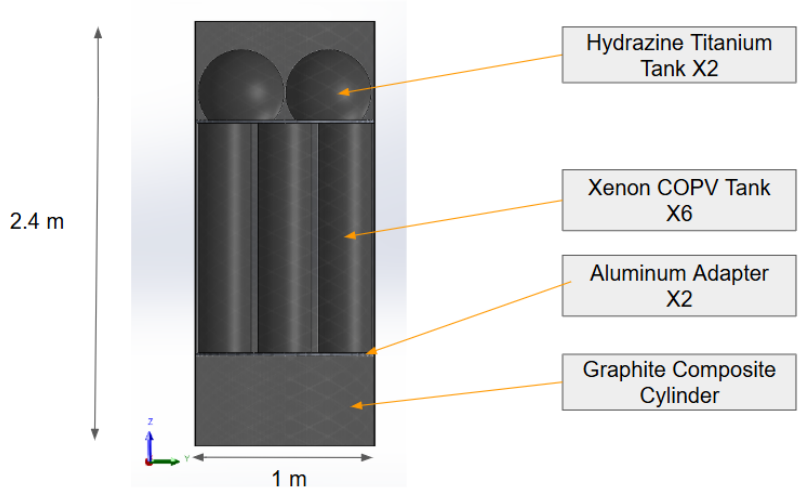


Figure 28: Component Placement - Central cylinder

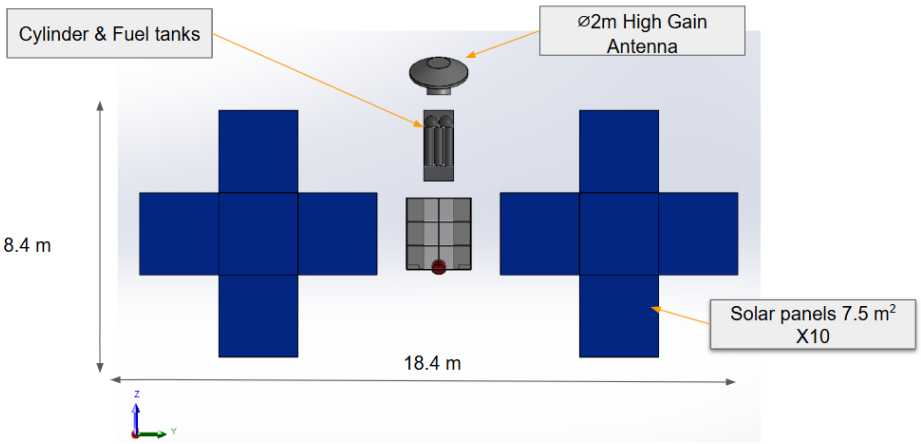


Figure 29: Component Placement - Deployed configuration

3.2.3 Fuel Tanks Design (Tamar Levin)

Xenon Tanks

The xenon propellant is stored in composite overwrapped pressure vessels (COPVs). Each tank consists of a thin metallic liner for sealing and a fiber composite overwrap for strength. COPVs are mass efficient compared to all-metal pressure vessels, with reported mass savings of approximately 25-30% compared to similar all-titanium spherical tanks [36].

The selected tank is the General Dynamics model 220165-1 xenon tank, with a volume of 82 L, dimensions of 0.3 m $\times$ 1.3 m, a MEOP of 185 bar, and a dry mass of 11 kg [37].

The Psyche mission used seven COPV xenon tanks of the same model [38]. Each tank stored approximately (155 kg) of xenon, giving a total xenon capacity of approximately (1085 kg). COPV tanks were also used on the Dawn mission [39]. Therefore, the tank has relevant flight heritage and is considered reliable for deep-space missions.

The xenon temperature must be controlled to keep the propellant in the required supercritical condition. Based on the Psyche mission, the preferred xenon temperature range is approximately 34–41°C, with the temperature kept below 45°C [40]. The absolute minimum temperature is approximately 20°C to keep xenon in the supercritical regime.

For Hydrazine, two Titanium blow down tanks were chosen, as detailed in the propulsion section.

Xenon Tanks Adapter

An adapter was designed to support the xenon tanks inside the spacecraft bus main cylinder. The adapter is made from Al 6061-T6, with an outer diameter of 1 m and a mass of 22 kg.

Al 6061-T6 was selected as a preliminary adapter material because it is commonly used for spacecraft structural adapters and brackets and provides a good balance between strength, stiffness and density. Al 6061-T6 has a yield strength of 276 MPa, modulus of 70 GPa, and density of 2700 kg/m^3, as shown in Table 20. A possible alternative is Al 7075-T6, which has a yield strength of 503 MPa, and similar modulus and density [41]. This would increase the yield strength by approximately 82%, while increasing the adapter mass by only about 4%. However, Al 7075-T6 is typically more expensive than Al 6061-T6.

Another alternative is Ti-6Al-4V, which has a yield strength of 880 MPa, modulus of 114 GPa, and density of 4430 kg/m^3. This would increase the yield strength by 219%, but would also increase the adapter mass by 64%. Ti-6Al-4V is also significantly more expensive and harder to machine than aluminum alloys. Therefore, Al 7075-T6 is the more mass-efficient higher-strength alternative, while Ti-6Al-4V should only be used if very high stresses are expected based on the analysis.

A static analysis was performed in Solidworks to minimize the adapter mass while ensuring that it can carry the launch loads. The perimeter of the adapter was fixed, and the adapter was loaded with the equivalent force of a full tank at 6g, corresponding to approximately 9

kN per tank. The maximum Von Mises stress was 113.8 MPa, giving a safety factor of 2.4. The maximum displacement was 6.5 mm.

This static analysis was used as an initial method to verify the design and does not fully represent the launch environment, which is dynamic. Therefore, sine vibration analysis were performed to evaluate the response of the entire structure, including the adapter, to launch vibration loads.

3.2.4 Structure Analysis

During launch, the spacecraft is exposed to both quasi-static acceleration loads and dynamic vibration loads. Those loads are transmitted to the spacecraft through the payload adapter. Static analysis to verify the structure does not yield under self weight during launch acceleration, and modal analysis was done to confirm that the natural frequencies satisfy launch vehicle requirements. Additionally sine vibration analysis was done to evaluate the stress response under the launch vibration environment and verify that dynamic amplification at the spacecraft natural frequencies does not result in structural failure.

Static and Buckling Analysis (Brandon Lee)

A static structural analysis and eigenvalue buckling analysis were performed to evaluate the preliminary spacecraft bus structure under Falcon Heavy launch loading conditions. The finite element model was generated in ANSYS Mechanical using a mechanical physics preference with quadratic elements. A global element size of 50 mm was applied, and sheet bodies were meshed using a quad-dominant method where appropriate. The central cylindrical tube was assigned a hard body sizing of 50 mm. The honeycomb side walls and shear panels were assigned hard body sizing of 65 mm, while the tank support brackets were assigned hard body sizing of 50 mm. The horizontal panels used 50 mm edge sizing around circular cutouts with soft behavior to improve mesh quality near curved features. The payload adapter was meshed with an 80 mm soft element size.

The model was constrained using a fixed support at the bottom of the payload adapter, representing the launch vehicle interface. Launch loads were applied as acceleration components to all bodies based on the Falcon Heavy quasi-static load cases ([34]). Additional nonstructural spacecraft masses were represented using point masses.

For the highest axial acceleration case of 6 g, the static and buckling results showed relatively small total deformation in the side honeycomb panels, central cylinder, and tank support brackets. The largest deformation occurred in the top and bottom panels, as well as in the interior horizontal honeycomb panels (Figure 30). A second important load case combined 3.5 g axial acceleration with 2.0 g lateral acceleration. Under this lateral acceleration case, the primary buckling modes appeared in similar locations, with additional buckling modes forming in the interior horizontal panels. However, the lateral acceleration cases are expected to be more transient than purely static during launch, so these results are most useful for identifying vulnerable structural regions rather than directly predicting final failure.

The primary predicted failure mode was buckling of the bottom panel, driven by bending

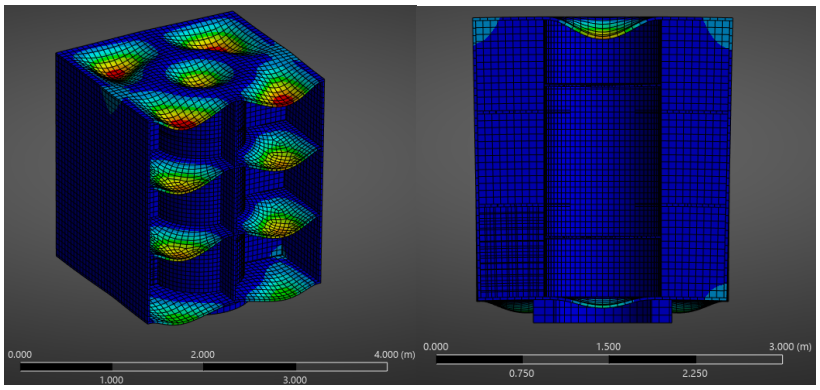


Figure 30: Total spacecraft deformation shapes with the cross section perpendicular to the X-axis (left) and from the central cross section perpendicular to the X-axis (right).

moments from the attached solar panels. The honeycomb panel layout and solar array attachment regions are more critical for buckling than the central cylindrical tube itself in the current structural configuration. However, the exact critical load multipliers for the panels should be treated as inconclusive at this stage. Orthotropic stress limits in the panel x- and y-directions were not available, so conservative placeholder values of 1 MPa were used. In addition, linear eigenvalue buckling analysis can overestimate the actual critical load because it does not fully capture geometric imperfections, residual stresses, contact nonlinearities, material nonlinearities, or post-buckling behavior. Therefore, the current results should be interpreted as a preliminary screening tool for locating likely buckling-sensitive regions, with more detailed nonlinear buckling analysis required in later design iterations.

Improvements to the primary spacecraft structure that can reduce the risk of buckling will be discussed in the Structural Risk Assessment section.

Modal Analysis (Tamar Levin)

A modal analysis was performed to evaluate the natural frequencies and corresponding mode shapes of the spacecraft structure. The analysis was conducted in the stowed configuration for launch. The model includes the primary structure, tanks, propellant, and solar arrays. The payload adapter was fixed (clamped), representing the launch vehicle boundary condition, such that the spacecraft behaves as a vertical cantilever beam. The payload adapter is shown in Figure 31.

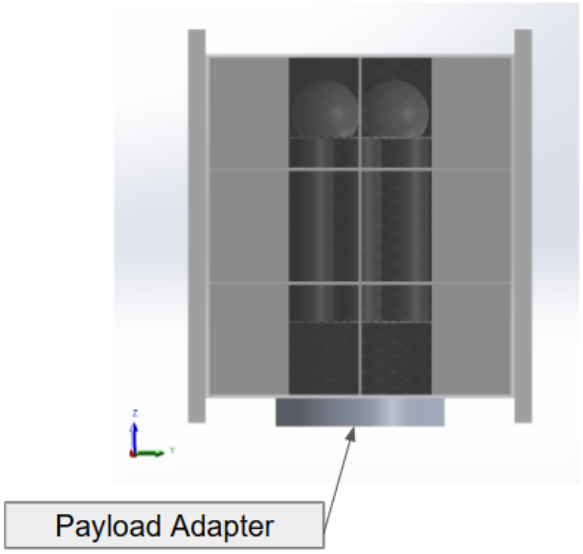


Figure 31: Payload adapter

The following mechanical properties were used for the modal and dynamic analysis. For simplicity, the composite materials were assumed to be isotropic in this analysis. This is a non-conservative assumption, since the differences in stiffness in different directions and may reduce the predicted natural frequencies.

Table 20: Mechanical properties used for the modal and dynamic analyses.

Material	**Modulus [GPa]**	**Strength [MPa]**	**Density [kg/m^3]**	**Reference**
Aluminum honeycomb 5052, 1/4 in cell and 0.003 in foil thickness	1.62	7.6	252	[42]
Graphite composite, used for the central cylinder	69	620	1380	[43]
Titanium alloy Ti–6Al–4V, used for the hydrazine tanks	114	880	4430	[44]
Aluminum 6061, used for the xenon tank adapters, launch payload adapter, and assumed COPV liner material	70	276	2700	[45]

The connections between components were modeled as infinitely rigid. This assumption increases the effective stiffness (non conservative). A low stiffness value of 1.6 GPa, corresponding to the aluminum honeycomb core, was used for the aluminum honeycomb material. This is a conservative assumption, since it neglects the increased effective stiffness provided

by the face sheets. The modal analysis was performed in SolidWorks Simulation using the FFEPlus iterative solver.

The first four mode shapes obtained from the analysis are summarized in Table 21.

Table 21: Modal analysis results for the stowed spacecraft configuration.

Mode No.	**Frequency [Hz]**	**Mode Shape**
1	42	Bending in Y
2	44	Bending in X
3	64	Torsion about Z
4	89	Axial deformation in Z

The first lateral mode occurs at 42 Hz, which is above the required 10 Hz. The first axial mode occurs at 89 Hz, which is above the required 25 Hz.

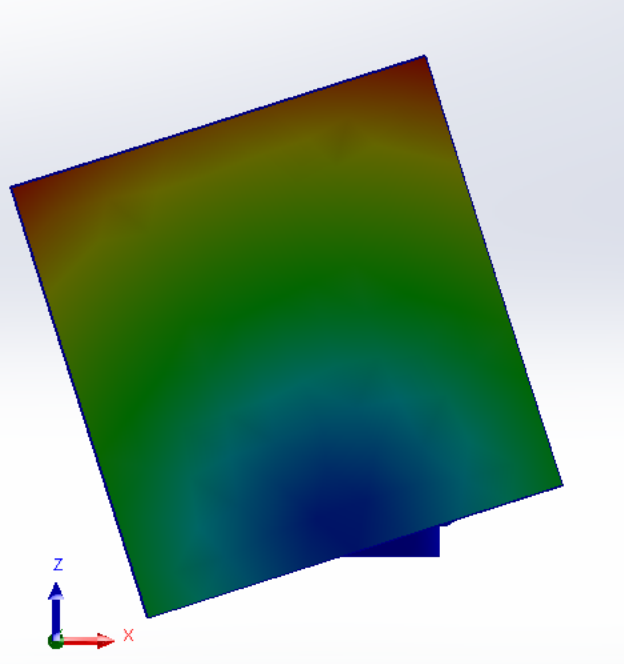


Figure 32: First bending mode in the X direction

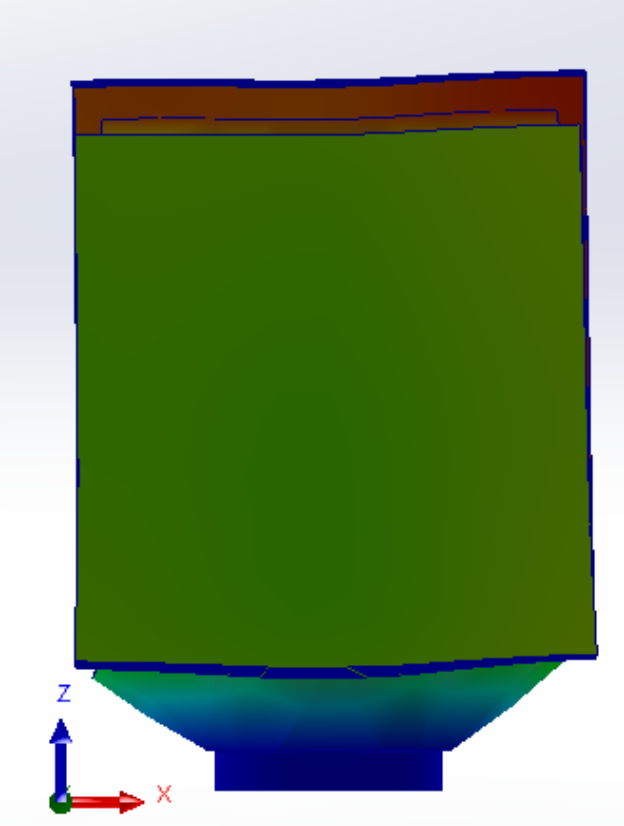


Figure 33: Axial mode in the Z direction.

The first modes are global spacecraft modes rather than local panel modes, validating that the design as an effective load-carrying structure. Although the preliminary results show sufficient frequency margins, the actual connections are not infinitely stiff, and the structure materials are non-isotropic. In addition, the solar panel mounts and payload boom were not implemented in the model and are expected to reduce the overall stiffness. Hence, the actual natural frequencies are expected to be lower than those predicted by the current model.

Dynamic Analysis (Tamar Levin)

A dynamic analysis was performed to evaluate the spacecraft structural response to the launch vehicle sine vibration environment. The goal of the analysis was to verify that the structure can withstand the required axial and lateral sine acceleration loads during launch [34].

The analysis was performed using a damping ratio of 2%. The applied sine vibration environment covered the frequency range from 5 Hz to 100 Hz, with acceleration amplitudes below $1g$. Excitations were applied independently in the axial direction and in the two lateral directions, X and Y. The Falcon 9 sine vibration environment used for the analysis is shown in Figure 34.

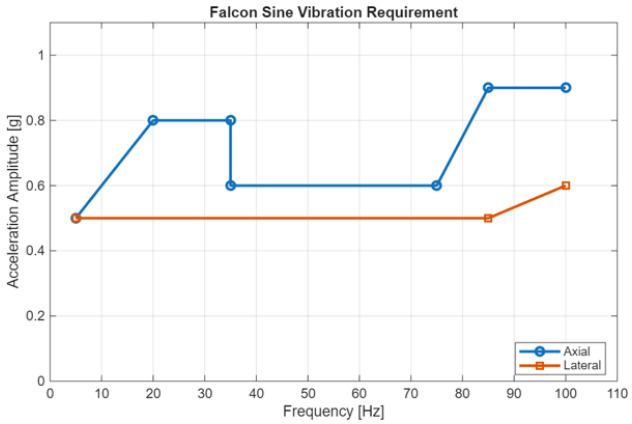


Figure 34: Falcon 9 sine vibration environment used for the dynamic analysis.

For all excitation directions, the maximum von Mises stress occurred at the bottom xenon tank adapter. The stress-frequency response was therefore evaluated at the node where the maximum von Mises stress was obtained.

The maximum response occurred at approximately 52–54 Hz, corresponding to the first two fundamental modes of the structure. The largest response was obtained for a $0.5g$ lateral acceleration in the Y direction. The maximum von Mises stress obtained in the analysis was 32 MPa, located at the bottom xenon tank adapter, as shown in Figure 35.

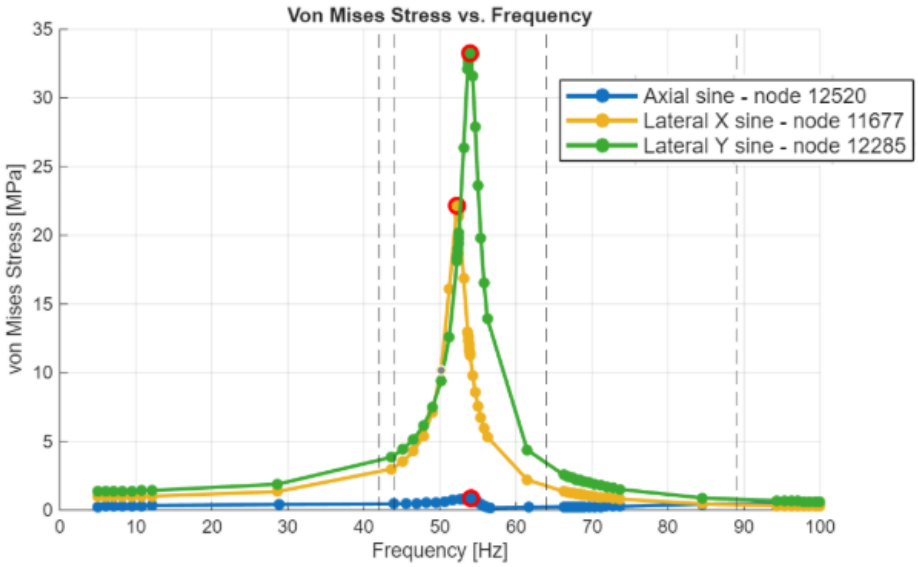


Figure 35: Von Mises stress distribution from the dynamic simulation.

The maximum stress is below the yield strength of the Al 6061-T6 adapter material:

$$\sigma_{\text{VM,max}} = 32 \text{ MPa} < S_y = 270 \text{ MPa}$$

Therefore, the bottom xenon tank adapter remains below yield stress, and the structure satisfies the sine vibration stress requirement. Similar to the modal analysis, the actual dynamic response may be higher due to the non-conservative assumptions of rigid connections and the omission of flexible components from the model.

Flexible Mode Analysis (Tamar Levin)

A flexible mode analysis was performed to evaluate the natural frequencies of the deployed spacecraft configuration for attitude control calculations. Unlike the launch modal analysis, this analysis considers the spacecraft in its deployed configuration and without support constraints, representing in space operations.

The first three flexible modes obtained from the analysis are summarized in Table 22.

Table 22: Flexible mode analysis results for the deployed spacecraft configuration.

Mode No.	Frequency [Hz]	Mode Shape
1	0.163	Bending in Z
2	0.478	Bending in Z
3	0.564	Torsion about Y

The first flexible mode is very low, and occurs at 0.163 Hz, as expected based on the flexibility of deployed solar arrays. The first flexible mode shape is shown in Figure 36.

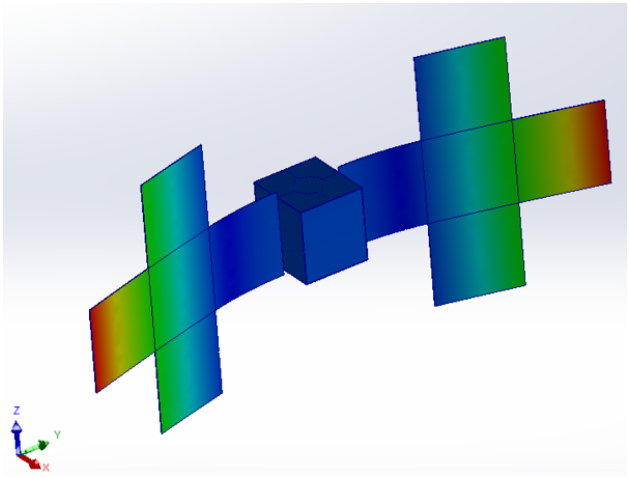


Figure 36: First flexible mode shape of the deployed spacecraft configuration.

Note that the first six modes obtained in the simulation were near-zero-frequency rigid-body modes, as expected for a free body. These modes were therefore disregarded, and only the subsequent flexible modes are reported.

Analysis Conclusion (Brandon Lee and Tamar Levin)

The static analysis showed that the structure can withstand a (6g) launch acceleration with sufficient safety margin from yield.

The dynamic analysis showed that the spacecraft can withstand the launch sine vibration environment over the 5–100 Hz frequency range, with a maximum von Mises stress of 32 MPa. This is below the Al 6061-T6 yield strength of 270 MPa, giving a yield safety factor of approximately 8.4.

Both the static and dynamic analyses identified the bottom xenon tank adapter as a critical component, since it connects the main spacecraft mass, propellant tanks, to the main load-carrying structure, the inner cylinder. The design of this adapter should therefore be revisited to ensure that it can withstand worst-case scenarios that were not yet examined in the current analysis.

The modal analysis showed that the lowest natural frequencies were above the launch vehicle requirements: the first lateral mode was 42 Hz, compared with the 10 Hz requirement, and the first axial mode was 89 Hz, compared with the 25 Hz requirement.

Note that the analysis was performed for a configuration with five xenon tanks. However, a recent increase in dry mass resulted in the need to add an additional tank, bringing the total to six tanks. The tank adapter should therefore be redesigned accordingly, and the analysis should be updated. To accommodate the higher propellant mass, the adapter may need to be increased in thickness or changed to a higher-strength and or higher-stiffness material.

3.2.5 Future Work (Brandon Lee and Tamar Levin)

Several areas of the spacecraft structural design and analysis remain for future work.

- **Component design:**
  - Design the solar panel mounting and deployment mechanism.
  - Design the payload boom and attachment interface.
  - Add missing components to the CAD model, including reaction wheels, attitude-control thrusters, heaters, batteries, avionics, and payload components.
- **Structural analysis:**
  - Repeat the modal and dynamic analyses using realistic connections instead of rigid connections and with the updated six tanks.
  - Obtain information about honeycomb tensile strengths to obtain critical buckling loads.
  - Perform a random vibration analysis.

3.3 Thermal

3.3.1 Design Rationale (Riley Foard)

The thermal architecture draws heavily from the Dawn and Psyche missions due to their similar mission profiles and thermal environments. Both spacecraft were designed for a hot case of 1 AU and a cold case ranging from 3 to 3.33 AU, while the expected trajectory for this mission will have a hot case of 1 AU and a cold case of roughly 3.5 AU. Both spacecraft used electric propulsion, indicating that the internal heat dissipation due to the propulsion system will also be similar, further justifying heritage choices for the thermal subsystem.

High energy radiation is not explicitly modeled in this thermal analysis. This is because the spacecraft does not travel closer to the Sun than Earth orbit, so the mission is not expected to encounter an unusually severe radiation environment beyond nominal deep space radiation.

3.3.2 Thermal Control System Hardware (Riley Foard)

Spacecraft thermal control will be achieved through a combination of passive and active elements. Passive elements are exceedingly simple to implement and require no power input, making them an attractive option for long-duration missions. The AFONSO thermal architecture is a largely passive system consisting of multilayer insulation (MLI) on the outside of the spacecraft to retain heat in the cold case, radiator panels with passively actuated mechanical louvers to reject heat in the hot case, internal heat pipes to conduct heat from internal components to the radiator panels, and electric resistive patch heaters on components requiring heating in the cold case. In addition, the fuel tanks will be covered in MLI to minimize radiative heat losses as they must be maintained at a higher temperature. The following is a brief overview of thermal subsystem hardware:

- **Multi-layer insulation (MLI)** is a thin, sheet-like insulation material that will cover the exterior of the spacecraft to minimize radiative heat losses and solar heat absorption. This will reduce heating requirements in the cold case and help to reject solar heating in the hot case. For our analysis, we use MLI comprised of aluminized Mylar, which is commonly used on spacecraft.
- **Passive mechanical louvers** consist of rows of radiative blades mounted on hinges. The hinges are driven by bimetal actuator springs with a set actuation temperature that automatically open the blades when hot and close when cold, increasing or decreasing the radiative surface area as necessary to maintain a target temperature. When mounted on a radiator panel on the spacecraft body, these passive louvers can autonomously regulate the spacecraft temperature provided the internal heat is properly conducted to the radiator. We base our louver properties on those of Sierra Space's 20-blade passive thermal louvers, which have extensive flight heritage on missions such as Juno and New Horizons.
- **Patch heaters** are flexible electric resistive heaters that can be customized to fit the shape of any component. They are implemented alongside thermostats or temperature sensors to control how much heater power each component receives, and typically have redundant heating circuits to ensure fault tolerance.

- **Heat pipes** are a passive means of conducting heat from a hot component to a radiator. They consist of an extruded tube containing a working fluid (commonly ammonia in aluminum tubing), with grooves on the inside surface to enable capillary action. The evaporator section at one end of the pipe is mounted to the hot component, and the condenser section at the other end is mounted to the radiator. The working fluid is vaporized in the evaporator and travels through the length of the heat pipe until it reaches the condenser section, where it condenses and deposits heat energy into the radiator. The fluid travels back toward the evaporator section via capillary action. Heat pipe parameters must be selected based on the required heat flux given by the preliminary thermal analysis, and sizing calculations are performed in Section [].

3.3.3 Component Operating Temperatures (Michael Hwang)

Each component of the spacecraft model must remain within its assigned operational temperature limits for the duration of the mission. The updated temperature limits for applicable spacecraft subsystems are listed in Table 23.

Node / Component	$\mathbf{T}_{\text{setpoint}}$ (°C)	$\mathbf{T}_{\text{min}}$ (°C)	$\mathbf{T}_{\text{max}}$ (°C)
Payload	0.0	-20	50
Battery	20.0	15	30
Avionics	-10.0	-30	70
Ion Thruster PPU	0.0	-20	60
RCS Propulsion	0.0	-20	60
Fuel Tanks	40.0	34	42
External Sensors	-10.0	-30	60

Table 23: Allowable temperature ranges for spacecraft components that are sensitive to temperature.

The most significant changes from the previous design are the addition of temperature limits for the battery and fuel tanks. These limits are much narrower than those of other subsystems and must be carefully tuned to ensure the battery and fuel tanks do not freeze or overheat.

3.3.4 Single-Node Steady-State Thermal Analysis (Michael Hwang)

To assess the feasibility of the preliminary thermal architecture and size the radiator and louvers, a single-node, steady-state thermal analysis was performed. In this analysis, the spacecraft was assumed to be a uniform-temperature cube with side length 2 meters. Based on the louver size for Dawn, which had 5.6 m^2 louver area, the spacecraft was assigned 6 m^2 of louver area distributed as two 3 m^2 radiators on two faces not facing the Sun.

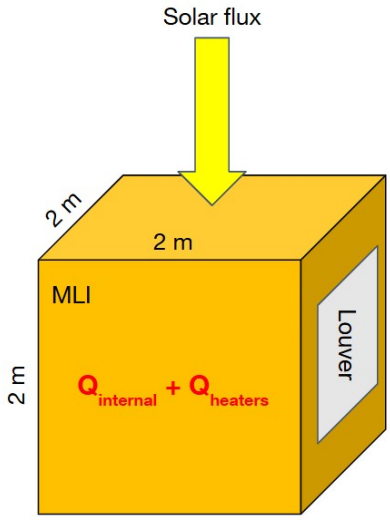


Figure 37: Simplified single-node thermal model of the spacecraft.

The remainder of the spacecraft surface area is covered in MLI, and the internal heat generation was roughly estimated to be 1000 W in the hot case and 500 W in the cold case. These values were obtained by adding an estimate of the propulsion waste heat to the tentative power budget of 200 W for the hot case, and adding additional power dissipation from avionics for the cold case. The thermal properties used in the single-node analysis (and also in the multi-node analysis) are listed in Table 24.

Surface	**Absorptivity,** α	**Emissivity,** ϵ	**Area,** m^2
MLI	0.15	0.03	18
Louvers, closed	–	0.14	6
Louvers, open	–	0.74	6

Table 24: Thermal optical properties used for the single-node thermal analysis.

The equilibrium temperature of the spacecraft is calculated by balancing the radiative heat rejection with the incoming solar heat flux at either 1 AU or 3.5 AU, as shown in Equation 19. This analysis assumes that one side of the spacecraft is always normal to the Sun, so $A_{\text{Sun-facing}} = 4\ \mathrm{m}^2$.

$$Q_{\text{solar}}(r) = A_{\text{Sun-facing}} \frac{S_{1\text{AU}}}{r^2}, \qquad S_{1\text{AU}} = 1361\ \mathrm{W/m}^2, \qquad r \text{ in AU} \tag{18}$$

$$Q_{\text{int}} + Q_{\text{solar}} + Q_{\text{heater}} = \sigma T^4 \left(\epsilon_{\text{louver}} A_{\text{louver}} + \epsilon_{\text{MLI}} A_{\text{MLI}}\right) \tag{19}$$

Case	Louver State	Distance (AU)	Q_{int} (W)	Q_{solar} (W)	Q_{heater} (W)	ϵ_{louver}	T (K)
Hot case	Open	1.0	1000	816.6	0	0.74	283.2
Hot case	Closed	1.0	1000	816.6	0	0.14	390.3
Cold case	Closed	3.5	500	66.7	10	0.14	293.0

Table 25: Single-node thermal analysis cases and resulting equilibrium temperatures.

For all cases, a target temperature of 293 K was chosen as it is approximately the average of the minimum and maximum operating temperatures for most components. The hot case was analyzed by calculating the equilibrium temperature for louvers open and closed. This yields a temperature range of 283 to 390 K (Table 25), which indicates that the louvers are capable of maintaining a target temperature of 293 K by opening and closing intermittently. Next, the cold case was analyzed by setting the spacecraft to the desired target temperature of 293 K and determining the required heater power to maintain this temperature. This results in $Q_{\text{heater}} = 10W$, which is a reasonable power consumption. These results indicate that the chosen louver size is appropriate for the spacecraft, and a more detailed analysis can be performed.

3.3.5 Multi-Node Transient Thermal Analysis Framework (Riley Foard)

Because each component has a different operational temperature range, a single-node model is not robust enough to ensure that every component in the spacecraft remains at an appropriate temperature. Though the average temperature might be close to the target, there may still be areas of localized heating or lack thereof that drive individual components above or below their safe operating temperatures. To ensure that individual components remain within their required temperature ranges, the single-node analysis was refined to a multi-node analysis in which groups of related components are lumped into a total of 18 interconnected nodes, each with distinct internal power generation, environmental heat flux, thermal capacitance, and conductive links to other nodes.

Node Definitions

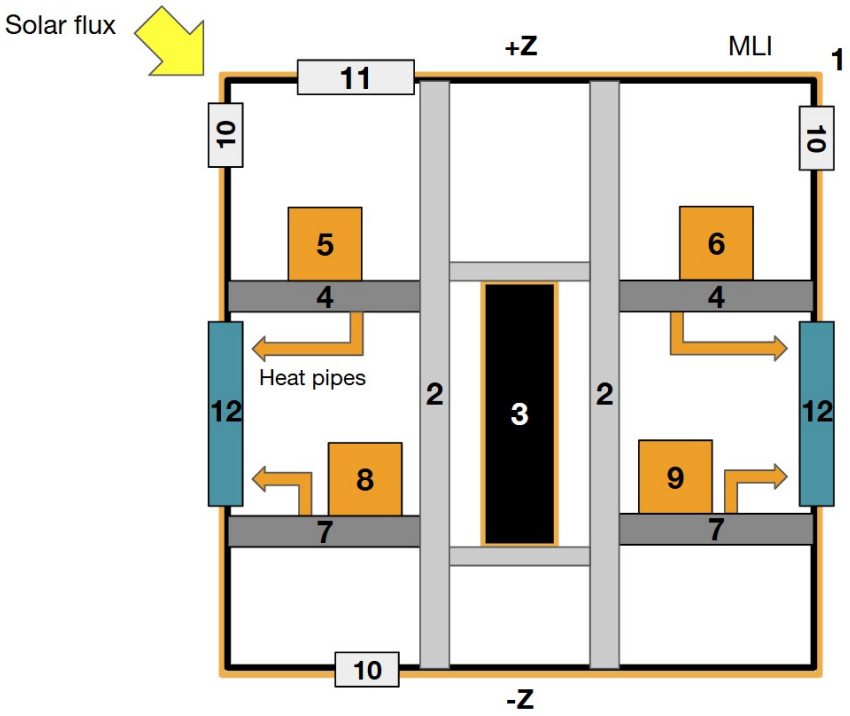


Figure 38: Cross-section of spacecraft showing updated multi-node thermal model: (1) $\pm X$, $\pm Y$, and $\pm Z$ structural panel nodes, covered in MLI; (2) central cylinder node; (3) fuel tank node, covered in MLI; (4) upper bench node; (5) payload node; (6) battery node; (7) lower bench node; (8) avionics node; (9) PPU node; (10) RCS nodes; (11) external sensor node; (12) radiator nodes.

The updated thermal model divides the spacecraft into 18 isothermal nodes. Since the midterm design, the node network has been refined to more accurately reflect the physical structure of the spacecraft. The main structure is divided into six structure-panel nodes corresponding to the $\pm X$, $\pm Y$, and $\pm Z$ faces, along with a central cylinder node corresponding to the main internal structural cylinder. The radiators are split into two separate radiator nodes mounted to the $+Y$ and $-Y$ panels. The fuel tanks are inside the central cylinder and mounted on two disks, with titanium interfaces for thermal isolation. The external sensors are mounted on the $+X$ structure panel, and the RCS propulsion node is divided among four panels, reflecting the location of each RCS cluster. Heat flux contributions from the solar array were neglected because the conduction through the solar array drive assembly is expected to be rather small, and the heat would be conducted into the outer structure which would have a negligible impact on the internal components.

Node Connectivity and Thermal Properties

The multi-node thermal analysis determines heat flux at each node based on conductivity to and from connected nodes. The updated model includes conduction through bolted aluminum interfaces, a titanium battery standoff, titanium fuel-tank mounting disks, aluminum

panel edges, the internal structural cylinder, and heat pipe links between the benches and radiators.

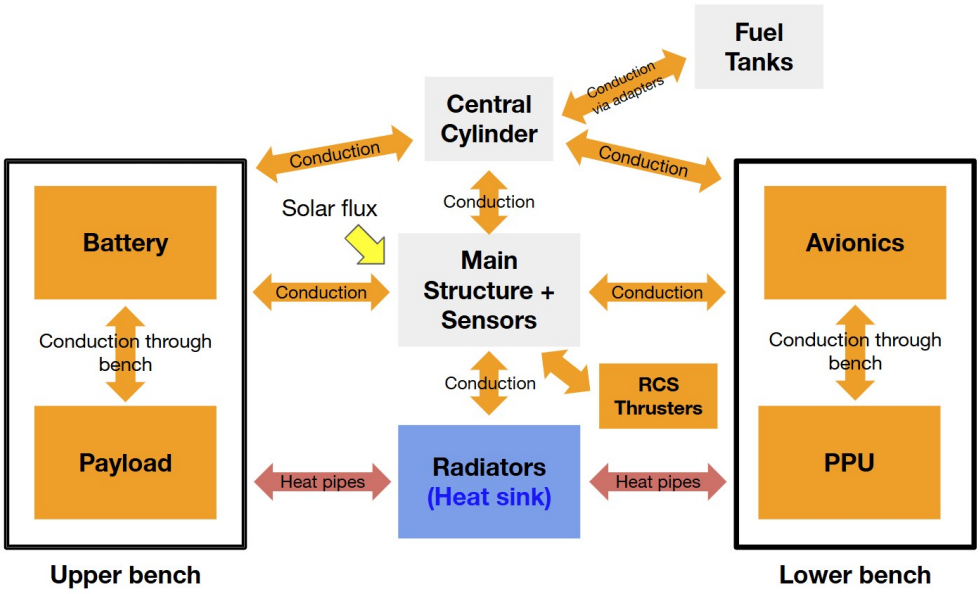


Figure 39: Diagram of the spacecraft showing conductive links between nodes.

For this analysis, the effect of internal radiative heat transfer was neglected because the spacecraft model is not yet detailed enough to determine view factors, and because internal conduction is expected to dominate radiative effects. Table 26 summarizes the constants, geometry, and optical properties used in the updated thermal model. Changes from the previous design include:

- Spacecraft dimensions have been updated to 2.2 by 2.4 by 2.4 meters.
- Louver radiating area is now 68.9% of the total area, while the mounting area is 31.1%.
- The Sun-facing area is taken to be the average projected area of the spacecraft, which is equal to one-fourth of the total surface area.
- MLI degradation occurs over the course of the mission, based on studies done on MLI retrieved from the Hubble Space Telescope [46]. MLI absorption increases linearly by 50% over the duration of the mission, while absorptance linearly decreases by 5%. Note that this is a worst-case estimate, as Hubble experienced harsher thermal cycling and environmental effects in LEO compared to the expected environment of AFONSO.
- Louver actuation temperature is set to 20 degrees Celsius, with a transition band of 5 degrees.

Parameter	Value	Units
σ	$5.670374419 \times 10^{-8}$	$\mathrm{W/(m^2\,K^4)}$
T_{space}	3	K
S_0	1361	$\mathrm{W/m^2}$
L	2.2	m
W	2.4	m
H	2.4	m
$A_{\mathrm{louver,total}}$	6.000	$\mathrm{m^2}$
$A_{\mathrm{louver,each}}$	3.000	$\mathrm{m^2}$
$f_{\mathrm{louver,rad}}$	0.689	–
$f_{\mathrm{louver,mount}}$	0.311	–
$A_{\mathrm{radiator,rad}}$	2.067	$\mathrm{m^2}$ per radiator
$A_{\mathrm{radiator,mount}}$	0.933	$\mathrm{m^2}$ per radiator
A_{total}	32.64	$\mathrm{m^2}$
A_{MLI}	26.64	$\mathrm{m^2}$
A_{Sun}	8.16	$\mathrm{m^2}$
$\alpha_{\mathrm{MLI,BOL}}$	0.140	–
$\alpha_{\mathrm{MLI,EOL}}$	0.210	–
$\epsilon_{\mathrm{MLI,BOL}}$	0.0300	–
$\epsilon_{\mathrm{MLI,EOL}}$	0.0285	–
$\epsilon_{\mathrm{louver,open}}$	0.74	–
$\epsilon_{\mathrm{louver,closed}}$	0.14	–
$T_{\mathrm{louver,actuate}}$	20	°C
$\Delta T_{\mathrm{louver,transition}}$	5	°C

Table 26: Constants, geometry, and thermal optical properties used in the updated multi-node thermal model.

Each node was assigned an internal heat generation based on component power consumption and duty cycles, as seen in Table 27. Payload heat is 15 W in standby and 30.7 W during science, while battery heat is 100 W during transfer and 25 W during science. Avionics heat is fixed at 167.2 W. PPU heat is scaled by transfer segment and multiplied by two in the updated analysis to account for two thrusters. RCS heat is zero except near arrival and departure events, where it is set to 50 W.

Node / Component	Transfer 1	Parthenope Science	Transfer 2	Psyche Science	Transfer 3	Themis Science
Payload	15.0	30.7	15.0	30.7	15.0	30.7
Battery	100.0	25.0	100.0	25.0	100.0	25.0
Upper Bench	0	0	0	0	0	0
Avionics	167.2	167.2	167.2	167.2	167.2	167.2
Ion Thruster PPU	294.9	20.0	59.2	20.0	122.4	20.0
RCS Propulsion	0 or 50	0 or 50	0 or 50	0 or 50	0 or 50	0 or 50
Lower Bench	0	0	0	0	0	0
Structure Panels	0	0	0	0	0	0
Internal Structural Cylinder	0	0	0	0	0	0
Fuel Tanks	0	0	0	0	0	0
Radiators	0	0	0	0	0	0
External Sensors	0	0	0	0	0	0

Table 27: Updated internal heat generation (in W) assigned to each thermal node by mission mode. RCS heat is 50 W only within 0.05 yr of arrival or departure events; otherwise it is 0 W.

Conductance between nodes is calculated from the specified contact geometry and material properties, derived from the structural model. Bolted aluminum interfaces use $h_{\text{bolted,Al}} = 1000\ \text{W}/(\text{m}^2\,\text{K})$ [47]. The battery is mounted on a square titanium standoff for thermal isolation, and the fuel tanks are mounted to the internal cylinder through two disks with titanium contacts. The updated conductance and mounting-area parameters are shown in Table 28, and the resulting conductances between nodes are shown in Table 29. We assume a total conductance of 50 W/K for the heat pipes between the benches and the radiators as a first-order estimate, and the validity of this estimate will be confirmed using the results of the thermal analysis and the sizing calculations in Section [].

Parameter	Value	Units
$h_{\text{bolted,Al}}$	1000	$\text{W}/(\text{m}^2\,\text{K})$
t_{panel}	0.0254	m
k_{panel}	167	$\text{W}/(\text{m}\,\text{K})$
k_{Ti}	7	$\text{W}/(\text{m}\,\text{K})$
$A_{\text{payload,mount}}$	0.200	m^2
$A_{\text{avionics,mount}}$	0.200	m^2
$A_{\text{PPU,mount}}$	0.150	m^2
$A_{\text{RCS,mount}}$	0.100	m^2
$A_{\text{extSensor,mount}}$	0.020	m^2
$A_{+X,\text{panel}}$	5.76	m^2
$A_{-X,\text{panel}}$	5.76	m^2
$A_{+Y,\text{panel}}$	5.28	m^2
$A_{-Y,\text{panel}}$	5.28	m^2
$A_{+Z,\text{panel}}$	5.28	m^2
$A_{-Z,\text{panel}}$	5.28	m^2
$A_{\text{MLI},+X}$	5.76	m^2
$A_{\text{MLI},-X}$	5.76	m^2
$A_{\text{MLI},+Y}$	2.28	m^2
$A_{\text{MLI},-Y}$	2.28	m^2
$A_{\text{MLI},+Z}$	5.28	m^2
$A_{\text{MLI},-Z}$	5.28	m^2
$A_{\text{sun},+X}$	1.44	m^2
$A_{\text{sun},-X}$	1.44	m^2
$A_{\text{sun},+Y}$	1.32	m^2
$A_{\text{sun},-Y}$	1.32	m^2
$A_{\text{sun},+Z}$	1.32	m^2
$A_{\text{sun},-Z}$	1.32	m^2
$A_{\text{upper bench}-\text{main structure}}$	0.23368	m^2
$A_{\text{lower bench}-\text{main structure}}$	0.23368	m^2
$n_{\text{battery standoff}}$	1	–
$w_{\text{battery standoff}}$	0.25	m
$L_{\text{battery standoff}}$	0.025	m
$A_{\text{battery standoff}}$	0.0625	m^2
$D_{\text{tank disk}}$	1.010	m
$t_{\text{tank disk}}$	0.015	m
$A_{\text{tank disk edge}}$	0.0476	m^2
$L_{\text{tank disk,radial}}$	0.505	m
$D_{\text{internal cylinder}}$	1.010	m
$w_{\text{cylinder end ring}}$	0.004	m
$A_{\text{cylinder end mount}}$	0.01264	m^2
$G_{\text{battery standoff}}$	17.50	W/K
$G_{\text{tank disk mount}}$	1.319	W/K
$G_{\text{cylinder mount}}$	12.642	W/K
$G_{\text{panel edge,XY}}$	8.852	W/K
$G_{\text{panel edge,XZ}}$	8.852	W/K
$G_{\text{panel edge,YZ}}$	7.777	W/K
$G_{\text{bench}-\text{panel}}$	58.42	W/K
$G_{\text{heat pipe,bench}-\text{rad}}$	25 per radiator link	W/K
$G_{\text{radiator mount}}$	933.0	W/K

Table 28: Conductance, mounting-area, and panel-area parameters used in the multi-node thermal model.

Node 1	**Node 2**	**G_{ij} (W/K)**
Payload	Upper Bench	200.000
Battery	Upper Bench	17.500
Avionics	Lower Bench	200.000
Ion Thruster PPU	Lower Bench	150.000
RCS Propulsion	$+X$ Structure Panel	25.000
RCS Propulsion	$-X$ Structure Panel	25.000
RCS Propulsion	$+Z$ Structure Panel	25.000
RCS Propulsion	$-Z$ Structure Panel	25.000
External Sensors	$+X$ Structure Panel	20.000
Upper Bench	$+Z$ Structure Panel	58.420
Upper Bench	$+X$ Structure Panel	58.420
Upper Bench	$-X$ Structure Panel	58.420
Upper Bench	Internal Structural Cylinder	12.642
Lower Bench	$-Z$ Structure Panel	58.420
Lower Bench	$+X$ Structure Panel	58.420
Lower Bench	$-X$ Structure Panel	58.420
Lower Bench	Internal Structural Cylinder	12.642
Internal Structural Cylinder	$+Z$ Structure Panel	12.642
Internal Structural Cylinder	$-Z$ Structure Panel	12.642
Fuel Tanks	Internal Structural Cylinder	1.319
$+X$ Structure Panel	$+Y$ Structure Panel	8.852
$+X$ Structure Panel	$-Y$ Structure Panel	8.852
$-X$ Structure Panel	$+Y$ Structure Panel	8.852
$-X$ Structure Panel	$-Y$ Structure Panel	8.852
$+X$ Structure Panel	$+Z$ Structure Panel	8.852
$+X$ Structure Panel	$-Z$ Structure Panel	8.852
$-X$ Structure Panel	$+Z$ Structure Panel	8.852
$-X$ Structure Panel	$-Z$ Structure Panel	8.852
$+Y$ Structure Panel	$+Z$ Structure Panel	7.777
$+Y$ Structure Panel	$-Z$ Structure Panel	7.777
$-Y$ Structure Panel	$+Z$ Structure Panel	7.777
$-Y$ Structure Panel	$-Z$ Structure Panel	7.777
Upper Bench	$+Y$ Radiator	25.000
Upper Bench	$-Y$ Radiator	25.000
Lower Bench	$+Y$ Radiator	25.000
Lower Bench	$-Y$ Radiator	25.000
$+Y$ Structure Panel	$+Y$ Radiator	933.000
$-Y$ Structure Panel	$-Y$ Radiator	933.000

Table 29: Conductive links used in the multi-node thermal model. Heat flow is modeled as $Q_{ij} = G_{ij}(T_i - T_j)$.

Node / Component	Heat Capacity, C (J/K)
Payload	1.0×10^5
Battery	1.0×10^5
Upper Bench	1.0×10^5
Avionics	1.0×10^5
Ion Thruster PPU	1.0×10^5
RCS Propulsion	1.0×10^5
Lower Bench	1.0×10^5
$+X$ Structure Panel	8.333×10^4
$-X$ Structure Panel	8.333×10^4
$+Y$ Structure Panel	8.333×10^4
$-Y$ Structure Panel	8.333×10^4
$+Z$ Structure Panel	8.333×10^4
$-Z$ Structure Panel	8.333×10^4
Internal Structural Cylinder	1.5×10^5
Fuel Tanks	3.0×10^5
$+Y$ Radiator	1.0×10^5
$-Y$ Radiator	1.0×10^5
External Sensors	1.0×10^5

Table 30: Assumed thermal capacitance values for each thermal node. The total structural heat capacity of 5.0×10^5 J/K is distributed equally across the six structure panels.

The last element needed to solve the transient heat balance equation is the thermal capacitance of each node, which determines how quickly or slowly the node changes temperature for a given heat flux. As seen in Table 30, most component nodes are assigned 1.0×10^5 J/K, while the six structure panels share a total structural capacitance of 5.0×10^5 J/K. This is a simple order-of-magnitude estimate based on the total mass and heat capacity of aluminum because it is difficult to determine actual node heat capacities without detailed datasheets or testing.

Heat Balance Equation

From Equation 20, adding all the heat flux contributions at each node and dividing by the corresponding thermal capacitance yields the rate of change of nodal temperature with time. Using MATLAB, this quantity can be integrated and the temperature histories of each node over time can be determined.

$$\frac{dT_i}{dt} = \frac{Q_{\text{net},i}}{C_i} = \frac{Q_{\text{int},i} + Q_{\text{heater},i} + Q_{\text{env},i} + \sum_j G_{ij}\,(T_j - T_i) - Q_{\text{rad},i}}{C_i} \tag{20}$$

The only nodes for which the radiated heat flux is nonzero are the external nodes: the six MLI-covered structure panels, the two radiators, and the external sensors.

$$
\begin{aligned}
Q_{\text{rad,panel},i} &= \epsilon_{\text{MLI}}(t)\sigma A_{\text{MLI},i}\left(T_i^4 - T_{\text{space}}^4\right), \qquad i \in \{\pm X, \pm Y, \pm Z\} \\
Q_{\text{rad,radiator},i} &= \epsilon_{\text{radiator},i}\sigma A_{\text{radiator,rad}}\left(T_i^4 - T_{\text{space}}^4\right), \qquad i \in \{+Y \text{ radiator}, -Y \text{ radiator}\} \\
Q_{\text{rad,sensors}} &= \epsilon_{\text{sensors}}\sigma A_{\text{sensors,rad}}\left(T_{\text{sensors}}^4 - T_{\text{space}}^4\right)
\end{aligned}
\tag{21}
$$

Similarly, the only nodes for which the environmental heat flux is nonzero are the six structure panels and the external sensors.

$$
\begin{aligned}
Q_{\text{env,panel},i} &= \alpha_{\text{MLI}}(t) S A_{\text{sun},i}, \qquad i \in \{\pm X, \pm Y, \pm Z\} \\
Q_{\text{env,sensors}} &= \alpha_{\text{sensors}} S A_{\text{sensors,sun}}
\end{aligned}
\tag{22}
$$

3.3.6 Multi-Node Transient Thermal Simulation (Riley Foard)

Implementing the Multi-Node Model into a MATLAB Script

The transient thermal model was implemented in MATLAB as a mission-time thermal simulation of the spacecraft bus. Rather than computing the steady state heat transfer at just hot and cold cases, this model evaluated how the major spacecraft thermal nodes evolve over the full mission trajectory. This results in a much more accurate model of the spacecraft, as it includes the duty cycles of every component and the varying solar flux on the spacecraft throughout the mission. The code implements the 18 isothermal node thermal network outlined in the previous sections and integrates the temperature of each node forward in time using the transient energy balance described in Section 3.3.5.

The MATLAB script begins by defining the spacecraft geometry, optical properties, node list, thermal limits, heat capacities, conductive links, mission timeline, heater properties, and louver-control parameters. The mission timeline is represented by a vector of elapsed mission time from launch through the end of the Themis science stay. Rather than approximating the heliocentric radius using only a few major mission events, the updated model imports the spacecraft position history from the trajectory CSV file and calculates the heliocentric distance directly from the spacecraft's Cartesian position. At each time step, the code interpolates this heliocentric distance history and calculates the corresponding nominal solar flux using the inverse-square relation,

$$
S(t) = \frac{S_{1\text{AU}}}{r_{\text{AU}}(t)^2}, \qquad S_{1\text{AU}} = 1361 \text{ W/m}^2,
\tag{23}
$$

where $r_{\text{AU}}(t)$ is the spacecraft's heliocentric distance in AU. This provides a solar-flux profile that follows the actual implemented trajectory more closely than the midterm linear-interpolation model.

Solar Flux Eclipse Case

In addition to the nominal solar-flux during transient phases, the MATLAB script also includes a simplified asteroid eclipse model during the Parthenope, Psyche, and Themis

orbits. These eclipse cases were included because while the spacecraft is in orbit around each asteroid, the asteroid will periodically block any direct solar energy from hitting the spacecraft. To model effectively model the worst case scenario of this effect, the code assumes that the spacecraft is in a circular orbit at an altitude of 75 km around each asteroid and that every orbit experiences one maximum-duration eclipse. Note that while the actual orbit trajectory includes multiple orbits at different altitudes at each asteroid, operations confirmed that the 75km assumption for this worst case scenario modeling was sufficient. The fraction of each asteroid orbit spent in eclipse was then approximated by comparing the asteroid diameter to the circumference of the spacecraft orbit:

$$f_{\text{eclipse}} = \frac{D_{\text{asteroid}}}{2\pi \left(R_{\text{asteroid}} + 75 \text{ km}\right)}. \tag{24}$$

During each eclipse window, the direct solar flux is set to zero, representing the loss of solar heating while the spacecraft is behind the asteroid. And outside of the eclipse window, the solar flux follows the nominal inverse-square solar-distance relation. This produces a short square wave in solar heating during each asteroid orbit.

Next, because these eclipse windows are fairly short compared to the full mission duration, the code evaluates the eclipse history on a much finer time grid during asteroid science operations. A one-minute time step is used for eclipse plotting and heater-power evaluation so that the short eclipse transients are not artificially stretched over multiple days in the final plots. The eclipse transition times are also passed into the thermal ODE solver as segment boundaries, allowing the simulation to integrate piecewise instead of stepping directly across discontinuities in the solar-flux input. While this change does increase the run time of the script to nearly 5 minutes from its previous 2 minutes, it improves the numerical representation of the short square-wave solar-flux drops.

This eclipse model is intentionally worst-case and first-order. It does not attempt to model detailed asteroid shape, spacecraft attitude, local terrain shadowing, or exact orbit geometry as is done in other sections by the Avionics and Operations groups. Instead, it captures the primary thermal effect needed for this level of analysis. The resulting solar-flux profile shows these eclipse events as short downward spikes during the Parthenope, Psyche, and Themis science phases, and the thermal results show that these events only minimally reduce solar heating and slightly increase the required heater power.

Solar Flux Profile

The plotted heliocentric radius and solar flux profiles used in the code are shown in Figure 40.

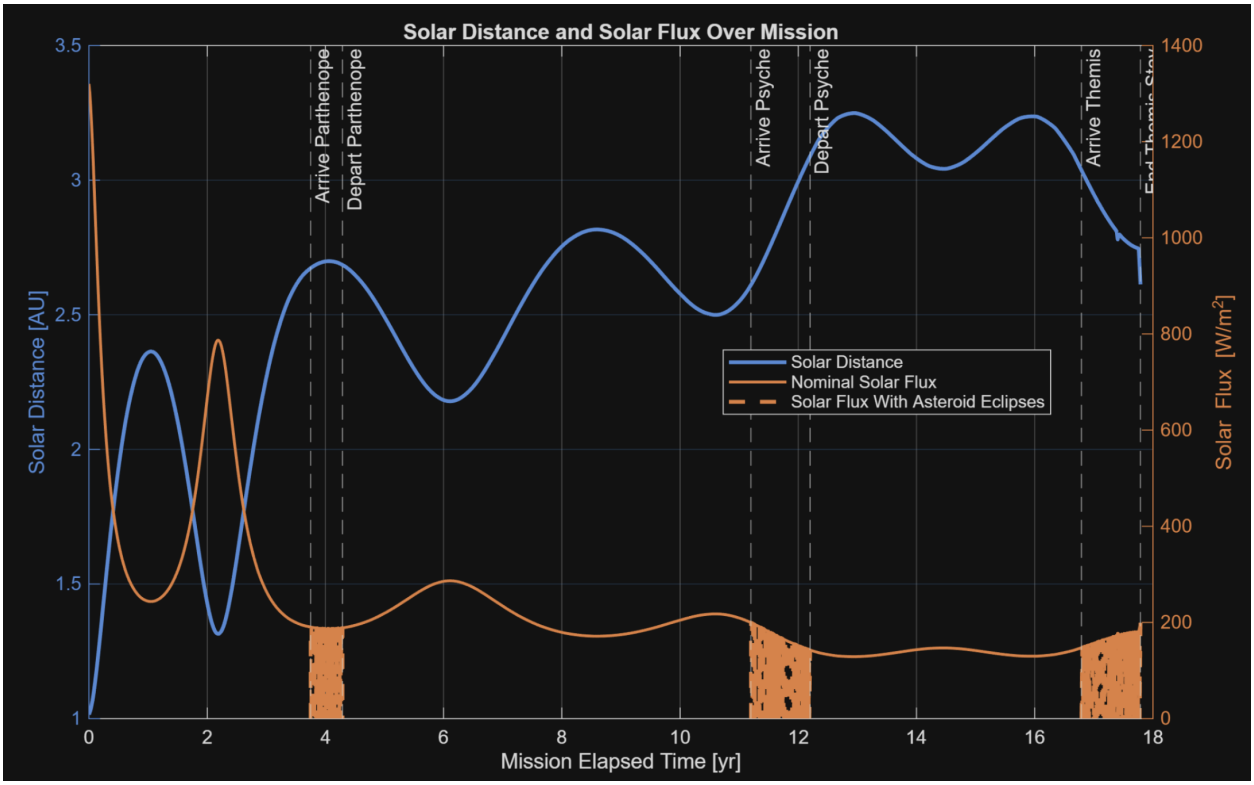


Figure 40: Heliocentric radius and solar flux plotted throughout the mission.

In this plot, the code also evaluates three different thermal-control cases:

1. **No control:** the louvers are fixed closed and heaters are disabled.
2. **Passive louvers only:** the louvers open and close passively based on radiator temperature, but heaters are disabled.
3. **Passive louvers + heaters:** the louvers regulate radiator emissivity and heaters activate when controlled nodes fall below their heater setpoints.

For each case, the transient thermal equations are solved using MATLAB's `ode15s` solver, which integrates stiff systems of ordinary differential equations using a variable-step, variable-order method. The simulation is solved piecewise across mission segments and asteroid eclipse transitions so that the solver does not integrate directly across short square-wave discontinuities in solar flux. At each time step, the script calculates the current mission mode, heliocentric distance, solar flux, eclipse state, internal heat generation, heater power, louver opening fraction, conductive heat transfer between nodes, and radiative heat rejection to space.

The two primary thermal-control elements in this model are the passive louvers and the heaters. The passive louver control is modeled by changing the effective radiator emissivity as a smooth function of radiator temperature using a hyperbolic tangent function:

$$f_{\text{open}} = \frac{1}{2}\left[1 + \tanh\left(\frac{T_{\text{sensor}} - T_{\text{actuate}}}{\Delta T_{\text{transition}}}\right)\right], \tag{25}$$

where T_{sensor} is the radiator-node temperature, T_{actuate} is the louver actuation temperature, and $\Delta T_{\text{transition}}$ controls how gradually the louvers open. The radiator emissivity is then computed as

$$\epsilon_{\text{radiator}} = \epsilon_{\text{closed}} + f_{\text{open}} \left(\epsilon_{\text{open}} - \epsilon_{\text{closed}} \right). \tag{26}$$

This allows the radiators to reject more heat when they are warm and reduce heat loss when they are cold.

The heaters are modeled using proportional control on selected temperature-sensitive nodes. Heater power is only applied to the payload, battery, avionics, ion thruster PPU, RCS propulsion, fuel tanks, and external sensor nodes. For most of these nodes, the heater setpoint is defined as a margin above the lower operating limit. The battery and fuel tank nodes are treated more carefully because they have narrower allowable temperature ranges; their heater setpoints are set directly to their target temperatures. If a controlled node falls below its heater setpoint, heater power increases in proportion to the temperature error, up to a specified maximum heater power. If the node is above its setpoint, the heater is turned off.

Ultimately, determining the required heater power is the main objective of progressing from a single-node analysis to a multi-node transient analysis. The heater history indicates whether the thermal-control system can keep individual components within their operational temperature ranges, while the peak heater power and integrated heater energy provide key inputs for the spacecraft power budget. These quantities are especially important for the power sub-team because they determine both the maximum instantaneous heater load and the total heater energy required over the mission.

Reaction Wheel Spin-Up Study

One concern that the thermal team had was whether reaction wheel spin-up events would have a noticeable effect on the spacecraft thermal response. This concern was based off of the fact that the reaction wheel chosen for this project was the Honeywell HR16-class wheel, and using the datasheet values available for maximum input electrical power and wheel performance, it could draw up to 195W, much larger than its steady state 22W draw.

Because the actual time-dependent spin-up power curve is not publicly available, the model did not attempt to reproduce the exact electrical or mechanical transient. Instead, the spin-up event was modeled as a simplified heat pulse with the same approximate integrated energy as the expected wheel spin-up event.

The reaction wheel was assumed to spin up to its maximum angular velocity over a short duration of approximately 10 s. The maximum electrical input power was taken to be 195 W, and the mechanical output efficiency was estimated to be approximately 64.5 percent by comparing the maximum calculated mechanical output power based off of data from the datasheet to the maximum electrical input power. The remaining power was treated as heat dissipated into the spacecraft. Because the true wheel power curve would likely ramp up and down rather than remain constant, the model approximated this transient as a step function with equivalent integrated area.

In the MATLAB script, this hypothetical reaction wheel heat pulse was inserted periodically throughout the mission, once every three months, to test whether the timing of the event would matter at different points along the trajectory. This allowed the analysis to compare the reaction wheel thermal disturbance against the much larger mission-scale thermal forcing from solar flux, asteroid science operations, and heater activity. Figure 41 shows the resulting response at three different time scales, moving from the broader mission context to a highly zoomed view of a single spin-up event.

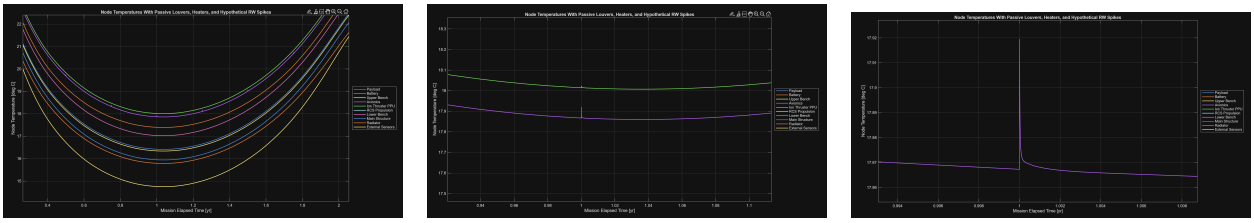


Figure 41: Reaction wheel spin-up thermal response shown at three different time scales. From left to right, the plots progress from a broader mission-scale view to a highly zoomed view of a single reaction wheel heat pulse.

The results show that the reaction wheel spin-up produces a visible but extremely small temperature perturbation. When viewed on a highly zoomed temperature scale, the heat pulse creates a sharp local spike in the affected node temperature. However, the feature is nearly 200 times smaller in magnitude than the features produced by solar flux. Thus, for the purposes of this mission level thermal model, the reaction wheel spin-up heat can be treated as negligible.

Assumptions, Estimations, and Code Limitations

For the MATLAB script to produce mission-wide transient results, several simplifying assumptions were made. The model, however, with these simplifications, still provides for a robust thermal analysis. Varying each of these assumptions and simplifications results in only a small, if not negligible change to the final plots and results. The most important assumptions and limitations are summarized in Table 31.

Table 31: Primary assumptions, placeholder estimates, and limitations in the updated multi-node transient thermal model.

Category	**Assumption / Placeholder**
Geometry and projected area	The spacecraft is modeled using the final MAXAR bus dimensions of 2.2 m × 2.4 m × 2.4 m. The solar-facing area is not attitude-resolved, and is instead approximated using the average projected area of the spacecraft, $A_{\text{proj}} = \frac{1}{2}(LW + LH + WH) = 8.16$ m^2.
Trajectory and solar flux	The updated model imports the spacecraft position history from the implemented trajectory CSV file and computes heliocentric distance from the Cartesian position history. The nominal solar flux is then computed using the inverse-square relation.
Asteroid eclipse model	During Parthenope, Psyche, and Themis orbits, the model includes a simplified worst-case eclipse model. The spacecraft is assumed to orbit each asteroid in a circular orbit at 75 km altitude, with one maximum-duration eclipse per asteroid orbit. During eclipse windows, direct solar flux is set to zero.
Lumped thermal nodes	The spacecraft is modeled as 18 isothermal nodes, as described in Section 3.3.5. This captures subsystem-level trends, but it does not resolve local temperature gradients within panels, instruments, tanks, radiators, or mounting interfaces.
Thermal capacitance	The thermal capacitance of each node is estimated rather than derived from finalized component masses and material properties. These values strongly affect the transient response time and should be refined when more detailed component data is available.
Conductive interfaces	Conductive links are modeled using simplified contact areas, assumed material conductivities, and estimated bolted-interface conductance. These links capture the expected heat-flow paths, but they do not replace detailed interface modeling.
Internal radiation and view factors	Internal radiative exchange between spacecraft components is neglected. This is reasonable for a first-order model because detailed internal view factors are not yet known, but internal radiation should be included in a future Thermal Desktop model.
Internal heat generation	The payload, avionics, battery, PPU, RCS, and other component heat loads are based on current power-budget and duty-cycle assumptions. These values should be updated as the power budget and operations timeline are finalized.
Reaction wheel heat spikes	Reaction wheel heat production is included as a simplified step-function heat input. The actual spin-up power curve is unknown, so the model approximates the transient as a short-duration equivalent-energy heat pulse.

Results

The transient multi-node model produced three main categories of results: node temperature histories, louver activity, and heater power demand throughout the mission. These results connect the thermal architecture directly to spacecraft operations. Rather than only evalu-

ating isolated hot and cold cases, the transient simulation shows how the spacecraft thermal environment evolves along the implemented trajectory, how the passive louvers respond to changing heat loads, and when active heater power is required to maintain temperature sensitive components within their allowable ranges.

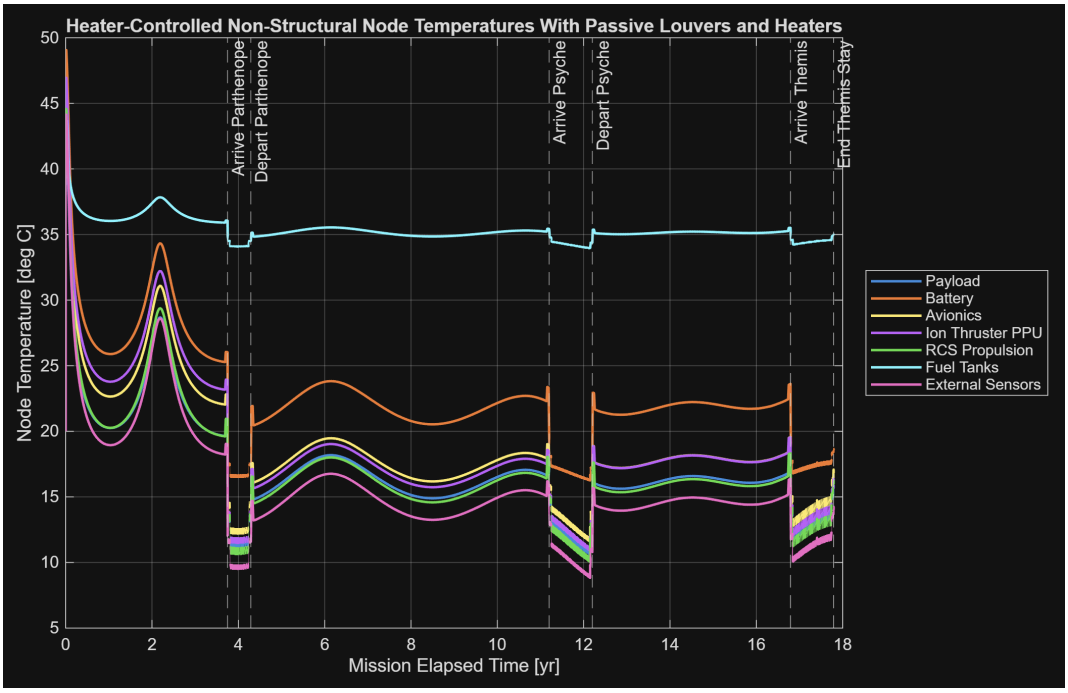


Figure 42: Heater-controlled relevant node temperatures with passive louvers and heaters throughout the mission.

Figure 42 shows the resulting node temperature histories from the updated thermal analysis. In general, each node follows the same broad shape as the solar flux profile. This trend is expected because the spacecraft receives the largest heat input during the early portion of the trajectory, then becomes progressively colder as it moves into the main asteroid belt. However, the louvers and heaters reduce the magnitude of these temperature swings by rejecting heat during warm portions of the mission and adding heat during colder portions of the mission.

A major feature of the temperature results is the presence of temperature troughs after asteroid rendezvous events. These troughs occur because the spacecraft enters a science-observing, lower-power operating mode during asteroid stays. During these phases, several components like the PPU radiate or dissipate less heat than they do during cruise, while the increase in payload science power is not large enough to fully replace the lost internal heat. As a result, each asteroid stay produces a colder thermal environment and requires additional heater power to maintain the controlled nodes within their allowable operating ranges.

The battery and fuel tank nodes required the most specific thermal-control accommodations. The battery has a much narrower operational temperature range than most of the other

spacecraft nodes, so the heater gain for the battery node was increased to keep it within its allowable range with margin. The fuel tanks also have a narrow allowable temperature range, but unlike most of the other nodes, their desired temperature is nearly 20°C higher. For this reason, the fuel tank heater control was also treated more aggressively, forcing the node toward a higher temperature setpoint near 40°C. These two nodes are shown separately in Figure 43.

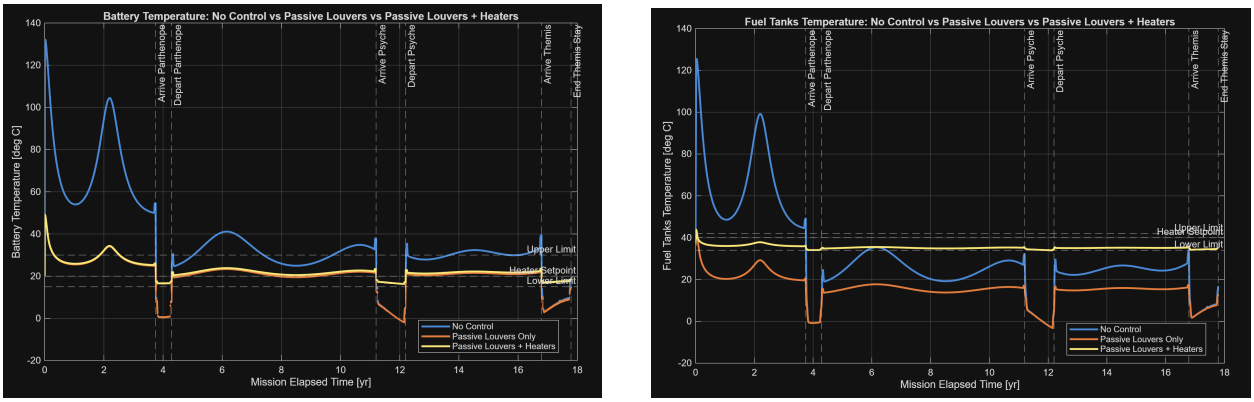


Figure 43: Battery and fuel tank node temperature results. The battery node requires tight temperature control due to its narrow operational margin, while the fuel tank node requires a higher controlled temperature near 40°C.

In both of these node temperature plots, the no-control and just louver-control effects on the node temperatures are also included as blue and orange lines. With no heaters or louvers implemented, it's quite visible how the node temperatures drastically exceed the operational temperature ranges. With just louver control, hot cases are handled, but with no heaters, the cold cases during the asteroid orbits dip below the operational temperature range. This proves why both louvers and heaters are required.

The total resulting louver response is shown in Figure 44. The louvers start nearly fully open during the initial mission hot case, when the spacecraft is closest to the Sun and must reject excess heat. After the spacecraft moves outward toward the main asteroid belt, the louver open fraction drops sharply. During the asteroid-belt portion of the mission, the louvers remain approximately closed for most of the time, with only small fluctuations. This behavior is consistent with the node temperature histories. Early in the mission, the main thermal problem is heat rejection, while later in the mission, the main thermal problem is heat retention.

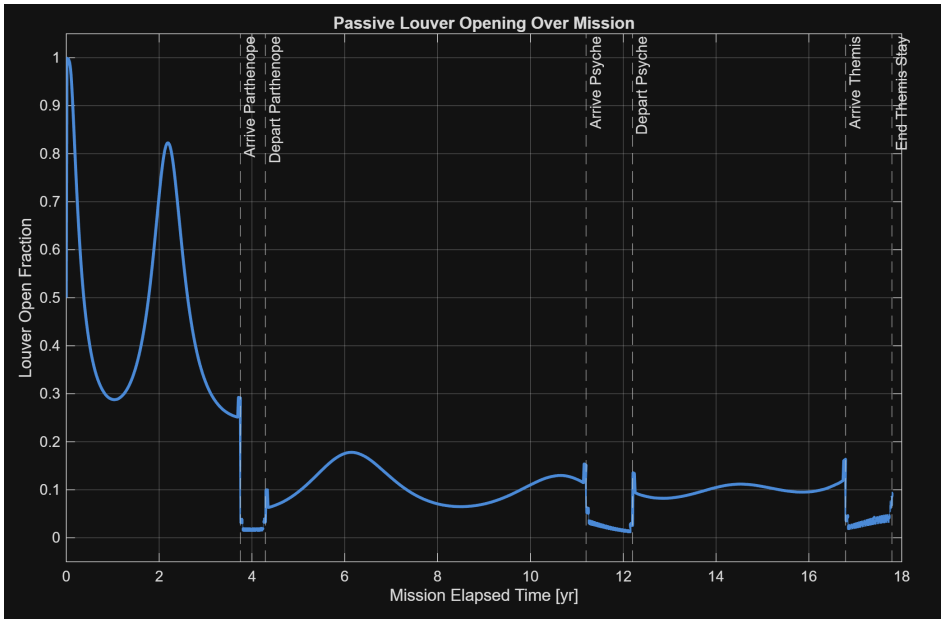


Figure 44: Passive louver opening fraction throughout the mission. The louvers open during the initial hot case and remain nearly closed throughout most of the asteroid-belt operations.

The final and most important output of the thermal-control analysis is the total heater power demand, shown in Figure 45. This plot is the culmination of the transient thermal model because it translates the temperature-control problem into a direct spacecraft power requirement. Heater power remains relatively modest during the early, high solar flux portion of the mission, then increases during colder asteroid belt operations. The largest heater power spikes occur during asteroid stays, when the spacecraft is in a lower power science mode and the internal heat generated by several subsystems is reduced.

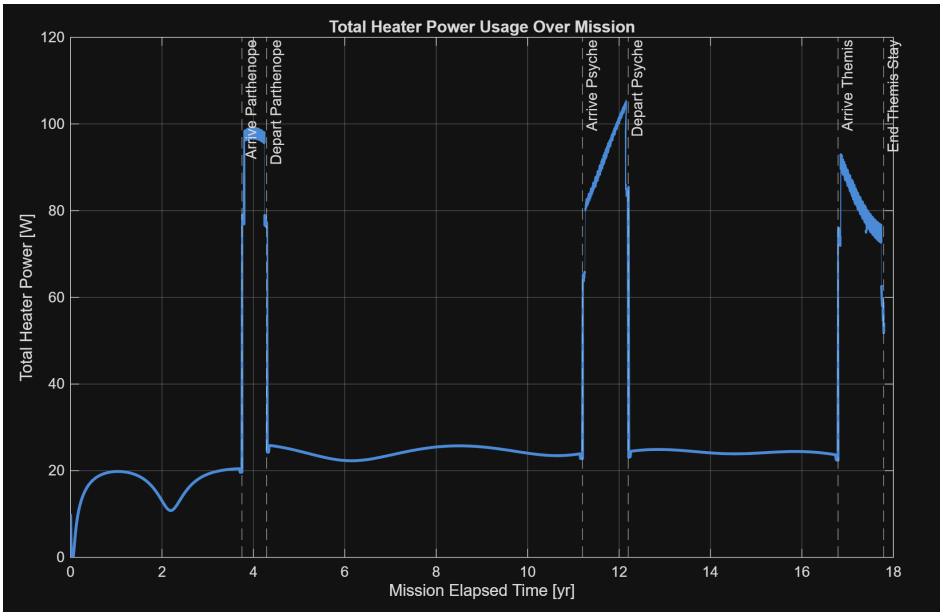


Figure 45: Total heater power usage throughout the mission. Heater demand increases during colder asteroid belt operations and spikes during asteroid stays.

The updated simulation predicts a peak heater power of approximately 105.27 W and a total integrated heater energy of approximately 4.9487×10^6 Wh. These are the most important thermal outputs for the power subsystem because they define both the maximum instantaneous heater load and the total heater energy required over the mission. Therefore, the heater power curve provides the direct interface between the thermal-control analysis and the spacecraft power budget.

3.3.7 Future Work (Riley Foard)

The current multi-node MATLAB model provides a very useful mission-level thermal estimate. However, if the 10 week quarter was not the limiting constraint, the next major step would be to move from a lumped-node model to a higher fidelity Thermal Desktop model. In the current analysis, each major spacecraft component or subsystem is represented as an isothermal node, and radiative exchange is simplified to external radiation from MLI-covered panels, radiator nodes, and external sensors. With more time, a Thermal Desktop model would allow the full spacecraft geometry to be represented directly, including the true spacecraft shape, radiator placement, component locations, and detailed surface properties.

The most important improvement from a Thermal Desktop model would be the ability to calculate geometric view factors and radiation exchange between surfaces. The MATLAB model does not explicitly compute which surfaces can see each other, which surfaces are shadowed, or how heat radiates between nearby spacecraft components.

The second major improvement would be refinement of heat paths. In the current model, conductive links are represented using estimated contact areas, simplified interface conductances, and approximate material properties. With additional time and more finalized data

on the locations and geometry of components, these links could be replaced with more precise geometric interfaces. This would improve the predicted transient response of each node, especially for thermally sensitive components such as the battery and fuel tanks.

3.4 Power (Miina Anvelt)

3.4.1 Solar Arrays

Solar Cell Tradeoffs

The solar array cells selected are perovskite and silicon tandem solar cells which have been shown in experimentation to have efficiency up to 35 percent and consist of a triple junction in the solar cell structure [48]. While these solar cells have yet to be used on a large-scale spacecraft mission, they were shown to be functional in space as a part of the Caltech Solar Power Project experimental payload [49]. Theoretically, the perovskite tandem solar cells are expected to reach up to 50 percent efficiency in a lab setting [48], [50]. Based on recent research into the solar cells, these solar cells are assumed to have advanced to the extent they can be incorporated into a larger scale solar panel system by the launch time in 2029. In comparison, heritage spacecraft typically use SolAero ZTJ triple junction cells. These have heritage use from the Psyche mission and are rated for an efficiency of 29 percent at a temperature of 28 degrees Celsius [51]. For the AFONSO spacecraft, the perovskite cells were selected because of their greater efficiency compared to other solar cells considered.

The major consideration for the solar arrays is ensuring they are taking enough power to provide charging to the batteries and the system throughout the trajectory to the asteroids. This means that one of the design factors considered was the size of the solar arrays. For the Psyche mission, the solar arrays had a 62.6 m^2 coverage area and for the DAWN mission the solar arrays had a 36.4 m^2 coverage area [51], [52]. The solar array size that was decided upon was a coverage area of 75 m^2 as this area ensured a closed power budget along the orbit. This is larger than the solar arrays used for the Psyche mission and the DAWN mission but matches the final power consumption from the electrical propulsion being slightly higher than past heritage missions. The maximum power consumption reached by the Psyche mission is 4.5 kW and for the Dawn mission is 2.5 kW while for the AFONSO mission the maximum power reaches up to 5 kW on an asteroid approach [51], [52]. The slightly larger size was then chosen as the additional mass of additional solar panels was not deemed harmful based on mass budget calculations while providing critical input power into the system.

Solar Input Power Calculations

The solar influx and power intake by the solar panels was determined by the following set of equations [53]:

Solar Flux:

$$I = \frac{L}{4 * \pi * d^2}$$

$$Power = Flux * Area$$

$$Luminosity : L = 3.85 * 10^26W$$

$$P = \frac{3.85 * 10^{26}}{4 * \pi * d^2} * 50 * 0.35$$

By then applying the power equation with an assumed luminosity of the sun and area of 75 meters squared for the solar panels, a solar array intake power over time plot was calculated for the entire trajectory in Matlab. The Matlab script takes in the values output by the trajectory team in the form of a csv and finds the position of the spacecraft relative to the sun. Then the power intake by the solar panels with 35 percent efficiency is plotted with respect to time. The solar panels are assumed to be facing towards the Sun at each calculated point in time for the above calculations. This assumption was completed by a combined Payload, Propulsion, and Operations team by determining the maximum angle of rotation of the solar panels that is required at each point along the trajectory to return to a sun-pointing position. Based on this calculation the ACS system adjusts the spacecraft about the thrust axis to return to a position of maximum sunlight and this assumption is applied for solar intake calculations. Since the maximum rotation required by the solar arrays to return to a position perpendicular a line from the sun to the spacecraft is 90 degrees, this achievable while still maintaining the same thrust vector direction and magnitude.

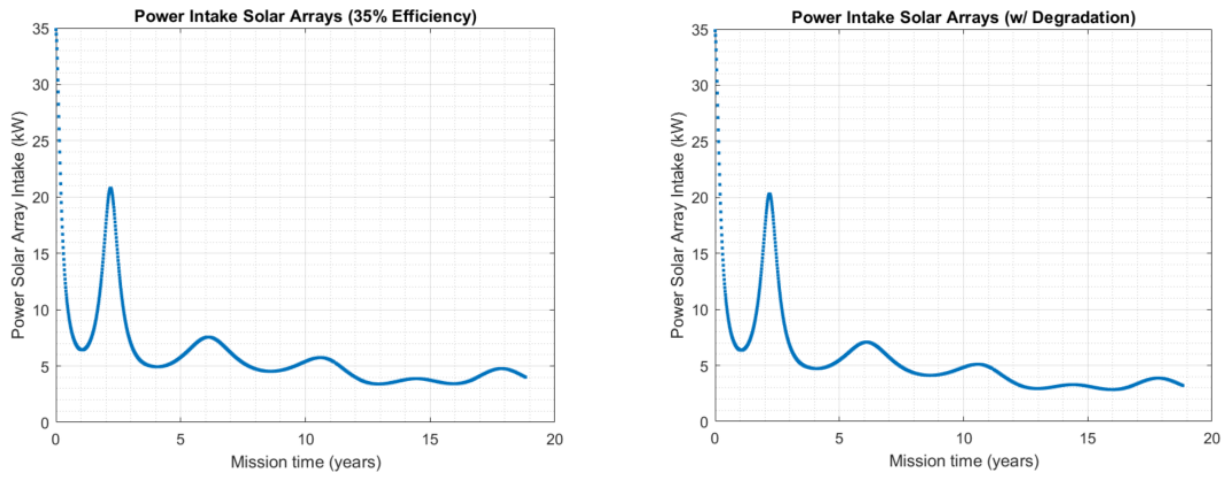


Figure 46: Solar Arrays input power without degradation and with 20 percent degradation over mission timeline

The plot in figure 46 indicates the intake power from the solar arrays, assuming 35 percent efficiency, for the whole trajectory path. The general trends followed are that intake from solar arrays decreases to around 3 kW at end of mission distances and the early mission phase (0 - 4 years) has power intake in the range of 5 - 35 kW. Since the chosen solar panel size follows heritage mission solar panel sizing, these power intake values also are similar to values predicted based on the Psyche mission trajectory which reaches a minimum incoming solar power around 2 kW as it reaches the Psyche asteroid [51]. The overall solar array

input power plot indicates that greater than 4 kW are incoming into the system at each point along the trajectory.

Solar Cell Degradation

Further analysis was completed on the effects of efficiency decrease over mission time. Based on long-term research which has been conducted into terrestrial solar panels, the degradation tends to be on the order of 0.7 percent to 1 percent every year [54]. Estimating the mission timeline to be around 17 years, the solar panels were assumed to then decrease 20 percent in total efficiency by the end of mission timeline. This means 1.17 percent degradation per year over the mission which represents a high estimate as the solar panels tested in research were used on Earth and faced degradation in various different conditions and aging conditions.

The predicated main causes of degradation in space will be extended use over more than ten years, and potential out-gassing from materials and components used on the spacecraft. While the spacecraft will be assembled in a clean room which ensures that all larger particles would not be on the solar panels, there is still the possibility of out-gassing from the materials themselves as they enter the low pressure, vacuum conditions in space. As described in a research article by Zilong Jiao et. al. the most common out-gassed molecule in space tends to be water and becomes present due to various ways the spacecraft are manufactured [55]. With this in mind, it is likely that the degradation of 20 percent over time will be met due to various factors impacting the solar cell performance. Based on the plot above for the solar array input power over time as a function of mission time with the 20 percent degradation accounted for, the solar arrays are still able to input greater than 2 kW over the whole mission timeline. The specific power budgets of the incoming power over the mission timeline compared to the power consumption by all systems were created to further analyze the available power over the mission.

3.4.2 Battery

Battery Tradeoffs

A major tradeoff considered in the design of the power system architecture of this spacecraft was a consideration of the use of an Radioisotope Thermoelectric Generator (RTG) or the use of solar array panels. This was a consideration as the main objective of the mission is to travel to three asteroids to meet scientific data objectives and the initial possible trajectories in the early design phase took extended periods of time between 15 to 20 years. The final choice that was decided upon were solar arrays instead of an RTG for two main reasons. Heritage from previous spacecraft missions proves that the mission is possible using solar arrays with a rechargeable battery and the subsystems such as the electrical propulsion system have high power requirements that vary across the mission profile [51], [56]. The first point is significant as using systems which have flight heritage means there is less risk associated with using the system. The second point is equally as important in the design consideration as an RTG is typically rated for continuous power release which would be difficult to use if the mission requirements are not continuously within a small range [57].

In terms of heritage, the two main missions that were analyzed were the Psyche mission and

the DAWN mission. In both missions the objective was to reach an asteroid, they used an electrical propulsion system, and they had solar arrays with a rechargeable battery. In the case of Psyche the battery was a MAXAR battery that had previously been rated and used in GEO orbits, and was a single 7170 Watt-hour Li-Ion battery. Based on the specifications provided in a study completed by Oh et. al, the main properties of the battery were a 13 Li-ion cell single string arrangement and a rating of 144 Ah after use over 15 years based on the MAXAR use data [51]. In comparison, the DAWN mission used a 35-amp-hour rechargeable nickel hydrogen battery [56]. For Psyche the planned mission duration is 8 years while for DAWN the mission is approximately 11 years [56], [51]. In both cases, the mission is a long duration mission and the power was provided through a system of solar arrays, batteries, and voltage converters. Based on these past missions, as well as other deep solar system missions, a system of solar arrays and batteries was similarly designed for this spacecraft mission.

Another tradeoff that went into the battery consideration was the use of Lithium-Ion batteries or Nickel Hydrogen batteries. This tradeoff was considered as the DAWN and Psyche missions used different batteries despite having similar mission power requirements. The Nickel Hydrogen battery has a long life cycle and an energy density between 40-75 Wh/kg. In comparison, the lithium-ion battery has an even high power density, around 150-280 Wh/kg, and has a higher reliability due to both flight heritage and the chemical processes involved in the battery [58], [59]. Based on the mission objectives, the goal of the spacecraft is to have continuous operations over a 16-17 year period while also being as lightweight as possible. Due to this, the lithium-ion battery was selected for its very high energy density. The battery selected is a 7858 Wh MA190-111 battery manufactured by GS YUASA (with a 122.3 Wh/kg power density) [60].

Battery Final Choice

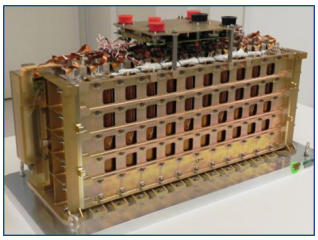


Figure 47: The MA190-111 battery manufactured by the company GY UASA (Image Source: [60])

The battery selected for this mission is a 7858 Wh MA190-111 battery (figure 47) manufactured by the company GY UASA [60]. While the Psyche mission used a 7170 Wh battery, there is limited publicly available information about the battery's dimensions and mass which led to the selection of the chosen battery. While it not only has a slightly higher Wh rating, its mass is around 60 kg and it has the dimensions of 0.643m $\times$ 0.234 m $\times$ 0.335 m (or a volume of 0.0503 meters cubed). Furthermore, it is rated at 4719 Wh after 18 years in GEO operations and is rated as being compatible for battery systems with voltages from

22 V to more than 100 V [60]. This means it has an energy density of 122.26 Wh/kg which is within the range expected from a space-rated lithium-ion battery. While it is not clear whether this battery has flight heritage, as a similar battery energy density and amount of energy (in Wh) was flown for the Psyche mission, this indicates the overall battery solution is feasible. For this mission, the main voltage bus is set to 80 V and based on the provided specifications, the battery will be compatible with the MAXAR power control unit system.

3.4.3 Power and Energy Budgets

Power Budgets

To verify that the solar panels and battery will be able to support the main spacecraft subsystems, a peak power consumption analysis over the trajectory of the mission was completed. Information was provided by each of the spacecraft subsystems about the components requiring power and the amount of power that is required (table 32). Then the assumption was made that there is 10 percent power loss in cables and thermal effects, based on typical margins of error considered in power contingency plans [61]. The results indicate a peak power consumption of 465.6 watts, or 512.16 watts assuming a 10 percent margin, across all subsystems, including scientific instruments, if they turned on continuously at maximum power and at the same time. In comparison, the standby power reaches 146.45 watts, or 161.1 watts with margin.

For all power budgets, a maximum power consumption state was assumed for the cruise phase of the mission as the maximum power accounts for the magnetometer being always on, the telecommunications being set into a mode where it can down-link, and the reaction wheels are adjusting the attitude. This allowed for the maximum power consumption to be compared to the solar array intake power, providing a worst case scenario power margin over time. For the phase around the asteroids, the power budgets also assume all systems are at maximum power to determine a worst case power scenario. The assumption is made that telecommunications is always in a state of down-link, which is not actually the case as there are specific down-link windows but provides an approximation for maximum system power consumption.

Component	**Max Power (W)**	**Standby Power (W)**
Magnetometer	4.4	4.4
VIR	17.6	8.8
Multispectral Imager	8.7	4.35
RAD750	14	14
IMU Unit	12	12
Star Tracker	12	12
Sun Sensors total	3.2	3.2
Total Reaction Wheels	195	66
Telecom	198.7	21.7

Table 32: Major subsystem components power consumption

Then the power required by the electrical propulsion system was calculated based on thrust and the thermal heater requirements were provided across the mission profile by the thermal team. These were considered initially independently of the subsystem power as they vary across the mission profile and make up the majority of the power consumption. In order to determine the maximum power required by the electrical propulsion system, the thrust values for each segment of the trajectory that were provided by Operations. The power required to result in the thrust segments was calculated based on the definition of power for an electrical propulsion system (equation below) [62]. The main assumptions that were made by this calculation were that the ISP of the propulsion system is 3100 to calculate the approximate exit velocity and the NEXT-C thruster efficiency is 73.9 percent, as defined by the Propulsion team in final design presentation ([63], [64]).

$$P = \frac{T * (Isp * g_0)}{2 * 0.739}$$

Applying the above formula to the thrust values that calculated by the Operations team, a plot was generated in Matlab for the total power consumption by the propulsion system (figure 48). The general formulation for the electrical propulsion power makes a couple of assumptions. First, the thrust provided by operations is a continuous thrust applied per segment which is then converted into a continuous power consumption. Second, the thrust is assumed for these calculations to have instantaneous increases between segments. These assumptions are present as each segment of the trajectory assumes a different constant thrust. Further analysis in future work would look into the power required to enable the instantaneous increases in thrust between the segments or whether these increases would need to occur over the period of a couple of days.

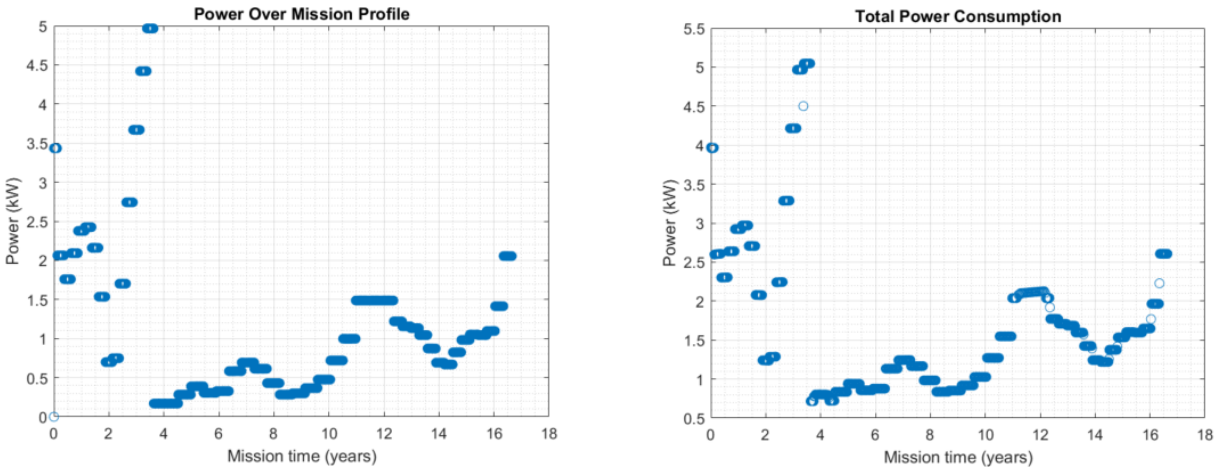


Figure 48: Electrical propulsion power consumption over mission (left) and total system power consumption including heaters and all subsystems (right)

The peak power consumption by the electrical propulsion system and the heater systems was then interpolated with respect to the time values considered in the solar array plot. By then

adding together the thermal peak power, the electrical propulsion peak power consumption, and the total subsystems power, the peak power consumption for specific points in time across the mission was calculated in Matlab. A plot was generated in Matlab of the intake solar array power subtracted by the peak power consumption by the total system (figure 49) considering regularly operating solar cells over time.

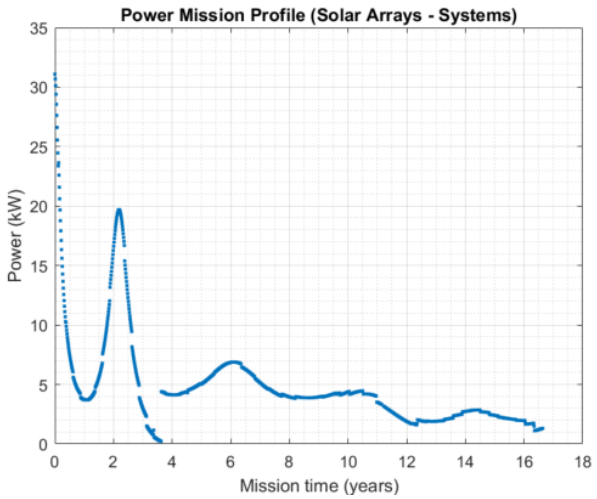
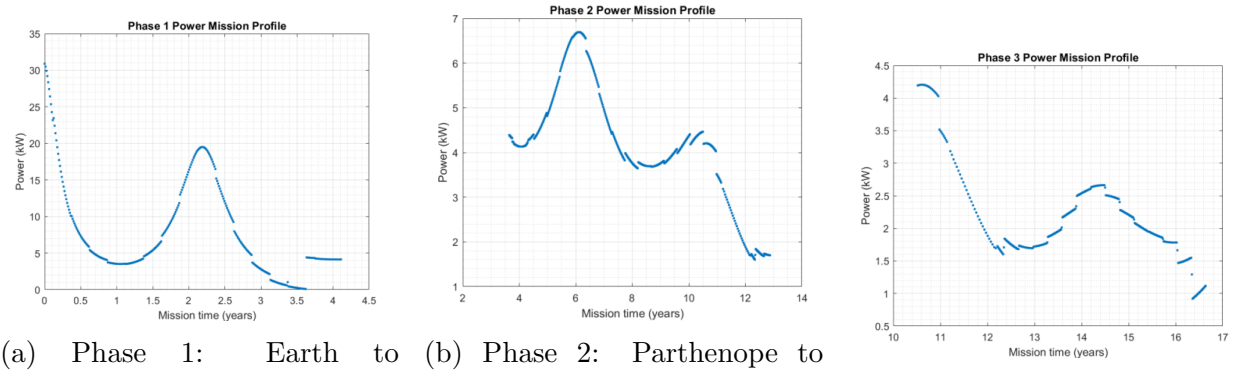


Figure 49: Power profile over mission timeline

The plot (figure 49) indicates that using solar intake power alone for the majority of the mission results in a positive power margin, which indicates that the power budget is closed. The area of concern, where the power margin drops to its lowest at around 0.1 kW, is on the approach to Parthenope. During this region of maximum power consumption, the total systems, will be put into standby, and telecommunications was assumed to have a duty cycle of operating every 38 days to ensure the margin is kept positive. Further analysis will need to be completed on orbit to ensure the power margin is maintained despite possible degradation in the thruster efficiency which is not accounted for in the current assumptions.

The chosen battery is also planned to be charged whenever its state of charge is less than 80 percent and there is additional input power from the solar arrays. This means that the battery will be used to store energy and help power the systems in the case the power margin drops negative for any reason. Since the battery capacity is 7 kWh, it can sustain 7 kW for a period of 1 hr or less power requirements for a longer time. In this case the general factor that is causing this large power consumption is the thrust in the approach to Parthenope and the electrical power consumption reaches up to 5 kW. The battery alone would only be able to sustain the 5 kW consumption by electrical propulsion and systems for less than 2 hours. However, the power margin has been calculated to be a positive value for the interval so the battery does not have to be used based on the margin. Hence, as the power margin is positive for the whole trajectory the battery will mainly be employed in anomaly scenarios.



(a) Phase 1: Earth to Parthenope

(b) Phase 2: Parthenope to Psyche

(c) Phase 3: Psyche to Themis

Figure 50: Comparison of power profile over the mission phases.

The power profile can also be further broken down into the power over each phase of the trajectory (figures 50a, 50b, 50c). Based on the plots for each phase, it is apparent how the overall power margin varies across each phase. For phase 1, from Earth to Parthenope, the spacecraft will experience the greatest range of power margin values. As the spacecraft takes off it has a power margin around 30 kW and reaches a minimum around 0.1 kW of margin around the approach to Parthenope. For phase 2, from Parthenope to Psyche, the spacecraft experiences a range of power margins from a maximum margin of 7 kW and a minimum around 1.75 kW. The overall power profile has slight variation but overall indicates that this phase will maintain above 1 kW of power margin at all points. For phase 3, form Pysche to Themis, the range of power margins is from a maximum around 4.25 kW to a minimum around 1 kW. This phase of the trajectory thus has the least variation in the power margin.

Overall, the power profile indicates the 75 meters squared solar arrays, along with the 7858 Wh battery, will be able to sustain the total system power requirements for the mission timeline. Future work could be completed to include degradation of the battery and the decrease in thruster efficiency over the mission.

Energy Budget

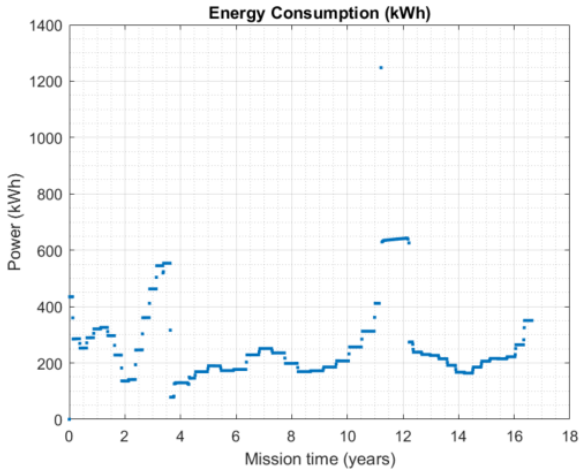


Figure 51: The total energy consumption by all subsystems over mission timeline.

Along with the power profile, further analysis was completed by numerically integrating the power consumption over each time segment (using Matlab) to get plots of the energy consumption over time. Each point on the energy consumption plots indicates the energy value that was determined for the time step considered. Based on the plots, the energy consumption values appear to be on the higher side but this is due to the extended length over which high power is being used. For example, for the thrust values that reach almost 5 kW of continuous power, they are being applied over a period of time which is greater than two months which indicates that the watts*hour breakdown where the continuous watt amount is multiplied by the duration in hours becomes a large value. Since these plots indicate the energy consumption, they follow a different general pattern of maximum and minimum than the power profiles which were based on the solar arrays subtracted by the power consumption by the systems. Overall, the energy consumption appears to increase on the approaches to the asteroids and this follows an expected pattern as the approaches are also when electrical propulsion power consumption increases.

3.4.4 Voltage Converters

The voltage converters chosen for this mission are the MAXAR power control unit used in the Psyche mission power system with modifications for the subsystem requirements of this mission [51]. The main modifications from the Psyche power control unit will be having the main voltage bus at 80 V instead of 100 V as the maximum voltage consumption by any subsystem in the mission is less than or equal to 80 V. A general block diagram of the power control unit is outlined in the block diagram below and follows the Psyche mission power control unit block diagram as the same components are planned to be used. The general idea is that the power incoming from the solar arrays is directed into solar array power controllers and then into shunt switch units. These units either divert any extra voltage incoming to ground or send the voltage into the boost converter tray [51]. In the boost converter tray, the incoming voltage, if less than 80 V, is boosted up to 80 V for the main voltage bus.

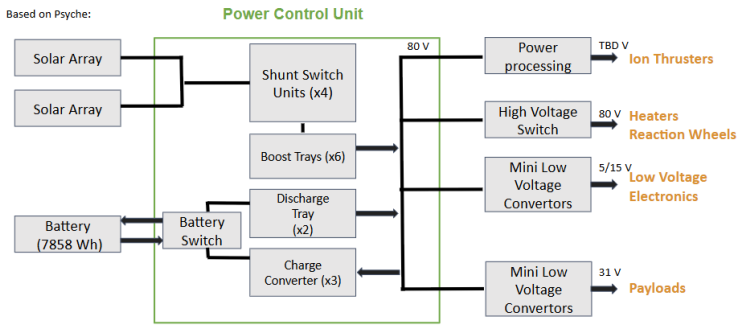


Figure 52: A power control unit block diagram simplified from Psyche power control unit diagram in Oh et. al. [51]

After this conversion the voltage is then available to the system either to be directed into the batteries or into the subsystems. Beyond the main voltage bus, the voltage is then converted into lower voltages for the various subsystems using mini low voltage converters. These converters have heritage use from the Psyche mission and the units are planned to be used for this mission as well. They allow for the voltage to be down-converted from 80 V to 5 V, 15 V, and 31 V as required by various subsystems.

Component	Voltage (V)	Current (A)
Magnetometer	31	0.1419
VIR	31	0.568
Multispectral Imager	31	0.281
RAD750	5	2.8
IMU Unit	15	0.8
Star Tracker	28	0.01429
Sun Sensors	28	3.2
Total Reaction Wheels	80	0.275

Table 33: Major subsystem components voltage and current.

Overall, the voltage converter units selected have heritage both being flown in the past as a part of the Psyche mission. They enable power to be taken in by the solar arrays, normalized to a general voltage bus, and then distributed to charge a battery or power the systems. Further future analysis will determine how the voltage conversion affects power loss and additional components that may aid power storage.

3.5 Propulsion

3.5.1 Major Components Overview (Citli)

The propulsion system utilizes three primary hardware components across the launch vehicle and spacecraft. Table 34 summarizes each component, its type, thrust level, quantity, and mission role.

Label	Component	Type	Thrust (N)	Qty	Mass (kg)	System
(A)	SpaceX Merlin 1D	Bipropellant engine	845,000 (SL)	27	470	Takeoff
(B)	NASA NEXT-C	Electric ion thruster & PPU	0.300	2	78	Deep space transit
(C)	Aerojet MR-111G	Monopropellant thruster	1.8–4.9	12	4.44	RCS

Table 34: Major propulsion components. SL = sea-level thrust.

This architecture separates propulsion into three systems for different mission phases. The Falcon Heavy provides the high thrust required for Earth escape and initial heliocentric injection. Once in deep space, the NEXT-C ion thrusters serve as the primary propulsion, utilizing their high specific impulse to achieve the 13.0701 km/s ΔV requirement within the spacecraft's mass budget. The MR-111G monopropellant thrusters act as an independent Reaction Control System (RCS) for attitude management and asteroid proximity operations.

3.5.2 Engine Selection (Citli)

The total required post-launch $\Delta V \geq 13.0701$ km/s drives a two-stage architecture.

Stage 1: Falcon Heavy (Launch Vehicle)

- **First stage:** Three reusable Falcon 9 engine cores, each with 9 SpaceX Merlin 1D liquid-fuel engines (27 engines total).
- **Second stage:** One Merlin 1D Vacuum restartable liquid-fuel engine.

The Falcon Heavy performs the impulsive launch phase. Achieving the necessary Earth escape and heliocentric injection energy (C_3, defined as the specific kinetic energy required to escape Earth's gravity well) within a 7.0 km/s ΔV budget requires high thrust to minimize gravity losses. The Merlin 1D engine suite provides this power, while the vacuum-optimized second stage ensures efficient orbital insertion. Using a commercial launch vehicle avoids the mass penalties associated with carrying a dedicated chemical Earth-escape stage.

The launch vehicle uses a triethylaluminum-triethylborane (TEA-TEB) pyrophoric ignition system. This system consists of a chemical mixture that ignites spontaneously upon contact with oxygen, eliminating the need for spark igniters. This ensures reliable vacuum restarts during cold staging (the process of shutting down the booster engines and physically separating before igniting the upper stage) to reach the required departure v_∞. Detailed launch sequencing and staging operations are outlined in the payload separation sections.

Stage 2: Spacecraft (Solar Electric Propulsion & RCS)

- **Primary propulsion:** 2× NEXT-C Gridded Electrostatic Ion Thrusters.
- **Reaction Control System:** 12× Aerojet MR-111G Monopropellant Thrusters.

A trade study was conducted to compare gridded ion and Hall-effect thrusters for the primary deep-space engine. The selection criteria prioritized high specific impulse (I_{sp}) to meet the 13.0701 km/s post-launch ΔV requirement, along with sufficient thrust to avoid a multi-engine cluster.

As shown in Table 35, Hall-effect thrusters like the Fakel SPT-140 provide high thrust-to-power ratios but have a lower I_{sp} of approximately 1800 s. Achieving a 13.0701 km/s ΔV with an I_{sp} of 1800 s on a 2518 kg spacecraft requires over 1300 kg of propellant. This results in a mass fraction exceeding 52%, which limits available payload and structural mass.

The NEXT-C operates at a nominal I_{sp} of 3100 s, reducing the required xenon propellant to approximately 880 kg (a 35% mass fraction). Compared to the NSTAR thruster used on the Dawn mission, the NEXT-C provides higher thrust capability. This allows the mission to use two NEXT-C units: one active and one cold spare.

The NEXT-C Power Processing Unit (PPU) allows for grid voltage tuning, which increases thrust at the expense of specific impulse. This capability, demonstrated on NASA's DART mission, enables the spacecraft to adapt to changing solar power availability in the outer asteroid belt without requiring additional engine types.

	Fakel SPT-140	**NASA NSTAR**	**NASA NEXT-C (Selected)**
Propulsion Type	Hall-Effect	Gridded Ion	Gridded Ion
Propellant	Xenon	Xenon	Xenon
Nominal Power	4.5 kW	2.3 kW	6.9 kW
Max Thrust	280 mN	92 mN	237 mN (up to 300 mN tuned)
Specific Impulse (I_{sp})	1800 s	3100 s	4170 s (3100 s tuned)
Nominal Mass Flow Rate	15.8 mg/s	3.0 mg/s	3.92 mg/s (calculated at 119 mN)
Dry Mass	8.5 kg	8.3 kg	39.0 kg
Propellant Mass Fraction (13.1 km/s)	52.3%	35.0%	35.0%
Units Needed for 300 mN Thrust	2	4	1 (Active) + 1 (Spare)
Primary Flight Heritage	Psyche (2023)	Dawn (2007)	DART (2021)

Table 35: Primary Electric Propulsion Candidate Specifications.

Twelve Aerojet MR-111G hydrazine thrusters were selected for attitude control, thrust vector alignment during NEXT-C burns, and momentum wheel desaturation. A detailed trade study justifying the selection of these thrusters over cold gas and bipropellant alternatives, alongside specific configuration mapping, is provided in Section 3.5.7 (Attitude Control System).

The spacecraft uses a distributed tank architecture. The primary xenon propellant ($\sim$ 880 kg) is distributed across 6 Composite Overwrapped Pressure Vessels (COPVs). These vessels are cylinders that utilize a thin metallic fluid barrier reinforced by a carbon-fiber structural wrap for lightweight, high-pressure storage. Each cylinder measures 0.3 m by 1.3 m. Using six vessels instead of a single tank improves structural packing within the Maxar 1300 bus, reduces required wall thickness, and allows for individual tank isolation in the event of a micro-meteoroid puncture.

The RCS hydrazine propellant is stored in 2 independent titanium tanks. The system uses a Nitrogen (N_2) blowdown pressurant for the hydrazine rather than Helium. In a blowdown system, the pressurant gas expands and pressure naturally decreases as propellant is consumed, requiring no active pressure regulators. Furthermore, because Helium is a highly volatile, small molecule, it presents a higher risk of permeation and leakage over the 16.81-year mission timeline. The hydrazine tanks are sized with a 25% ullage volume (empty

internal headspace) to accommodate the larger Nitrogen gas. This results in a mass penalty of less than 1 kg but mitigates the risk of long-term pressurant loss.

3.5.3 Propellant Selection (Sophia)

Stage 1 — Falcon Heavy

- Propellants: Liquid oxygen (LOX) and kerosene (RP-1). This combination was selected by SpaceX for its high energy density, relatively low cost, and mature handling infrastructure. RP-1 is a refined form of kerosene with tightly controlled composition to prevent coking in the turbopump and injector.
- Ignition: TEA-TEB (triethylaluminum–triethylborane) pyrophoric ignition system. TEA-TEB ignites spontaneously on contact with oxygen, eliminating the need for a separate ignition system and enabling reliable multi-engine restarts. This is critical for the second stage, which must relight in vacuum to achieve the required C_3 for the mission.
- First stage tanks (6 total): Each of the three reusable booster cores carries one LOX tank and one RP-1 tank, for a total of six tanks across the first stage. The LOX tank is positioned above the RP-1 tank in each core to keep the center of mass stable as propellant is consumed.
- Second stage tanks (2 total): A single LOX tank and a single RP-1 tank feed the Merlin 1D Vacuum engine. The second stage is expendable and sized to provide the ΔV needed to reach the required departure $v_\infty = 7.0$ km/s after separation from the first stage.

Stage 2 — Spacecraft

- Ion thruster propellant — Xenon: The NEXT-C ion thruster uses xenon as its propellant due to its high atomic mass, chemical inertness, and low ionization energy. High atomic mass maximizes momentum transfer per ion, directly improving thrust efficiency. Xenon's inertness eliminates corrosion risk to the ion optics and spacecraft surfaces over the 17-year mission lifetime. The thruster is operated at $I_{sp} = 3100$ s rather than its rated maximum of 4,190 s because the PPU is tuned to a lower beam voltage, trading some specific impulse for higher thrust. This operating point was fixed as a constraint by the trajectory optimizer to satisfy the mission ΔV budget within the allowed time of flight.
- Xenon storage — COPVs: Xenon is stored supercritically at approximately 100 bar, achieving a storage density of $\rho_{Xe} \approx 1000$ kg/m^3. Composite Overwrapped Pressure Vessels (COPVs) are used rather than metallic tanks to minimize structural mass at this storage pressure. The total xenon propellant mass of 885 kg is distributed across 6 COPVs connected in parallel, providing redundancy and allowing propellant feed to continue if one vessel develops a leak or valve fault.
- RCS propellant — Hydrazine: The 12 Aerojet MR-111G thrusters are fed by hydrazine monopropellant stored in 2 redundant tanks. Hydrazine was selected for three reasons. First, it is storable at room temperature with no active thermal control required, which is critical for a mission lasting over 17 years. Second, the monopropellant configuration significantly reduces feed system complexity compared to a bipropellant RCS — there is only one propellant line per thruster rather than two, halving the number of valves,

regulators, and potential leak points. Third, the MR-111G has extensive flight heritage on Atlas V in this exact configuration (12 thrusters), reducing development risk. The tradeoff is a lower I_{sp} of approximately 220–230 s compared to a bipropellant system, but given that the RCS handles only attitude control and not translational ΔV, mass flow efficiency is a secondary concern.

3.5.4 Assumptions and Propellant Mass Calculations (Sophia)

Key Assumptions

- Nominal O/F ratio for MMH/NTO: $O/F = 1.65$
- Ion thruster thrust cap: $F_{max} = 300$ mN
- Ion thruster specific impulse: $I_{sp} = 3100$ s
- Standard gravity: $g_0 = 9.81$ m/s^2
- LT chain initial mass: $m_0 = 2518.0$ kg (from verified trajectory report)

Tsiolkovsky Rocket Equation

$$\frac{m_0}{m_f} = \exp\left(\frac{\Delta v}{I_{sp} \cdot g_0}\right) \quad \Longrightarrow \quad m_{prop} = m_0\left(1 - \exp\left(\frac{-\Delta v}{I_{sp} \cdot g_0}\right)\right) \tag{27}$$

Per-Leg Propellant Mass

Leg 1 — Earth → Parthenope ($\Delta v = 6147.1$ m/s):

$$\begin{aligned} m_f &= 2518.0 \times \exp\left(\tfrac{-6147.1}{3100\times 9.81}\right) = 2518.0 \times e^{-0.2021} = 2057.2 \text{ kg} \\ m_{prop,1} &= 2518.0 - 2057.2 = \mathbf{460.8} \text{ kg} \end{aligned} \tag{28}$$

Leg 2 — Parthenope → Psyche ($\Delta v = 2475.8$ m/s):

$$\begin{aligned} m_f &= 2057.2 \times \exp\left(\tfrac{-2475.8}{3100\times 9.81}\right) = 2057.2 \times e^{-0.0814} = 1896.3 \text{ kg} \\ m_{prop,2} &= 2057.2 - 1896.3 = \mathbf{160.9} \text{ kg} \end{aligned} \tag{29}$$

Leg 3 — Psyche → Themis ($\Delta v = 4447.2$ m/s):

$$\begin{aligned} m_f &= 1896.3 \times \exp\left(\tfrac{-4447.2}{3100\times 9.81}\right) = 1896.3 \times e^{-0.1462} = 1638.3 \text{ kg} \\ m_{prop,3} &= 1896.3 - 1638.3 = \mathbf{258.0} \text{ kg} \end{aligned} \tag{30}$$

Total xenon propellant:

$$m_{Xe,\text{total}} = 460.8 + 160.9 + 258.0 = \mathbf{879.7} \text{ kg} \tag{31}$$

Mass Flow Rate

The instantaneous thruster mass flow rate while firing:

$$\dot{m} = \frac{F}{I_{sp} \cdot g_0} \tag{32}$$

Per-leg values using mean thrust from the verified Sims–Flanagan throttle profiles:

Leg	F_{mean} (mN)	I_{sp} (s)	$\dot{m}$ (kg/s)
Earth → Parthenope	119.21	3100	3.92×10^{-6}
Parthenope → Psyche	22.49	3100	7.40×10^{-7}
Psyche → Themis	54.26	3100	1.78×10^{-6}

Table 36: Ion thruster mass flow rates per leg. F_{mean} values are trajectory optimizer outputs; the hardware cap is $F_{max} = 0.3$ N.

3.5.5 Ion Thruster Power and Efficiency (Sophia and Citli)

Jet Power:

The useful kinetic power delivered to the exhaust beam is:

$$P_{jet} = \frac{F^2}{2\dot{m}} = \frac{F \cdot I_{sp} \cdot g_0}{2} \tag{33}$$

Since $\dot{m} = F/(I_{sp} \cdot g_0)$, Equation 33 simplifies to a function of thrust alone, given fixed $I_{sp} = 3100$ s and $g_0 = 9.81$ m/s^2. Per-leg jet power is computed using the mean thrust from the verified Sims–Flanagan throttle profiles:

Leg 1 — Earth → Parthenope ($F_{mean} = 119.21$ mN, $\dot{m} = 3.92 \times 10^{-6}$ kg/s):

$$P_{jet,1} = \frac{(0.11921)^2}{2 \times 3.92 \times 10^{-6}} = \mathbf{1813}\ \text{W} \tag{34}$$

Leg 2 — Parthenope → Psyche ($F_{mean} = 22.49$ mN, $\dot{m} = 7.40 \times 10^{-7}$ kg/s):

$$P_{jet,2} = \frac{(0.02249)^2}{2 \times 7.40 \times 10^{-7}} = \mathbf{342}\ \text{W} \tag{35}$$

Leg 3 — Psyche → Themis ($F_{mean} = 54.26$ mN, $\dot{m} = 1.78 \times 10^{-6}$ kg/s):

$$P_{jet,3} = \frac{(0.05426)^2}{2 \times 1.78 \times 10^{-6}} = \mathbf{825}\ \text{W} \tag{36}$$

Thruster Efficiency:

Thruster efficiency relates jet power to electrical input power at the thruster:

$$\eta_{\text{thruster}} = \frac{P_{jet}}{P_{\text{input,thruster}}} \quad \Longrightarrow \quad P_{\text{input,thruster}} = \frac{P_{jet}}{\eta_{\text{thruster}}} \tag{37}$$

η_{thruster} is throttle-level dependent and must be read from the published NEXT-C throttle table (Shastry et al., AIAA 2020-3604) at the operating throttle fraction $|u|$ for each leg. The per-leg thruster input power is therefore:

$$P_{\text{input,thruster,1}} = \frac{1813}{\eta_{\text{thruster}}(|u| = 0.397)} \tag{38}$$

$$P_{\text{input,thruster,2}} = \frac{342}{\eta_{\text{thruster}}(|u| = 0.075)} \tag{39}$$

$$P_{\text{input,thruster,3}} = \frac{825}{\eta_{\text{thruster}}(|u| = 0.181)} \tag{40}$$

where $|u|_{mean} = F_{mean}/F_{max}$ and $F_{max} = 0.3$ N.

PPU Efficiency:

The Power Processing Unit (PPU) converts unregulated spacecraft bus power (80–160 V DC) to the voltages required by the thruster (beam supply, discharge, neutralizer cathode). PPU efficiency η_{PPU} is likewise throttle-level dependent per the NEXT-C throttle table:

$$P_{\text{input,PPU}} = \frac{P_{\text{input,thruster}}}{\eta_{PPU}} = \frac{P_{jet}}{\eta_{\text{thruster}} \cdot \eta_{PPU}} \tag{41}$$

Total System Efficiency:

$$\eta_{\text{total}} = \eta_{\text{thruster}} \times \eta_{PPU} = \frac{P_{jet}}{P_{\text{input,PPU}}} \tag{42}$$

The per-leg jet power, throttle fractions, and the expressions for input power are summarized in Table 37.

Leg	$\boldsymbol{F_{mean}}$ **(mN)**	$\boldsymbol{\lvert u\rvert_{mean}}$	$\boldsymbol{P_{jet}}$ **(W)**	$\boldsymbol{P_{\text{in,thr}}}$ **(W)**	$\boldsymbol{P_{\text{in,PPU}}}$ **(W)**
Earth → Parthenope	119.21	0.397	1813	$1813\,/\,\eta_{\text{thr}}(0.397)$	$1813\,/\,\eta_{\text{total}}(0.397)$
Parthenope → Psyche	22.49	0.075	342	$342\,/\,\eta_{\text{thr}}(0.075)$	$342\,/\,\eta_{\text{total}}(0.075)$
Psyche → Themis	54.26	0.181	825	$825\,/\,\eta_{\text{thr}}(0.181)$	$825\,/\,\eta_{\text{total}}(0.181)$

Table 37: Per-leg power budget. P_{jet} is fully determined from verified trajectory data. $P_{\text{in,thr}}$ and $P_{\text{in,PPU}}$ require η_{thruster} and η_{PPU} from the NEXT-C throttle table (Shastry et al., AIAA 2020-3604) at the listed $|u|_{mean}$ values.

PPU Throttle Relationship:

The PPU tunes thrust by adjusting beam voltage V_b and beam current I_b:

$$P_{\text{input}} = V_b \cdot I_b, \qquad \eta_{\text{thruster}} = \frac{F^2}{2\,\dot{m}\,V_b\,I_b} \tag{43}$$

The Sims–Flanagan throttle fraction $|u|$ relates to hardware thrust as:

$$F = |u| \times F_{max}, \qquad F_{max} = 0.3 \text{ N} \tag{44}$$

$|u|$ is a trajectory optimizer output determined by mission geometry and time of flight, not a fixed hardware setting.

3.5.6 Stage Separation and Jettison Details (Citli and Sophia)

The primary staging event for the spacecraft is the separation from the Falcon Heavy. Following the launch vehicle's staging sequence, the upper stage will deliver the spacecraft to the required Earth-escape trajectory. Spacecraft separation is achieved through a pyrotechnic release mechanism. Following separation, the launch vehicle's role is complete and the spacecraft relies entirely on its onboard solar electric propulsion and reaction control systems.

Falcon Heavy itself has the following staging events:

1. Side booster separation (from center core)
2. Center core separation (from upper stage)
3. Payload fairing separation (technically not a stage but a jettison event)
4. Payload separation from upper stage (spacecraft's interface)

1) Side Booster Separation

The side boosters jettison shortly after max-q, once the vehicle reaches approximately Mach 6–7 at 50–70 km altitude. They throttle down before separation to reduce structural loads, then pneumatic pushers physically detach them from the central core, which continues burning until MECO.

2) Central Core Separation and Hot Staging

Falcon Heavy uses cold staging for central core separation: the center core engines shut down (MECO), there is a brief coast, and then the upper stage Merlin Vacuum ignites. This differs from hot staging, where the upper stage fires during the throttle-down of the lower stage to eliminate the unpowered coast and reduce gravity losses. Therefore, this would reduce velocity losses and the amount of propellant needed to account for these losses. Thus, cold staging is slightly less efficient than the hot staging used in Super Heavy used for SpaceX's Starship vehicle. However, the architecture in Falcon Heavy currently only supports cold staging.

3) Fairing Separation

During ascent, the nose cone (fairing) protects the payload from immense aerodynamic pressure, thermal loads, and acoustic vibrations as it travels through the atmosphere. Once the atmosphere is thin enough, these forces are negligible and the fairing becomes dead mass and should be shed as quickly as possible to improve mass performance.

Once the vehicle reaches approximately 110–120 km altitude and dynamic pressure drops to negligible levels, pneumatic struts split the fairing into two halves, which follow a ballistic

trajectory away from the vehicle. SpaceX recovers each half using steerable parafoils, as each fairing costs approximately $3 million and is reusable.

4) Payload Separation

The payload is now ready to be fully deployed. The upper stage Merlin Vacuum completes the Trans-Asteroid Injection burn, delivering the spacecraft to $v_\infty = 7.0$ km/s. A pyrotechnic Marman clamp band releases, and separation springs push the spacecraft away at 0.3–0.5 m/s. The spacecraft then autonomously deploys its solar arrays, acquires the Sun using RCS thrusters, and establishes initial contact with the Deep Space Network before the first NEXT-C engine ignition is commanded.

Additionally, a trade-study was conducted to assess the viability of jettisoning depleted tanks mid-mission.

By the time the spacecraft arrives at Parthenope, the NEXT-C ion thruster will have completed its largest continuous burn, consuming 460.8 of the 879.7 kg total xenon budget. This consumption equates to completely emptying two of the five 82 L General Dynamics COPVs (each loaded with approximately 155 kg of xenon). The theoretical jettison mass would then consist of the two empty tanks and one structural tank adapter:

$$m_{\text{jettison}} = 3(11.1 \text{ kg}) + 1(22.0 \text{ kg}) = 55.3 \text{ kg}$$

While jettisoning 55.3 kg reduces the spacecraft's dry mass, the mass penalty of the required separation hardware voids these benefits. Jettisoning the tanks in deep space would require the integration of pyrotechnic separation nuts and bolts, helical compression springs, cable cutters, and explosive isolation valves on the high pressure xenon feed lines. The additional hardware would also introduce critical new risks such as explosive misfires or propellant feed line leaks that could jeopardize the mission.

Trajectory analysis confirmed that the mission is not thrust-limited, during the final two transit legs to Psyche and Themis, the engine only needs to output between 22.5 mN and 54.3 mN. This falls under the thruster's 300mN maximum limit. Because the engine is within the operational margin, the mass reduction provides no necessary trajectory advantage.

Given the minor mass savings and severe risks introduced, the mid-mission concept is rejected, and the spacecraft will retain all five Xenon tanks for the duration of the mission.

3.5.7 Attitude Control System (Mahak Mathur)

The purpose of the Attitude Control System (ACS) is to measure and regulate the spacecraft's orientation, pointing direction, and stability. Therefore, the ACS will utilize a reaction control system along with Proportional-Integral-Derivative (PID) controller logic to maintain the spacecraft's position and properly accomplish the mission.

System	Heritage	Type	Thrust (N)	Isp (s)	Minimum Pulse Bit (Ns)	Mass (kg)	Trade-Offs
Aerojet MR-103G /J	Dawn	Hydrazine	0.19-1.13	202-224	0.01333	0.33	High reliability and precision; moderate efficiency Limited fuel supply; fuel requires tanks = greater system mass; potentially low thrust = slow slew rates
Moog 58E151 Thrusters	Psyche	Cold Gas	0.5	70	N/A	0.12	Cold gas prevents contamination of the instrumentation and simple, immediate, high-torque attitude control Lower specific impulse = more propellant mass (less efficient), warmer space conditions may decrease reliability
Aerojet MR-111G	New Horizons (Pluto), etc.	Hydrazine	1.8-4.9	219-229	0.076 Ns	0.37	High reliability; greater thrust for faster spacecraft slews; greater fuel efficiency Lower precision (greater MPB)
Aerojet MR-106L	MAVEN, etc.	Hydrazine	4-10	228-235	0.015	0.59	High reliability and precision; high efficiency Heavier than other options; thrust is potentially too high

Figure 53: Trade Study of Reaction Control Systems

Reaction Control System (Mahak Mathur)

A trade study was used to select an RCS configuration that will provide sufficient control for spacecraft operations while minimizing system mass and avoiding interference with on-board instrumentation. Thus, the main technical considerations were impulse precision and capability as well as system complexity.

A starting point for the investigation was the R-4D Hypergolic Rocket Engine used by the Apollo Lunar Module. The RCS using the R-4D engines consists of four such engines in quadruple form to properly balance the force around the spacecraft's center powered using hypergolic fuels (hydrazine and dinitrogen tetroxide). While the R-4D system provides a fairly efficient specific impulse of around 312 s, its high nominal thrust of 490 N is disproportionate to the more modest needs of the mission. Therefore, the investigation moved to consider other options, which are given in the table below. Note that the two main types of RCS under consideration are hydrazine monopropellant thrusters - which use pulsed bursts of hydrazine gas to maneuver the spacecraft and maintain stability - and cold gas thrusters - which expel cold gas (like nitrogen) to provide thrust for maneuvering and orientation.

Based on the above trade study, it quickly became clear that the cold gas solution would not be feasible for this mission, primarily due its significantly lower fuel efficiency. Comparing the specific impulse of the Moog thrusters with the Aerojet ones, it is clear that the cold gas system would require over three times the propellant mass of the hydrazine system for the same maneuvers, which is too high of a mass penalty for the relatively low gains.

Therefore, the spacecraft will use hydrazine monopropellant thrusters for the Reaction Control System. Of the options considered, the Aerojet MR-111G provides the best mix of thrust, fuel efficiency, mass, and minimum pulse bit, meaning the spacecraft will be capable of performing both the coarse trajectory correction maneuvers while travelling and the fine attitude control maneuvers about the asteroids.

To properly perform maneuvers, a minimum of six thrusters are needed to generate pure rotation about the x, y, and z-axes - two thrusters per axis. However, to control for unexpected faults during the mission, it is preferable to have at least twelve thrusters on board the spacecraft for redundancy. Therefore, the spacecraft will carry twelve hydrazine monopropellant thrusters for steering and attitude control.

While these thrusters will provide additional stability and control, reaction wheels are also critical to allow the spacecraft to rotate and stabilize without using propellant. Therefore, the RCS will include four Honeywell HR16-100 reaction wheels arranged in a pyramid formation. These have been selected due to their effectiveness in past missions and their suitability for the current one.

which will be divided into four clusters of three orthogonal thrusters each. Based on the results from the trade study, each thruster will be set to provide a fixed nominal thrust of up to 4.5 N to accommodate thruster tolerances. The thruster placements are corrected against the spacecraft center of mass, defined as

$$\mathbf{r}_{\text{CoM_offset}} = \begin{bmatrix} 0.00 \\ 0.00 \\ 0.95 \end{bmatrix} \text{ m} \tag{45}$$

Subtracting this offset from the structural coordinates yields the final operational position vectors $\mathbf{r}_i$ used in the thrust allocation matrix:

$$\mathbf{r}_1 = \begin{bmatrix} 1.2 \\ 1.1 \\ 1.2 \end{bmatrix} - \begin{bmatrix} 0.00 \\ 0.00 \\ 0.95 \end{bmatrix} = \begin{bmatrix} 1.20 \\ 1.10 \\ 0.25 \end{bmatrix} \text{ m} \tag{46}$$

$$\mathbf{r}_2 = \begin{bmatrix} -1.2 \\ 1.1 \\ -1.2 \end{bmatrix} - \begin{bmatrix} 0.00 \\ 0.00 \\ 0.95 \end{bmatrix} = \begin{bmatrix} -1.20 \\ 1.10 \\ -2.15 \end{bmatrix} \text{ m} \tag{47}$$

$$\mathbf{r}_3 = \begin{bmatrix} -1.2 \\ -1.1 \\ 1.2 \end{bmatrix} - \begin{bmatrix} 0.00 \\ 0.00 \\ 0.95 \end{bmatrix} = \begin{bmatrix} -1.20 \\ -1.10 \\ 0.25 \end{bmatrix} \text{ m} \tag{48}$$

$$\mathbf{r}_4 = \begin{bmatrix} 1.2 \\ -1.1 \\ -1.2 \end{bmatrix} - \begin{bmatrix} 0.00 \\ 0.00 \\ 0.95 \end{bmatrix} = \begin{bmatrix} 1.20 \\ -1.10 \\ -2.15 \end{bmatrix} \text{ m} \tag{49}$$

Each cluster contains three orthogonal nozzle direction vectors $(\mathbf{t}_{i,X}, \mathbf{t}_{i,Y}, \mathbf{t}_{i,Z})$ creating the

following mapping:

$$\begin{aligned}
\text{Cluster 1:} \quad & \mathbf{t}_{1,X} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{t}_{1,Y} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{t}_{1,Z} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
\text{Cluster 2:} \quad & \mathbf{t}_{2,X} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{t}_{2,Y} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{t}_{2,Z} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \\
\text{Cluster 3:} \quad & \mathbf{t}_{3,X} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{t}_{3,Y} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{t}_{3,Z} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
\text{Cluster 4:} \quad & \mathbf{t}_{4,X} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{t}_{4,Y} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \quad \mathbf{t}_{4,Z} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}
\end{aligned} \tag{50}$$

Momentum Desaturation Control Logic (Mahak Mathur)

Momentum desaturation is the process by which excess angular momentum built up in the reaction wheels is unloaded using the RCS while maintaining spacecraft attitude control. The RCS interfaces with Avionics to use a threshold-driven dumping routine as follows:

1. The trigger threshold is activated when the absolute stored momentum on any reaction wheel is greater than 40.0 Nms.

2. The routine fires thrusters continuously until the internal momentum of all wheels is below 10.0 Nms.

3. When active, the system transforms wheel momentum into the body frame ($h_{\text{body}} = W_{\text{wheels}} * h_{\text{wheels}}$) and commands a counter-torque pulse of $-0.4 \times \text{sign}(h_{\text{body}})$ Nm.

4. Individual nozzle throttling is calculated using a bounded linear least-squares solver (`lsqlin`) to enforce the fact that the thrust values must be strictly positive.

5. Once a desat cycle ends, a minimum pulse lockout timer is forced to avoid chattering and save fuel resources.

6. Total fuel mass utilization is derived from the mass flow rate equation:

$$\dot{m} = \frac{F}{I_{sp} \cdot g_0} \tag{51}$$

   Values of $I_{sp} = 220.0s$ and $g_0 = 9.80665 m/s^2$ were used in the calculation.

ACS Control Logic Flow (Mahak Mathur)

Here is the general logic flow for the ACS:

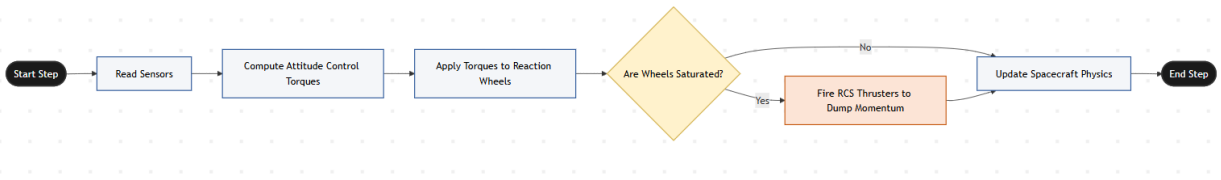


Figure 54: ACS Logic Flowchart

The primary attitude tracking loop uses a proportional-integral-derivative (PID) controller using quaternion feedback error. The current quaternion orientation $q_{current}$ is continuously evaluated against the target orientation $q_{orientation}$ to generate a quaternion error vector ϵ representing q_{error}. A commanded body velocity $\omega_{commanded}$ is generated from the error vector and constrained by a heritage limit of 0.1°/s to ensure structural safety. The final commanding torque profile τ_{cmd} computes independent axis control signals via

$$\tau_{cmd} = K_d * \omega_{error} + K_i * \int \epsilon dt \tag{52}$$

Controller gains are dynamically decoupled using the diagonal components of the vehicle's inertia tensor J based on a target natural frequency (ω_n):

1. Proportional Gain (K_p) scales the current error: $J_{diag} * \omega_n^2$

2. Derivative Gain (K_d) predicts future error by considering the rate of change of error: $2 * J_{diag} * \omega_n$

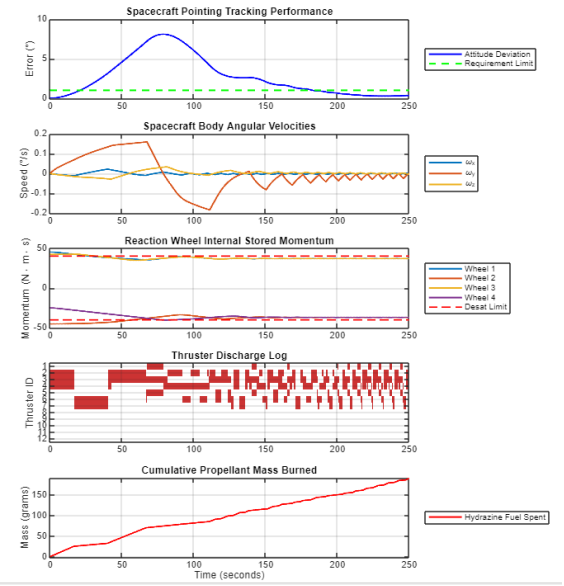
3. Integral Gain (K_i) eliminates steady-state error by correcting accumulated error: $0.01 * K_p$

A PID controller for attitude control that takes into account the mass properties, actuators, and reaction wheels was written in MATLAB to verify the feasibility of the RCS components. Various components such as the spacecraft tracking accuracy, body angular velocities, and mass of propellant consumed were computed and plotted to ensure that the ACS works. A sample series of plots is presented below

The hallmarks of success are that all the conditions are met - the spacecraft tracks within 1 degree, there is no slew rate higher than 0.1 degree/second, and the reaction wheels' momentum is consistently less than 40 Nms. As an example, this particular run "pre-loaded" the reaction wheels with greater than 40 Nms of momentum, triggering an immediate desaturation maneuver. Through these and other tests, the Attitude Control System is shown to be an effective means of regulating and stabilizing the spacecraft while both in cruise and in orbit about the asteroids.

Important Considerations and Future Work (Mahak Mathur)

While the ACS numerical simulation maintains excellent fidelity to the actual spacecraft, there are some considerations and parameters that can be further solidified as work on the



(a) ACS Verification Plots

```
=== MISSION VERIFICATION ===
Calculated Proportional Gains (Kp): [43.42, 7.66, 37.44]
Calculated Derivative Gains (Kd): [1736.62, 306.42, 1497.49]
Calculated Integral Gains (Ki): [0.4342, 0.0766, 0.3744]
Final Pointing Error: 0.3492°
Total Fuel Expended:  189.64 grams
STATUS: MISSION SUCCESS (Within Bounds)
```

(b) ACS Verification Stats

Figure 55: Overall ACS Verification Results

mission continues.

Rigidity of the Spacecraft For the purposes of the simulation, the spacecraft is assumed to be perfectly rigid, although in reality there are vibrational modes that may affect the controls system. However, these modes are relatively insignificant in terms of the additional requirements on the thrusters and reaction wheels, particularly given the mass margin included in the hydrazine fuel. Regardless, future work would involve implementing spacecraft non-rigidity into the simulation. Note, however, that the most up-to-date inertia matrix is used for the spacecraft, meaning the Euler equations are as accurate as possible.

RCS Component Degradation In this simulation, the RCS thrusters are assumed to continually and instantaneously output a maximum thrust of 4.5 N. While degradation is somewhat accounted for by setting said thrust to 4.5 N instead of the nominal maximum of 4.9 N, future work could ensure the long-term success of the mission by more thoroughly analyzing the performance of the Aerojet thrusters and Honeywell reaction wheels over time.

3.5.8 Hydrazine Desaturation Budget and Tank Selection (Jason Kamau)

The analysis here considers two sources of reaction-wheel momentum buildup. The first is the cruise thrust-vector steering history, represented by changes in θ_{e1}. The second is the asteroid proximity attitude history, represented by the rotation angle θ_{rot} and the corresponding rotation axis $\hat{\mathbf{n}}$ from the proximity attitude files. Environmental disturbances such as solar radiation pressure are not added as a separate momentum source in the final budget because the attitude optimizer already accounts for their influence in the attitude solution.

Reaction-Wheel Momentum and Desaturation Model (Jason Kamau)

The ACS desaturation logic unloads stored reaction-wheel momentum using the RCS. In the flight-control logic, desaturation begins when wheel momentum exceeds 40 Nms and continues until it is reduced below 10 Nms. For this propellant-sizing analysis, the same behavior is represented with a simplified impulse-equivalent MATLAB model. The model tracks wheel momentum buildup from each commanded slew and computes the hydrazine required to unload the wheel once the desaturation threshold is reached. A 50 Nms threshold was used for this sizing case because the selected wheels have a usable momentum range of approximately 12 Nms to 150 Nms. After each unload, the wheel momentum is reset to 5 Nms.

For each commanded slew, the wheel momentum contribution is computed from the spacecraft inertia tensor and commanded angular velocity:

$$\mathbf{H}_{\text{slew}} = \mathbf{I}_{\text{body}}\boldsymbol{\omega}. \tag{53}$$

The angular velocity is modeled as

$$\boldsymbol{\omega} = \dot{\theta}_{\text{slew}}\hat{\mathbf{a}}, \tag{54}$$

where $\dot{\theta}_{\text{slew}}$ is the slew rate and $\hat{\mathbf{a}}$ is the slew axis. The cruise case assumes body-Z steering from θ_{e1}, while the proximity cases use the rotation-axis vector $\hat{\mathbf{n}}$ from each asteroid attitude file. The assumed slew rate is

$$\dot{\theta}_{\text{slew}} = 0.1\,\text{deg/s} = 1.7453 \times 10^{-3}\,\text{rad/s}. \tag{55}$$

Each body-axis wheel is modeled independently. When a wheel reaches $H_{\text{sat}} = 50\,\text{Nms}$, it is unloaded to $H_{\text{target}} = 5\,\text{Nms}$. The dumped angular momentum is

$$\Delta H_i = H_{\text{pre},i} - H_{\text{post},i}, \tag{56}$$

where i denotes the wheel axis. The hydrazine required for each desaturation is then estimated as

$$m_i = \frac{|\Delta H_i|}{I_{\text{sp}} g_0}, \tag{57}$$

with $I_{\text{sp}} = 220\,\text{s}$ and $g_0 = 9.806\,65\,\text{m/s}^2$.

This impulse-equivalent model is appropriate for preliminary tank sizing. It does not yet include detailed nozzle allocation, thruster moment arms, pulse timing, minimum impulse bit, or valve transients, which should be included in a later closed-loop ACS/RCS simulation.

Cruise Slew Desaturation (Jason Kamau)

The cruise calculation uses the sequential θ_{e1} slew history because it is the dominant thrust-vector steering term. The θ_{e2} history was evaluated but neglected because its incremental changes were much smaller and had little effect on the total ACS hydrazine budget. This introduces a small non-conservative error, but cruise desaturation is only a minor part of

the total propellant demand. A numerical deadband of 0.001 deg is applied so that very small attitude changes are not counted as physical maneuvers. The cruise file contains 1019 trajectory timesteps over a mission duration of 16.81 yr. The change in steering angle at each timestep is $\Delta\theta_{e1,k} = \theta_{e1,k+1} - \theta_{e1,k}$. The corresponding slew duration is

$$t_{\text{slew},k} = \frac{|\Delta\theta_{e1,k}|}{\dot{\theta}_{\text{slew}}}.$$

The cruise-only desaturation model produced 165 counted slew events and 82 wheel desaturation events. The resulting hydrazine consumption was $m_{\text{cruise}} = 1.9735\,\text{kg}$.

Table 38: Cruise θ_{e1} reaction wheel desaturation summary.

Quantity	**Value**
Trajectory timesteps	1019
Valid timesteps	850
Invalid timesteps	169
Mission duration	16.81 yr
Trajectory timestep	520 483.40 s
Deadband threshold	0.001 deg
Assumed slew rate	0.100 deg/s
Counted θ_{e1} slew events	165
Reaction wheel saturation limit	50 Nms
Post-desaturation target	5 Nms
Desaturation events	82
Peak wheel momentum	56.985 Nms
Cruise hydrazine used	1.9735 kg

Asteroid Proximity Desaturation (Jason Kamau)

The asteroid proximity model was applied to eight proximity cases: two Parthenope cases, three Psyche cases, and three Themis cases. Each case file contains a high-rate attitude-rotation history. Rather than using evenly spaced downsampling, the model keeps the largest 1019 absolute θ_{rot} changes from each case and sorts those selected intervals back into chronological order before propagating reaction-wheel momentum. This preserves the attitude changes most likely to drive wheel saturation while avoiding the unrealistic assumption that every high-rate attitude-output sample is an independent full-body slew.

A maneuver is counted only if

$$|\theta_{\text{rot}}| > \theta_{\text{min}}, \tag{58}$$

where the selected threshold is $\theta_{\text{min}} = 5\,\text{deg}$.

The 5 deg threshold is a modeling assumption used to separate major retargeting maneuvers from small pointing corrections. Rotations below this value are treated as minor tracking updates and are not counted as separate desaturation-driving slews. Increasing the threshold would reduce the hydrazine estimate by filtering out more attitude changes, while decreasing it would increase the number of counted maneuvers and wheel desaturation events. The 5 deg

value was selected as a middle-ground assumption that preserves the dominant proximity retargeting events without over-counting small attitude updates.

Table 39: Proximity-orbit reaction wheel desaturation and hydrazine budget by asteroid case using the largest 1019 attitude-rotation intervals and a 5 deg maneuver threshold.

Case	Altitude [km]	Total Desats	X Desats	Y Desats	Z Desats	Peak X [Nms]	Peak Y [Nms]	Peak Z [Nms]	X Hyd. [kg]	Y Hyd. [kg]	Z Hyd. [kg]	Case Hyd. [kg]	Asteroid Total [kg]
Parthenope A	309	336	274	54	8	76.945	54.451	69.975	6.952	1.174	0.202	8.327	17.708
Parthenope B	78	395	336	41	18	78.020	53.709	68.112	8.071	0.881	0.428	9.381	
Psyche A	300	287	204	63	20	71.066	53.595	72.072	4.788	1.368	0.462	6.618	
Psyche B	175	217	100	24	93	70.170	53.596	74.888	2.442	0.518	2.348	5.309	18.621
Psyche C	75	265	201	11	53	78.546	53.404	72.382	5.173	0.240	1.282	6.695	
Themis A	170	153	135	17	1	72.398	52.854	74.994	3.240	0.365	0.032	3.637	
Themis B	75	301	197	50	54	73.328	54.266	70.204	4.835	1.082	1.271	7.188	18.918
Themis C	25	327	212	9	106	76.619	53.356	75.553	5.310	0.194	2.589	8.093	
Total	–	**2281**	**1659**	**269**	**353**	–	–	–	**40.811**	**5.823**	**8.614**	**55.248**	**55.248**

The asteroid-operation hydrazine mass is therefore 55.248 kg.

The asteroid cases dominate the ACS propellant budget because proximity operations require repeated surface-relative retargeting at multiple altitudes. Lower-altitude cases tend to produce more aggressive pointing histories, since apparent surface motion increases as orbital altitude decreases.

ACS Hydrazine Budget (Jason Kamau)

The unmarginized ACS hydrazine budget is summarized in the table below:

Table 40: Unmarginized ACS hydrazine budget contributions from cruise and asteroid operations.

Mission Phase	Contribution	Hydrazine [kg]
Cruise	θ_{e1} thrust-vector slew desaturation only	1.974
Asteroid Operations	Parthenope proximity operations	17.708
Asteroid Operations	Psyche proximity operations	18.621
Asteroid Operations	Themis proximity operations	18.918
Asteroid Operations Total	Proximity operations subtotal	**55.248**
Total ACS Hydrazine	Cruise + asteroid operations	**57.222**

$$\text{Asteroid operations contribution} \quad \frac{m_{\text{asteroid}}}{m_{\text{ACS}}} = \frac{55.248}{57.222} = 0.966 \approx 96.6\,\%. \tag{59}$$

$$\text{Cruise slew contribution} \quad \frac{m_{\text{cruise}}}{m_{\text{ACS}}} = \frac{1.974}{57.222} = 0.034 \approx 3.4\,\%. \tag{60}$$

Thus, proximity operations are the primary driver for hydrazine tank sizing. This result is consistent with the mission profile: cruise slews are sparse and largely associated with thrust-vector steering, while asteroid operations require repeated retargeting and wheel desaturation across multiple proximity altitudes.

Assumptions, System-Level Implications, and Future Work (Jason Kamau)

The hydrazine budget was developed as a preliminary feed-system sizing estimate using the available cruise and proximity attitude histories. These attitude rotations were converted into reaction-wheel momentum buildup and desaturation events, rather than simulated with a full closed-loop ACS/RCS controller. As a result, the model captures the main propellant-driving behavior but relies on several simplifying assumptions.

A constant slew rate of 0.1 deg/s was assumed for all counted maneuvers. Since wheel momentum scales as Equation 53, higher slew rates would increase momentum buildup and likely increase hydrazine use, while slower slews would reduce propellant demand but lengthen retargeting time.

For asteroid proximity operations, the largest 1019 attitude rotations from each case were retained. This preserves the rotations most likely to drive wheel saturation, making it more conservative than evenly spaced downsampling. At the same time, it avoids treating every high-rate attitude sample as a separate full-body slew. Rotations below 5 deg were treated as small pointing corrections rather than discrete retargeting maneuvers.

The model also uses three independent body-axis wheel states. The actual spacecraft is planned to use four reaction wheels, with one included for redundancy. A four-wheel layout could improve momentum distribution and provide single-wheel-failure tolerance, but the final wheel geometry and control allocation matrix were not included at this stage.

Finally, the hydrazine mass was computed using an impulse-equivalent momentum dump. This does not yet include detailed RCS moment arms, thruster pulse-width limits, minimum impulse bit, valve transients, or exact thruster placement. These effects could either increase the required hydrazine through losses or reduce it if the final thruster geometry provides stronger torque authority.

Future work should couple this desaturation model to a full proximity-operations timeline and closed-loop ACS/RCS simulation. This would allow observation duration, attitude deadband, slew-rate limits, instrument sequencing, solar-array pointing, Earth-communication geometry, retargeting frequency, wheel allocation, and nozzle-level thrust allocation to be included before final feed-system sizing.

Hydrazine Tank Selection Basis (Jason Kamau)

From our final unmarginalized hydrazine mass budget of 57.222 kg, we can apply a 10 % margin (62.944 kg.) to account for uncertainty in the final proximity operations timeline, attitude-control implementation (See 3.5.8), thruster duty cycle, pulse-mode losses, and final RCS thruster placement. This margin-added value is used as the minimum required hydrazine load for tank sizing.

To provide adequate storage volume and maintain heritage with a demonstrated deep-space hydrazine system, the selected design uses two Dawn-heritage Orbital ATK (now Northrop Grumman) 80274-1 hydrazine tanks. Each 80274-1 tank has a listed total volume of 60 L, a qualified propellant volume of 45.1 L, and a dry mass of 6.7 kg. The same tank family is

listed with flight heritage including Dawn, OCO, OCO-2, MSL, TESS, and other spacecraft [65], [66].

Using two tanks gives a total qualified hydrazine capacity of $V_{N_2H_4,\text{qual}} = 2(45.1) = 90.2\,\text{L}$.

Using a representative hydrazine density of approximately $\rho_{N_2H_4} \approx 1.02\,\text{kg/L}$, , the corresponding maximum hydrazine mass capacity is approximately $m_{N_2H_4,\text{max}} = \rho_{N_2H_4} V_{N_2H_4,\text{qual}} = 1.02(90.2) = 92.0\,\text{kg}$.

Thus, the two-tank configuration can accommodate up to approximately 92 kg of hydrazine if additional contingency is required. Relative to the margin-added ACS estimate, this maximum tank capacity provides 29.056 kg of additional capacity, corresponding to 46.2 % reserve above the margin-added estimate. Although the tanks can physically support this volume, the nominal hydrazine allocation is selected as 80 kg. This nominal load is above the margin-added ACS requirement by 17.056 kg, which corresponds to 27.1 % reserve above the margin-added estimate. The 80 kg value is therefore adopted as the nominal ACS hydrazine fill, while the 92 kg tank capacity is retained as a maximum-growth case for final operational uncertainty.

The nitrogen pressurant is tracked separately from the hydrazine consumption budget (Ideal Gas Law). The current feed-system pressurant allocation is $m_{N_2} = 0.793\,\text{kg}$.

Therefore,

$m_{\text{loaded,nominal}} = 80.000 + 0.793 = 80.793\,\text{kg}.$ $m_{\text{loaded,max}} = 92.000 + 0.913 = 92.913\,\text{kg}.$

Table 41: Comparison of Dawn RCS hydrazine usage and current mission tank selection

Quantity	**Dawn Heritage**	**Current Mission**
Hydrazine tank type	ATK/Orbital ATK 80274-1	ATK/Orbital ATK 80274-1
Number of hydrazine tanks	1	2
Qualified propellant volume per tank	45.1 L	45.1 L
Total qualified propellant volume	45.1 L	90.2 L
Hydrazine used / required	45.3 kg depleted at EOM	62.944 kg with margin
Maximum hydrazine capacity	∼45.3 kg	∼92 kg
Primary sizing driver	Extended asteroid operations	Multi-asteroid proximity operations

This larger hydrazine capacity gives the RCS more margin than Dawn's single-tank configuration. The reserve covers nominal desaturation demand while leaving propellant for wheel unloads, pointing corrections, safe-mode recovery, trajectory cleanup, and proximity retargeting. This is important for a multi-asteroid mission, where attitude-control activity is expected to drive most of the hydrazine use.

Dawn is a useful comparison because its mission ended after depletion of its (45.3 kg) hydrazine supply. Its reaction wheel failures also shifted more attitude-control responsibility

to the RCS, making hydrazine conservation a major late-mission constraint. That heritage supports using a larger two-tank architecture for this mission [65].

3.5.9 Feed System (Jason Kamau)

The propulsion feed system is split into two main sections: the xenon ion propulsion system (IPS) feed network and the hydrazine attitude control system (ACS) feed network. Each system uses independent routing and plumbing because the IPS and ACS serve different mission functions. The IPS feed system supplies xenon to the NEXT-C ion thrusters for primary low-thrust trajectory control, while the ACS feed system supplies hydrazine to the RCS thrusters for attitude control, reaction-wheel desaturation, and contingency stabilization. The integrated feed architecture is shown in Fig. 56.

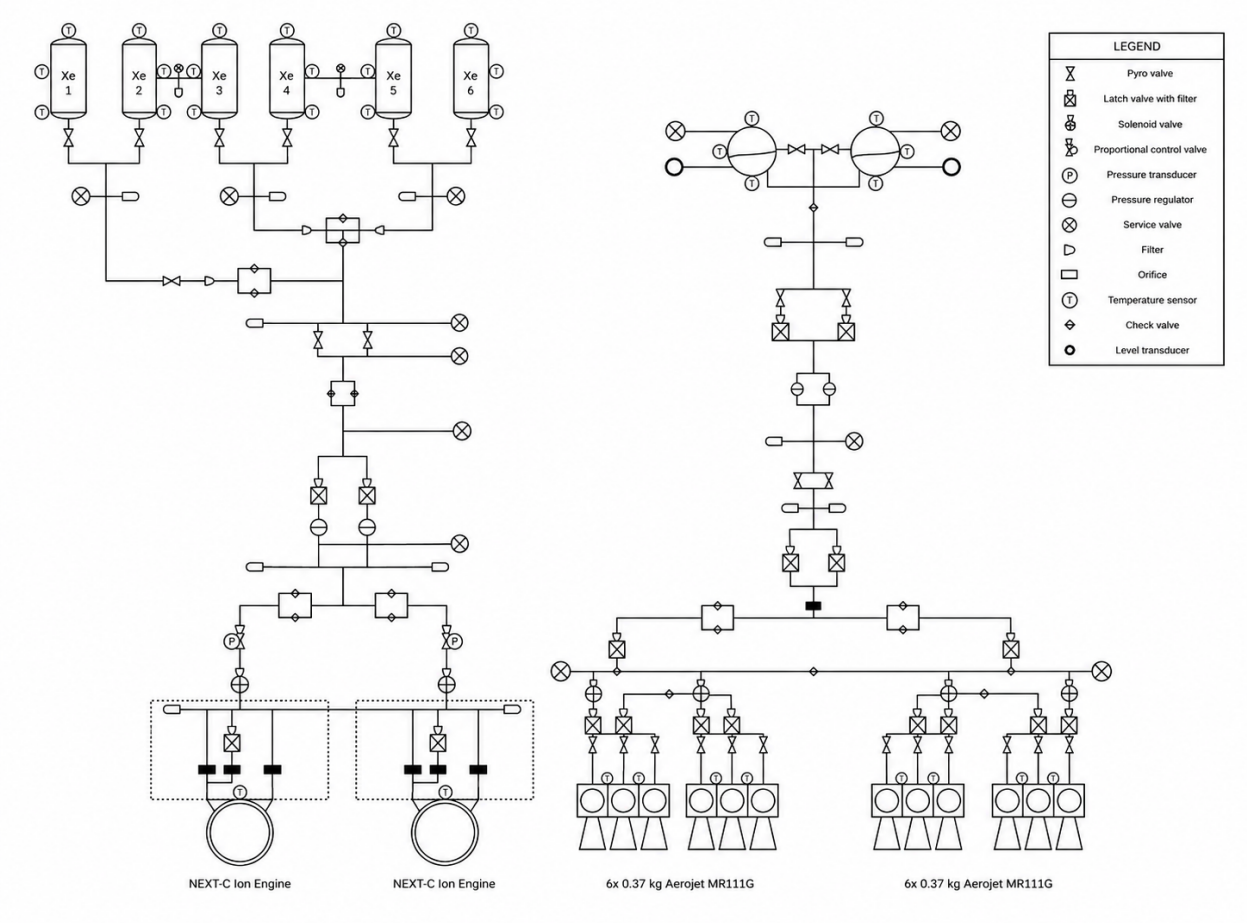
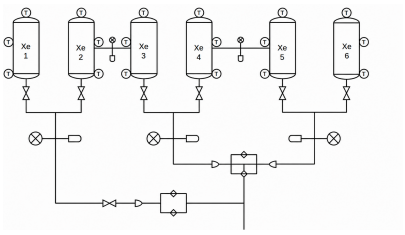


Figure 56: Integrated propulsion feed system architecture, including the xenon IPS feed network and the hydrazine ACS feed system.

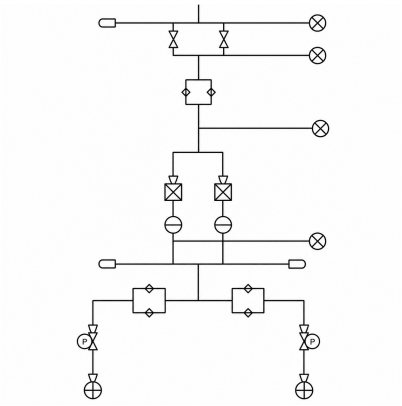
Xenon Electric Propulsion Feed System (Jason Kamau)

The IPS feed system supplies xenon to the NEXT-C ion thrusters through a regulated, multi-branch xenon distribution network. The selected architecture uses six xenon tanks

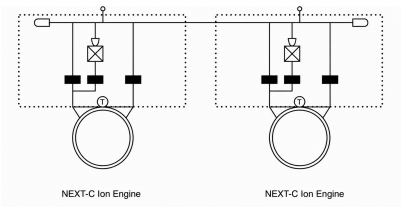
feeding a redundant manifold through isolation valves, filters, pressure regulation, flow-control hardware, and pressure instrumentation before propellant reaches the thruster feed assemblies. The IPS layout is shown in Fig. 57.



(a) IPS top high-pressure xenon storage section.



(b) IPS pressure management assembly (PMA).



(c) IPS xenon flow control (XFC) and thruster branch.

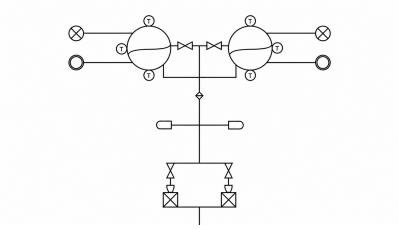
Figure 57: Ion propulsion system (IPS) feed architecture divided into the three major functional sections: the high-pressure xenon storage section, the pressure management assembly (PMA), and the xenon flow control (XFC) / thruster interface section.

The IPS feed architecture follows the same system-level approach as Psyche: distributed xenon storage, regulated flow control, throttled electric propulsion operation, and redundancy at the thruster and power-processing level [9]. Psyche used the Maxar SPT-140 Hall thruster system, while this mission uses NEXT-C gridded ion thrusters. Although the thruster technologies are different, both systems require stable xenon delivery over long-duration deep-space operation.

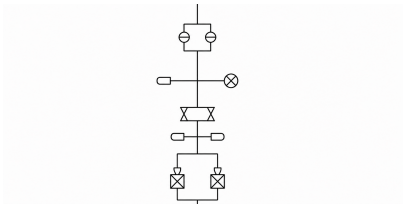
The main design difference is that the present system adopts gridded ion propulsion, similar in purpose to Dawn's ion-propulsion architecture, to improve propellant efficiency for a high-Δv asteroid tour. Because ion thruster performance depends on controlled xenon flow to the discharge chamber and cathode assemblies, this customized feed system includes pressure regulation, distributed pressure sensing, filters, and a xenon flow-control unit to reduce transients and contamination before the propellant reaches the thruster branches [9].

ACS Feed System (Jason Kamau)

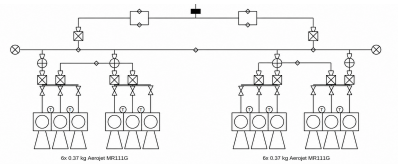
The ACS feed system supplies hydrazine to the attitude-control thrusters used for reaction-wheel desaturation, contingency control, and safe-mode recovery. The hydrazine architecture is shown in Fig. 58. It uses the two Dawn-heritage OA 80274-1 tanks feeding redundant ACS manifolds and 12 Aerojet Rocketdyne MR-111G thrusters.



(a) ACS top pressurization and tank section.



(b) ACS pressure management assembly (PMA).



(c) ACS dual-manifold thruster distribution system.

Figure 58: Attitude control system (ACS) feed architecture divided into the three major functional sections: the upper pressurization and tank section, the pressure management assembly (PMA), and the dual-manifold hydrazine thruster distribution system.

The ACS side carries direct Dawn heritage through the selected tank family and pressure-fed monopropellant architecture. Dawn used a conventional blowdown hydrazine RCS with nitrogen pressurant and ended its mission when its hydrazine supply was depleted [65]. That operational history motivated a larger hydrazine capacity for this mission (See Section 3.5.8 on Hydrazine Tank Selection).

The ACS lower feed system section is separated into dual redundant branches (manifolds) so that individual thruster clusters or feed paths can be isolated if a valve, line, or thruster fault occurs. Pyro valves provide positive isolation for launch and long-duration dormant periods. Latch valves and solenoid valves provide operational isolation and command authority during mission operations. Filters protect the valves and the system from particulate contamination. Pressure transducers and temperature sensors provide health monitoring for leak detection, regulator performance assessment, and thermal control. Check valves prevent backflow and limit fault propagation between branches.

Thermal Instrumentation (Jason Kamau)

Temperature sensors are included throughout the xenon and hydrazine feed systems to monitor tanks, valves, regulators, and thruster interfaces. Type K thermocouples were selected for the preliminary design because they provide a wide operating range, common aerospace use, and sufficient accuracy for feed-system health monitoring. As summarized in Table 42, Type K offers the best overall balance compared with the other thermocouple options considered.

Table 42: Thermocouple trade for feed-system temperature sensing.

Type	Material	Temperature Range	Standard Limit	Selection Note
K	Chromel/Alumel	−280 °C to 1260 °C	±2.2 °C	Selected; broad range and common aerospace use
J	Iron/Constantan	0 °C to 750 °C	±2.2 °C	Rejected; iron corrosion concern below ambient
E	Chromel/Constantan	−200 °C to 900 °C	±1.7 °C	Rejected; less ideal in vacuum
T	Copper/Constantan	−270 °C to 350 °C	±1 °C	Rejected; limited high-temperature range

Redundancy and Fault Tolerance (Jason Kamau)

The feed system is designed around graceful degradation. On the IPS side, the xenon tanks are arranged in parallel with branch isolation so that a single tank or feed-path failure does not remove all xenon availability. Redundant regulators, distributed pressure sensing, check valves, and service valves improve fault detection and allow isolation of suspect branches. The thruster-side architecture supports sequential operation, so the spacecraft does not rely on simultaneous operation of multiple electric thrusters. This reduces thermal loading, simplifies power management, and allows the remaining thruster branches to serve as operational spares.

On the ACS side, two hydrazine tanks and dual thruster manifolds preserve attitude-control capability if one branch is isolated. This is important because the ACS is required not only for planned reaction-wheel desaturation, but also for safe-mode recovery and emergency stabilization. The ACS is a mission-preserving subsystem that maintains spacecraft control if reaction wheels, sensors, or electric propulsion pointing modes degrade. Its reliability is therefore critical to the success of the mission.

Mass Summary (Jason Kamau)

Table 43 summarizes the updated feed-system component masses. The IPS and ACS feed-hardware masses are taken from the current feed-system mass worksheet cross-checked with the master mass budget spreadsheet.

Table 43: Combined IPS and ACS feed-system mass estimate.

Component	**IPS Xenon Feed System**			**ACS Hydrazine Feed System**		
	Qty.	**Mass Each [kg]**	**Total [kg]**	**Qty.**	**Mass Each [kg]**	**Total [kg]**
Pyro valve	18	0.200	3.600	9	0.200	1.800
Latch valve	14	0.320	1.920	4	0.320	1.280
Solenoid valve	4	0.200	0.800	2	0.200	0.400
Proportional control valve	–	–	–	2	0.115	0.230
Pressure transducer	5	0.200	1.000	10	0.200	2.000
Pressure regulator	2	0.450	0.900	2	0.450	0.900
Service valve	5	0.209	1.045	9	0.209	1.881
Filter	18	0.180	1.080	11	0.180	1.980
Orifice	1	0.050	0.050	6	0.050	0.300
Temperature sensor	14	–	–	17	–	–
Check valve	10	0.200	2.000	10	0.200	2.000
Level transducer	2	–	–	–	–	–
Feed hardware subtotal	–	–	**12.395**	–	–	**12.771**
NEXT-C ion thruster	2	14.000	28.000	–	–	–
Xenon tank	6	11.100	66.600	–	–	–
MR-111G thruster	–	–	–	12	0.370	4.440
80274-1 hydrazine tank	–	–	–	2	6.700	13.400
Major hardware subtotal	–	–	**94.600**	–	–	**17.840**
Subsystem subtotal	–	–	**106.995**	–	–	**30.611**
			Total propulsion feed-system mass			**137.606**

3.6 Avionics (Alexis Herfurth)

The avionics suite is heavily derived from flight heritage, particularly the Psyche mission. This spacecraft incorporates hardware and interfaces from the Lanteris Space Systems (formerly Maxar Space Solutions) 1300 model satellite bus along with additional heritage electronics. To ensure robustness and single-fault tolerance, the design employs cold-spare block redundancy with cross-strapped interfaces between hardware, including all interfaces between the Lanteris-provided components and the other selected hardware. The primary avionics subsystems highlighted in this section are the Command & Data Handling (C&DH) subsystem and the Attitude Control Subsystem (ACS). Avionics supporting other subsystems, such as power and telecommunications, are discussed in detail within their respective sections. For the complete component list for C&DH and ACS, see Figure 60.

3.6.1 Command & Data Handling

The Command & Data Handling (C&DH) subsystem serves as the "brains" of the spacecraft. This subsystem is responsible for managing, processing, and storing scientific data, as well as executing both autonomous and ground-issued commands. At the core of the subsystem is a single board computer (SBC), which hosts the flight software. A trade study comparing the two leading SBC options, BAE Systems RAD750 6U CompactPCI RAD750 and BAE Systems RAD6000 6U CompactPCI, was conducted and is summarized in Figure 59. Both SBCs have significant flight heritage with the RAD750 flown on more than 150 NASA missions and the RAD6000 on over 200 missions. For asteroid orbiter missions similar to this one, the RAD750 has been successfully used on the Psyche, OSIRIS-REx, and Lucy missions, while the RAD6000 was used on Dawn. Both SBCs are radiation-hardened. The main difference occurs primarily in processing capability and transistor count. The RAD6000 operates at clock speeds ranging from 2.5 to 66 MHz and delivers approximately 35 MIPS (Million Instructions Per Second), whereas the RAD750 operates between 110 and 200 MHz and achieves up to 266 MIPS. The RAD6000 contains approximately 1.1 million transistors, whereas the RAD750 contains about 10.4 million. This substantial increase in transistor count (which provides more processing power) and clock speed gives the RAD750 immensely greater computational performance. Although the RAD750 requires slightly more power consumption than the RAD6000 (10 W versus 5 W), the improvement in processing capability and overall performance represents a worthwhile trade-off. Furthermore, the RAD750 offers enhanced fault recovery features, making it the preferred option for spacecraft fault protection and mission reliability.

Consideration was also given to real-time operating systems (RTOS) compatible with the RAD750. VxWorks, which was used on the OSIRIS-REx and Dawn missions, provides high reliability and real-time guarantees. It has flown on over 20 NASA missions, including the most recent Artemis II. The INTEGRITY RTOS, used on the Psyche mission, offers high reliability, robust security, and strong real-time performance; however, it does not have as extensive a flight heritage as VxWorks. Real-Time Executive for Multiprocessor Systems (RTEMS), used on missions such as DART and Hera, has a history of use in many European Space Agency missions, though it may not provide precise time updates required for some spacecraft operations. Therefore, VxWorks was selected.

SBC	Heritage Missions	Processor Speed (Clock Rate / MIPS)	# of Transistors	Power	Pros	Cons
RAD6000	Dawn (+200 NASA missions)	2.5 to 66 MHz / 35 MIPS	1.1 million	~5 W	- Radiation-hardened - Used in over 200 NASA missions	- Has not been used since 2015 - Largely replaced by RAD750 for NASA missions
RAD750	Psyche, OSIRIS-REx, Lucy (+150 NASA missions)	110 to 200 MHz / 266 MIPS	10.4 million	~10 W	- Radiation-hardened - Higher processor speed - More transistors, more computing power - Improved fault recovery - Used in over 150 NASA missions	- More power consumption

Figure 59: Trade Study of Single Board Computers

Besides the RAD750, the only additional hardware components within the C&DH subsystem is Lanteris's proprietary router and interface hardware, which are discussed further in the Interfaces section.

3.6.2 Attitude Control Subsystem

The Attitude Control Subsystem (ACS) includes the avionics responsible for determining and controlling the spacecraft's orientation. Attitude sensors on this spacecraft include Northrop Grumman LN-200HPS Internal Measurement Units (IMUs), Jena-Optronik ASTRO APS star trackers, and Redwire digital sun sensors. The subsystem also incorporates Honeywell HR16-100 reaction wheels for attitude control. All parts are radiation-hardened. Trade studies were performed on each component to determine the most suitable option for the mission.

Starting with the IMUs (also referred to as Internal Reference Units or IRUs), four leading candidates were evaluated: Kearfott Two-Axis Rate Assembly IRU, Honeywell Miniature IMU, Northrop Grumman LN-200HPS IMU, and InnaLabs IMU. The Kearfott Two-Axis Rate Assembly, which flew on the Dawn mission, demonstrated high accuracy and precision. However, it has more heritage in military applications (i.e. missiles) than in spacecrafts. The Honeywell Miniature IMU, used on Psyche, OSIRIS-REx, and Lucy, offered a compact, lightweight design, measuring 233.7 mm in diameter by 169.9 mm in height with a mass of 4.6 kg. It contains no moving parts and boasts extensive flight heritage with over 300 IMUs utilized in space missions. Despite these advantages, its micro-electro-mechanical systems (MEMS)-based architecture results in greater bias, temperature-dependent accuracy, and relatively high power consumption of 25 W. The Northrop Grumman LN-200HPS IMU, which flew on Hayabusa2, provided an even more compact and lightweight solution with dimensions of 88.9 mm in diameter by 85.1 mm in height and a mass of 0.748 kg. This fiber-optic gyroscope (FOG)-based system also operates without moving parts, while providing lower random walk and reduced accelerometer and gyroscope bias than the Honeywell model. In addition, it requires less power at 12 W and has extensive heritage across deep-space and Mars rover missions. Finally, the InnaLabs IMU, utilized for the Hera mission, presented

another small and lightweight option, but lacked substantial flight heritage. After assessing all four candidates, the Northrop Grumman LN-200HPS IMU was selected as the optimal choice.

Moving on to star trackers, the top contenders were Sodern HYDRA, Jena-Optronik ASTRO APS, and Leonardo (formerly SELEX-EX) AA-STR. Each option represented a strong candidate, as these three manufacturers collectively account for more than 75% of the global star tracker market. Furthermore, each star tracker has a comparable size and mass, measuring approximately 150 x 150 x 200 mm and weighing about 2 kg. The Sodern HYDRA star tracker, flown on missions such as Dawn, Lucy, and Hera, demonstrated high precision and accuracy with a proven 100% mission success rate across 60 spacecraft. It also offers a low power consumption at 9.3 W, but it is known to exhibit some sensitivity to noise. The Jena-Optronik ASTRO APS, selected specifically for the Psyche mission, has strong flight heritage across NASA, JAXA, and ESA programs, including the recent Artemis II mission. In addition to its proven reliability, it provides straightforward spacecraft integration, however it may exhibit some sensitivity to noise and requires slightly higher power consumption at 12 W. Lastly, the Leonardo AA-STR, flown on missions such as OSIRIS-REx and DART, while it offered many of the same integration advantages and similar power requirements as the Jena-Optronik ASTRO APS, it possessed less extensive flight heritage compared to the other candidates. After evaluating all three options, the Jena-Optronik ASTRO APS star tracker was selected due to its ease of integration and demonstrated heritage across multiple international space missions.

For the last attitude sensor, the coarse sun sensors, each option was a different model from Redwire (formerly Adcole), the leading sun sensor company. The three candidates considered were the Redwire $\pm 64^{\circ}$ Digital, Redwire Pyramid, and Redwire Analog. The Redwire Digital sensor, flown on missions such as Psyche, OSIRIS-REx, DART, and Hera, offers high precision and accuracy with built-in processing capabilities. It also has extensive flight heritage, having been used on more than 70 NASA missions. However, these advantages come at the cost of increased complexity, risking a slightly higher chance of failure. The Redwire Pyramid coarse sun sensor, chosen for the Lucy mission, utilizes a two-axis sensor configuration with four detectors on each axis to provide a large field-of-view. This design offers reliable sun acquisition capability, while maintaining relatively simple integration. In addition, the sensor has demonstrated heritage on more than 18 NASA missions. Despite these benefits, it has less overall flight heritage than the Redwire Digital model and introduces a potential single-point failure risk due to its detector arrangement. Finally, the Redwire Analog coarse sun sensor, flown on the Hayabusa2 mission, provides a low-cost and highly simplified solution. Its analog architecture minimizes system complexity. However, the sensor exhibits lower precision and accuracy than the digital alternatives and is more sensitive to noise. After conducting this trade study, the Redwire Digital coarse sun sensor was selected due to its superior precision, processing capability, and extensive flight heritage across numerous deep space missions.

For the reaction wheels, the primary candidates considered were the Collins (formerly Goodrich) Ithaco TW-16B32 and the Honeywell HR16-100. The Collins Ithaco TW-16B32 wheels, which flew on the Dawn mission, have demonstrated heritage on previous deep space mis-

Component	Subsys.	Qty	Asteroid Heritage Missions	Location on Spacecraft	Rad-Had	Size / Mass	Operational Temp.	Max Power
BAE Systems RAD750 6U CompactPCI (VxWorks RTOS + 128 GB NVM card)	C&DH	2	Psyche, OSIRIS-REx, Lucy	Internal	Yes	160x233 mm / 1.22 kg	-55° to 70°C	14 W
Northrop Grumman LN-200HPS IMU	ACS	2	Hayabusa2	Inertial	Yes	88.9x85.1 mm / 0.748 kg	-54 to 71°C	12 W
Jena-Optronik ASTRO APS Star Tracker	ACS	2	Psyche	External	Yes	154x154x237 mm / 2 kg	-30 to 60°C	12 W
Redwire Digital Sun Sensor (±64°)	ACS	8	Psyche, OSIRIS-REx, DART	External	Yes	81×81×20 mm / 0.25 kg	-30 to 55°C	0.4 W
Honeywell HR16-100 Reaction Wheel	ACS	4	Psyche, OSIRIS-REx, Lucy	Internal	Yes	178x418 mm/ 12 kg	-30 to 70°C	22 W

Figure 60: Avionics Components List for C&DH and ACS

sions. However, concerns were raised regarding reliability, as both the Dawn and Hayabusa1 missions experienced reaction wheel failures. Moreover, these wheels have seen less recent use on modern spacecraft missions. In contrast, the Honeywell HR16-100 wheels, selected for missions such as Psyche, OSIRIS-REx, and Lucy, provide a more compact and efficient solution with extensive recent flight heritage. The Honeywell HR16-100 also provides significantly greater torque capability and saturation capacity with a nominal torque limit of approximately 0.1 to 0.2 Nm, an extended torque limit of 0.4 Nm at 3000 RPM, a maximum wheel speed of 6000 RPM, and momentum storage of 100 Nms. The Collins wheels are limited to 0.032 Nm, 5100 RPM, and 16.5 Nms, resulting in less precise attitude control and slower slewing capability. The Honeywell HR16-100 wheels' proven reliability, stronger performance, and continued use on current deep-space missions made them the preferred choice for this spacecraft.

3.6.3 Data Storage

Over the duration of this mission, approximately 837 GB of compressed science data is expected to be collected and downlinked to ground control over a total asteroid stay time of 832 days. To determine the minimum onboard storage requirement, the maximum volume of data accumulated before downlinking was found. The data collected during each asteroid observation orbit was analyzed for three cases: before the first telecommunications window, after the first telecommunications window, and the total data collected over the entire orbit. A summary of these results is presented in Tables 44, 45, and 46. Additional details regarding telecommunications and payload operations during data collection are discussed in their respective sections.

Table 44: Data Collected Before First Telecommunication Window

Asteroid Orbit	**Data Rate**	**Data Collecting Duration**	**Data Collected**
Parthenope A	0.01743 MB/s	35.0 days	52.708 GB
Parthenope B	0.02703 MB/s	40.0 days	93.416 GB
Psyche A	0.01743 MB/s	30.0 days	45.179 GB
Psyche B	0.0217 MB/s	40.0 days	74.995 GB
Psyche C	0.02703 MB/s	20.0 days	46.708 GB
Themis A	0.01743 MB/s	55.0 days	82.827 GB
Themis B	0.02703 MB/s	10.0 days	23.354 GB
Themis C	0.0217 MB/s	35.0 days	65.621 GB

Table 45: Data Collected After First Telecommunication Window

Asteroid Orbit	**Data Collecting Duration**	**Data Collected**
Parthenope A	27.9 days	42.016 GB
Parthenope B	33.8 days	78.936 GB
Psyche A	0.6 days	0.90357 GB
Psyche B	5.1 days	9.5619 GB
Psyche C	18.1 days	42.271 GB
Themis A	0.0 days	0.0 GB
Themis B	16.9 days	39.468 GB
Themis C	18.0 days	33.748 GB

Table 46: Total Data Collected for Each Asteroid Orbit

Asteroid Orbit	**Total Data Collecting Duration**	**Total Data Collected**
Parthenope A	62.9 days	94.724 GB
Parthenope B	73.8 days	172.352 GB
Psyche A	30.6 days	46.083 GB
Psyche B	45.1 days	84.557 GB
Psyche C	38.1 days	88.979 GB
Themis A	55.0 days	82.827 GB
Themis B	26.9 days	62.822 GB
Themis C	53.0 days	99.369 GB

From these results, the maximum data collected occurs during the Parthenope B orbit prior to the first telecommunications window, reaching approximately 94 GB over the 40 day period. Thus, the spacecraft must provide at least 94 GB of onboard storage capacity, sufficient for 40 days of operations, to accommodate the maximum data volume between downlink opportunities. To incorporate additional redundancy into the data storage architecture, the assumption was made that all data accumulated during an observation orbit remains onboard until after completion of the orbit before downlink begins. Under this worst case scenario, which also occurs during the Parthenope B orbit, the spacecraft would accumulate approximately 172 GB of data over 74 days. Therefore, a minimum onboard storage capacity of 172 GB is required to accommodate the maximum expected data collected before downlinking.

To achieve this capability, the spacecraft will rely on the multispectral imager's dedicated 8 GB of storage and the VIR spectrometer's dedicated 6 GB, supplemented with 256 GB of NAND flash memory connected to each RAD750 SBC through non-volatile memory (NVM) cards. In total, this provides 270 GB of onboard storage, 156% more storage than required. This data storage architecture is similar to that of the Psyche mission, which has a comparable total asteroid stay time of 817 days. Psyche also uses its imager's (same one used in this mission) 8 GB of storage and augments it with 128 GB of NAND flash connected to each RAD750 through NVM cards, sufficient for 56 days of operations. Since this mission's data storage design has flight heritage in Psyche, this architecture adds confidence in its ability to support long duration deep space data collection under similar mission constraints.

3.6.4 Interfaces

As mentioned before, the avionics architecture is based on an enhanced and modified version of Lanteris's 1300 modular spacecraft chassis in order to reduce development time and cost. Consequently, several mission interfaces are proprietary to Lanteris. A complete block diagram of these interfaces and each subsystem's components is provided in Figure 61.

The RAD750 interfaces with two Lanteris-supplied units via two separate RS485 serial data lines utilizing Lanteris's proprietary M-Bus protocol: the router and the Attitude Control Electronics (ACE) interface. The ACE interface, in turn, provides command and telemetry connectivity to the ACS sensors. Internally, the Lanteris chassis uses its proprietary S-Bus protocol to interface with the thermal, propulsion, power, and ACS subsystems. Another RS485 serial data line (not using Lanteris's M-Bus protocol) is used to interconnect each of the three payload instruments. A 1553 bus provides the interface to the star trackers and telecommunications components.

3.6.5 Redundancy & Fault Protection

The avionics were designed with careful attention to robustness. Both the C&DH and ACS are fully redundant, and the suite is engineered to be single-fault tolerant, allowing the mission to continue without impact following any single hardware failure. Cold-spare block redundancy is implemented with cross-strapped interfaces between all redundant hardware elements, including interfaces between Lanteris-provided and the other selected components. Error monitoring and fault response logic are hardcoded within the flight software, which

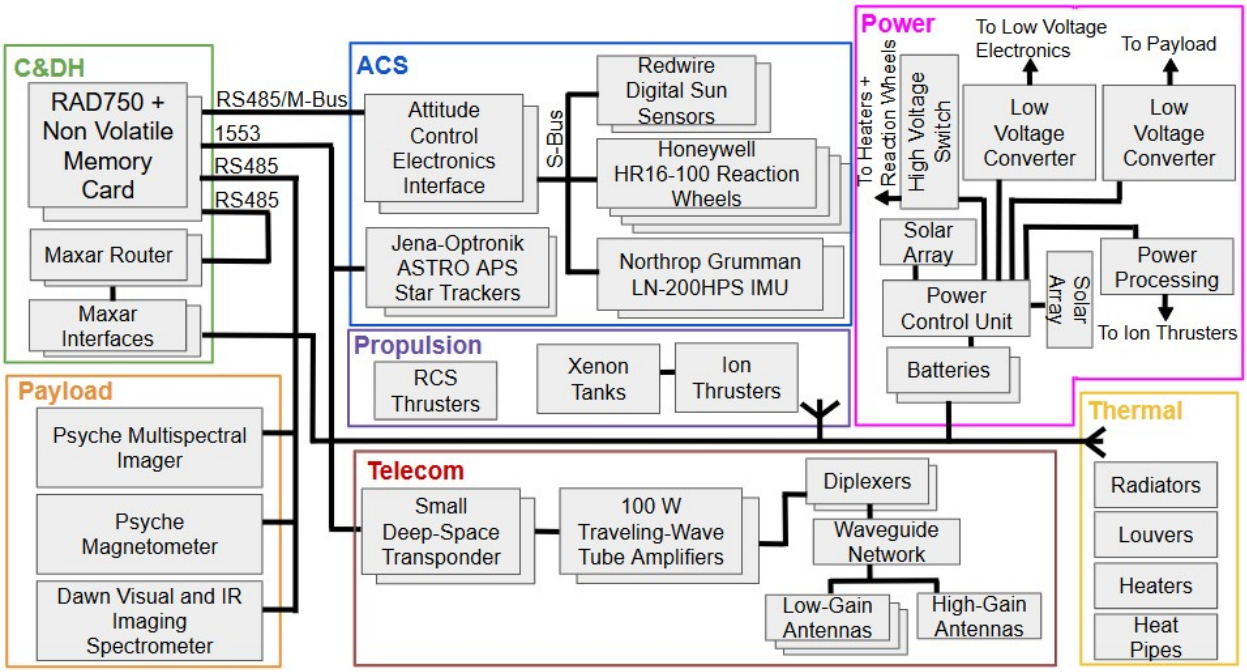


Figure 61: Block Diagram of Avionics Suite

detects anomalous behavior by comparing measured system performance against expected performance. Sun sensors and star trackers are distributed across the exterior of the spacecraft to provide full 360-degree sky coverage under all spacecraft orientations. The reaction wheel assembly is configured in a four-wheel tetrahedral geometry, allowing three-axis attitude control to be maintained even in the event of a single wheel failure.

The spacecraft supports three operational modes: nominal, standby, and safe. The spacecraft can autonomously transition between these modes in response to detected faults, while ground operators also retain the ability to initiate mode transitions. During all mode transitions, all safety-critical devices remain powered on, nonessential hardware is powered off, and the spacecraft maintains a predefined safe attitude configuration. Nominal mode is the default operating state of the spacecraft. In this mode, attitude knowledge is derived primarily from star trackers combined with a bias-corrected IMU, while attitude control is maintained using reaction wheels. Autonomous spacecraft behaviors and onboard sequences remain active, although spacecraft operations may also be directed through ground-managed sequences. Standby mode is a precautionary operating state intended for fault conditions with lower severity than those requiring safe mode. For this mode, attitude knowledge is obtained using coarse sun sensors combined with a bias-corrected IMU, while reaction wheels continue to provide attitude control. The spacecraft points the high-gain antenna toward Earth to establish communications and support rapid recovery operations. Autonomous behaviors and onboard sequences are terminated while in standby mode. Safe mode is the spacecraft's most protective operational state and is entered during severe fault conditions, such as a reaction wheel failure or a deep power system fault. Due to the reaction wheels' four-wheel tetrahedral configuration, only two or more wheel failures will cause the the spacecraft to

enter safe mode. For a single wheel failure, the spacecraft will remain in nominal mode. In safe mode, all nonessential systems, including science instruments, are powered down. Attitude knowledge is provided by sun sensors combined with an uncorrected IMU, while attitude control is performed using reaction control system (RCS) thrusters. The spacecraft points the low-gain antenna and solar arrays toward the Sun to maintain power generation and establish communications with Earth. In the event of a fault involving the RAD750 SBC, the most critical component of the spacecraft, flight software functionality will become temporarily unavailable while the spacecraft swaps to the backup RAD750. Therefore, during this transition, the spacecraft will automatically enter into safe mode.

3.7 Telecommunications (Edward Ju)

All spacecraft-to-ground data exchange runs through NASA's Deep Space Network at X-band with hardware from Psyche: a 2.0 m gimbaled high-gain antenna, three low-gain antennas, two redundant Small Deep Space Transponders, and two redundant 100 W traveling-wave tube amplifiers, cross-strapped through a seven-element waveguide-switch network for single-fault tolerance (Figure 62). The dish and supporting hardware are mounted on the +Z of the spacecraft.

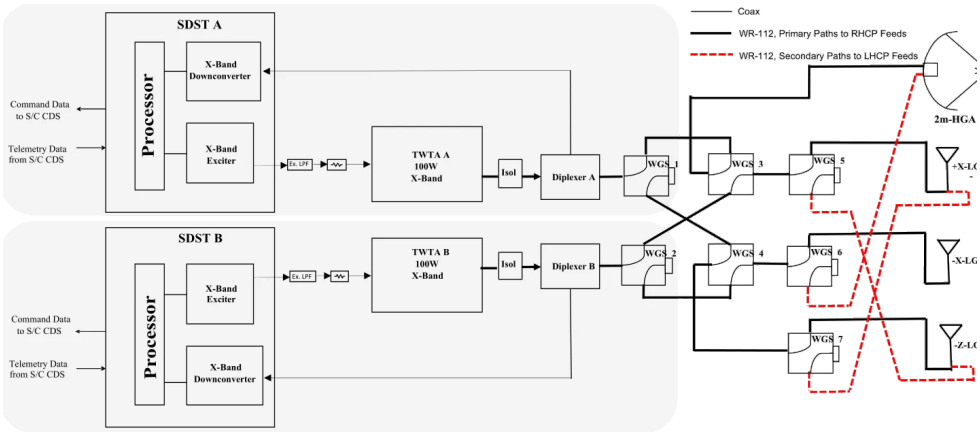


Figure 62: Telecom signal-chain block diagram. The cross-strapped transponders and amplifiers give single-fault tolerance on every path.

3.7.1 Antennas

The high-gain antenna is a 2.0 m gimbaled parabolic dish with an integrated dual-band feed mounted at the center of the top deck. During DSN passes the two-axis gimbal points the dish at Earth, so it can track Earth independently of the attitude the spacecraft holds for instrument work. The gimbal travels $-135°$ to $+45°$ in azimuth measured from the body $+X$ axis, which clears the magnetometer boom, and $66°$ to $90°$ in elevation. The lower elevation limit is a placeholder that can be reduced toward $0°$ if a taller boom is adopted (Structures team). Though both Psyche and Dawn fly body-fixed high-gain antennas, like some other heritage mission such as Mars Science Orbiter, AFONSO instead mounts its dish on a gimbal.

The peak gain of a parabolic dish follows the standard aperture relationship

$$G_{\mathrm{HGA}} = \eta \left(\frac{\pi D}{\lambda} \right)^2, \tag{61}$$

Here $D = 2.0$ m is the dish diameter and $\lambda = c/f = 0.0356$ m is the wavelength at the downlink frequency $f = 8.425$ GHz. The aperture efficiency $\eta = 0.60$ is a conservative engineering estimate (Typical X-band parabolic-dish efficiencies span 0.55 to 0.70). This gives $G_{\mathrm{HGA}} = 42.71$ dBi. The half-power beamwidth is

$$\theta_{\mathrm{HPBW}} = 1.22 \, \frac{\lambda}{D} = 1.24^\circ, \tag{62}$$

which is narrow enough that the dish must be pointed accurately. The reaction wheels hold the body to about 1°, and the gimbal, commanded from the star-tracker attitude knowledge, points the dish to within about 0.1°, well inside the 1.24° beam, so the pointing loss

$$L_{\mathrm{pt}}(\theta) = 12 \left(\frac{\theta}{\theta_{\mathrm{HPBW}}} \right)^2 \quad [\mathrm{dB}] \tag{63}$$

sits at only 0.08 dB and is absorbed into the feeder-loss line of the link budget.

Three low-gain antennas are mounted along the $+X$, $+Z$, and $-Z$ axes of the bus, matching the Dawn arrangement documented in DESCANSO Article 13 [8]. Each LGA produces a peak gain of about 6 dBi within ±45° of its boresight. Three perpendicular axes guarantee that at least one LGA has Earth somewhere inside its 90° cone for any spacecraft attitude. Figure 63 compares the two antenna patterns.

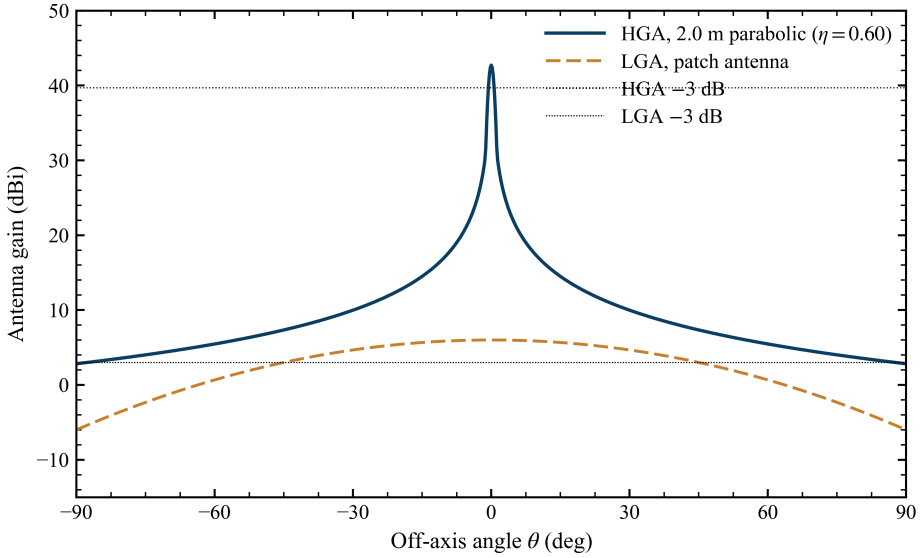


Figure 63: Gain patterns for the 2.0 m HGA (42.71 dBi, 1.24° beamwidth) and the LGA (about 6 dBi) at 8.425 GHz.

3.7.2 Transmitter and Amplifier

Each SDST modulates the downlink at 8.425 GHz and demodulates the uplink at 7.1675 GHz. The TWTA amplifies the SDST output from about 1 W to 100 W. A passive waveguide diplexer routes both directions through the same antenna, and a seven-element waveguide transfer switch network cross-straps either SDST and either TWTA to any antenna for single-fault tolerance.

3.8 Link Budget Methodology

A link budget accounts for every gain and loss the radio signal sees between the transmitter and the receiver. The effective isotropic radiated power is

$$\mathrm{EIRP} = P_{\mathrm{tx}} + G_{\mathrm{tx}} \quad [\mathrm{dBm}], \tag{64}$$

where P_{tx} is the transmitter output in dBm and G_{tx} is the transmit antenna gain in dBi. The free-space path loss is the geometric reduction of signal intensity over distance,

$$L_{\mathrm{FSPL}} = 20\log_{10}\!\left(\frac{4\pi d}{\lambda}\right) \quad [\mathrm{dB}], \tag{65}$$

which at 4.14 AU and 8.425 GHz is 286.79 dB. The signal power at the demodulator is

$$P_{\mathrm{rx}} = \mathrm{EIRP} - L_{\mathrm{feed}} - L_{\mathrm{FSPL}} - L_{\mathrm{atm}} + G_{\mathrm{rx,ant}} + G_{\mathrm{rx,chain}} \quad [\mathrm{dBm}], \tag{66}$$

with $G_{\mathrm{rx,ant}}$ the receive antenna gain and $G_{\mathrm{rx,chain}}$ the cumulative gain of the receiver electronics. The noise floor is set by thermal noise,

$$N_0 = k_B T_{\mathrm{sys}} \quad [\mathrm{W/Hz}], \tag{67}$$

with $k_B = 1.38 \times 10^{-23}$ J/K and T_{sys} the system noise temperature. The energy per bit to noise density ratio at bit rate R_b is

$$\frac{E_b}{N_0} = \frac{C}{N_0} - 10\log_{10}(R_b) \quad [\mathrm{dB}], \tag{68}$$

where $C/N_0 = P_{\mathrm{rx}} - 10\log_{10}(N_0/1\,\mathrm{mW})$ is the carrier-to-noise ratio. The DSN 34 m beam-waveguide receive parameters used here are $D_{\mathrm{DSN}} = 34$ m and $\eta_{\mathrm{DSN}} = 0.685$, giving $G_{\mathrm{DSN}} = 67.90$ dBi at 8.425 GHz [67].

3.9 Worst-Case Downlink Budget

The design worst case is the Themis conjunction at 4.14 AU, through the DSN 34 m antenna under the 90-percentile weather case (CD90), which fixes the receive system noise temperature at $T_{\mathrm{sys}} = 34$ K. The downlink closes with +1.64 dB of margin at 120 kbps (Table 47).

Quantity	Value
Transmit side (spacecraft)	
TWTA output power $10\log_{10}(100\,\mathrm{W}/1\,\mathrm{mW})$	+50.00 dBm
HGA peak gain at 8.425 GHz	+42.71 dBi
EIRP	+92.71 dBm
Feeder loss (incl. 0.08 dB HGA pointing)	−1.50 dB
Path	
Free-space path loss at 4.14 AU	−286.79 dB
Atmospheric loss (CD90 weather)	−0.40 dB
Receive side (DSN 34 m BWG)	
DSN 34 m receive gain	+67.90 dBi
Receiver chain gain (LNA, downconverter, IF amp)	+78.50 dB
Signal power at demodulator	−49.57 dBm
System noise temperature	34 K
Noise power in symbol bandwidth	−57.00 dBm
Result	
SNR	+7.43 dB
E_b/N_0 at 120 kbps	+4.43 dB
Required E_b/N_0 $(6.79 - 7.00 + 3.00)$	+2.79 dB
Margin (7 dB coding)	**+1.64 dB**
Uncoded BER at 120 kbps	1.06×10^{-2}
Achieved BER with rate-1/6 turbo (7 dB)	6.8×10^{-8}
Design target BER	10^{-3}

Table 47: Downlink link budget at the Themis worst case (4.14 AU, CD90 weather, 120 kbps), closing with +1.64 dB of margin.

3.9.1 Worst-Case Uplink Budget

The uplink runs at 7.1675 GHz from the DSN 34 m transmitter at 20 kW into the SDST receiver at $T_{\mathrm{sys}} = 500$ K, at the same geometry. The 2 kbps command rate closes with +25.48 dB of margin (Table 48), so the uplink would not be a limiting direction.

Quantity	Value
DSN transmitter power $10\log_{10}(20{,}000\,\mathrm{W}/1\,\mathrm{mW})$	+73.01 dBm
DSN 34 m antenna gain at 7.1675 GHz	+66.49 dBi
EIRP	+139.51 dBm
DSN feeder loss	−0.60 dB
Free-space path loss at 4.14 AU, 7.1675 GHz	−285.39 dB
Atmospheric loss	−0.40 dB
HGA receive gain at 7.1675 GHz	+41.25 dBi
Receiver chain gain	+78.00 dB
Spacecraft system noise temperature	500 K
E_b/N_0 at 2 kbps	+32.99 dB
Required E_b/N_0 $(9.6 - 5.1 + 3.0)$	+7.50 dB
Margin	**+25.48 dB**

Table 48: Uplink link budget at the Themis worst case, where the 2 kbps command link closes with +25.48 dB of margin.

3.9.2 Per-Target Performance

The mission holds a fixed 120 kbps downlink rate and keeps the surplus as margin (Table 49). The margin grows quickly at closer ranges, from +1.64 dB at the Themis far point to more than +10 dB at Parthenope opposition, and the achievable rate could be stepped well above 120 kbps at most geometries (Figure 64).

Target	Near (AU)	Near margin	Mid (AU)	Mid margin	Far (AU)	Far margin
Parthenope	1.45	+10.75 dB	2.45	+6.19 dB	3.45	+3.22 dB
Psyche	1.92	+8.31 dB	2.92	+4.67 dB	3.92	+2.11 dB
Themis	2.14	+7.37 dB	3.14	+4.04 dB	4.14	+1.64 dB

Table 49: Downlink margin at each target, with the Themis far point (4.14 AU) the worst case that sets the 120 kbps floor.

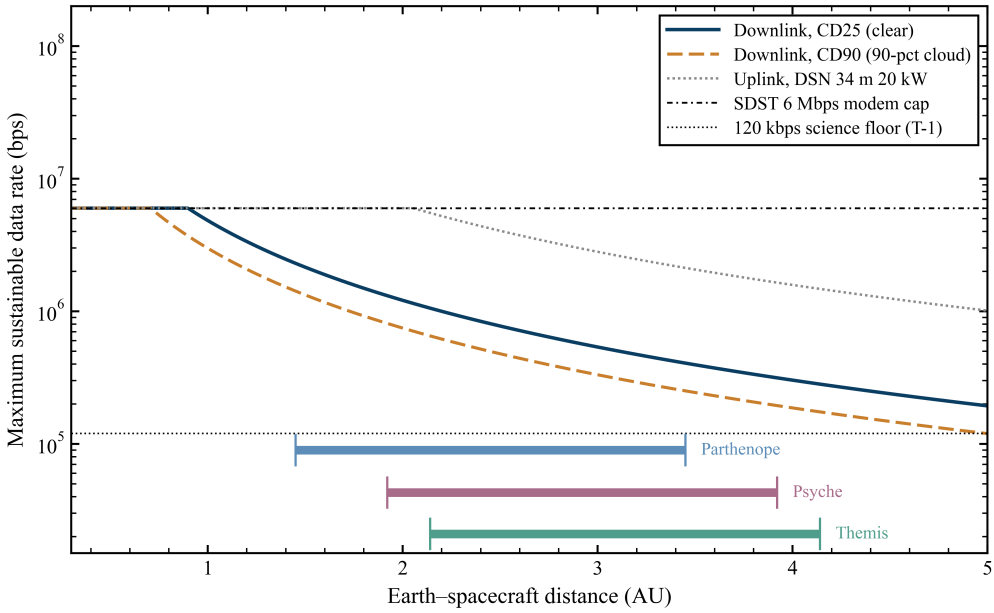


Figure 64: Maximum downlink rate versus Earth-spacecraft distance, staying above the 120 kbps requirement

3.9.3 Total Data Return

Running the fixed 120 kbps rate whenever the link is geometrically feasible delivers about 7.8 TB across the mission, from launch through the end of operations in 2047 (Figure 65). Most of it builds up during the cruise, where downlink time is plentiful, and the eight proximity stays contribute around 0.6 TB. That 7.8 TB is roughly 9 times the 837 GB the Payload team committed to return, giving enough margin for science. Stepping the rate up adaptively (instead of fixed 120 kbps) would raise the total data return to about 15 TB.

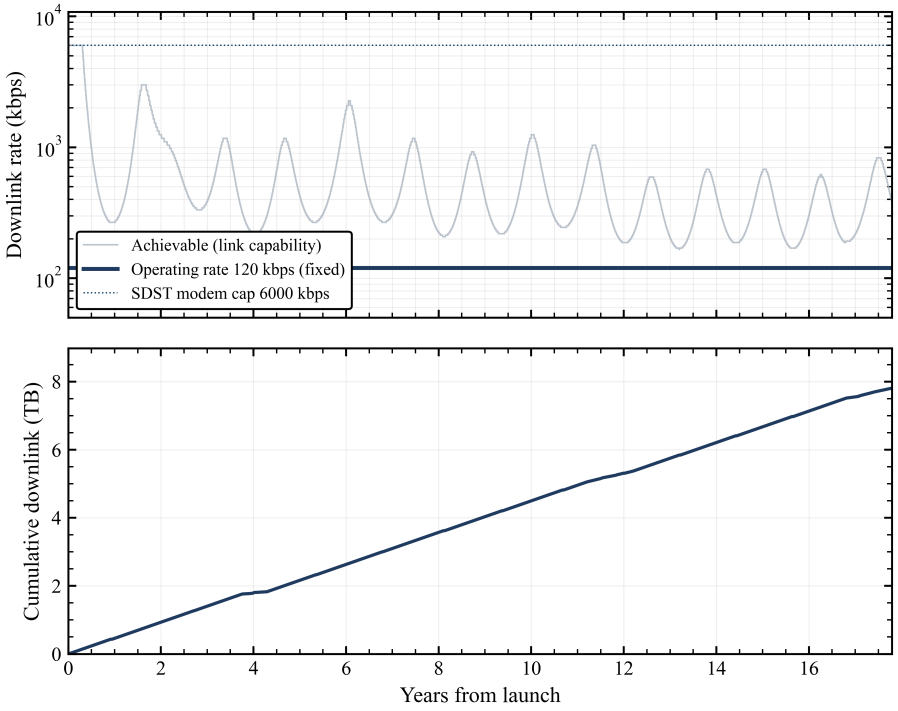


Figure 65: Link capability versus the fixed 120 kbps operating rate (top), and cumulative downlink reaching about 7.8 TB at end of mission (bottom).

3.9.4 Windows and Occultation

Two effects interrupt the link. Solar conjunction blocks it when the Sun-Earth-Probe angle drops below about 2 degrees, in eight events totaling about 46 days over the mission, the longest about 7 days. Asteroid occultation during the proximity stays removes about 78 more days. The per-orbit occultation has an edge-on geometric limit

$$f_{\mathrm{occ}} \leq \frac{2\arcsin(R/r)}{360^\circ}, \tag{69}$$

where R is the asteroid radius and r is the orbital radius, and ranges from near zero at the 300 km orbits to roughly 16 to 20 percent at the lowest orbits. Together the two effects leave the link available about 98 percent of the time. The full mission timeline of cruise, stays, and blackouts is shown in Figures 66 and 67, and every conjunction is short enough to ride out on the recorder.

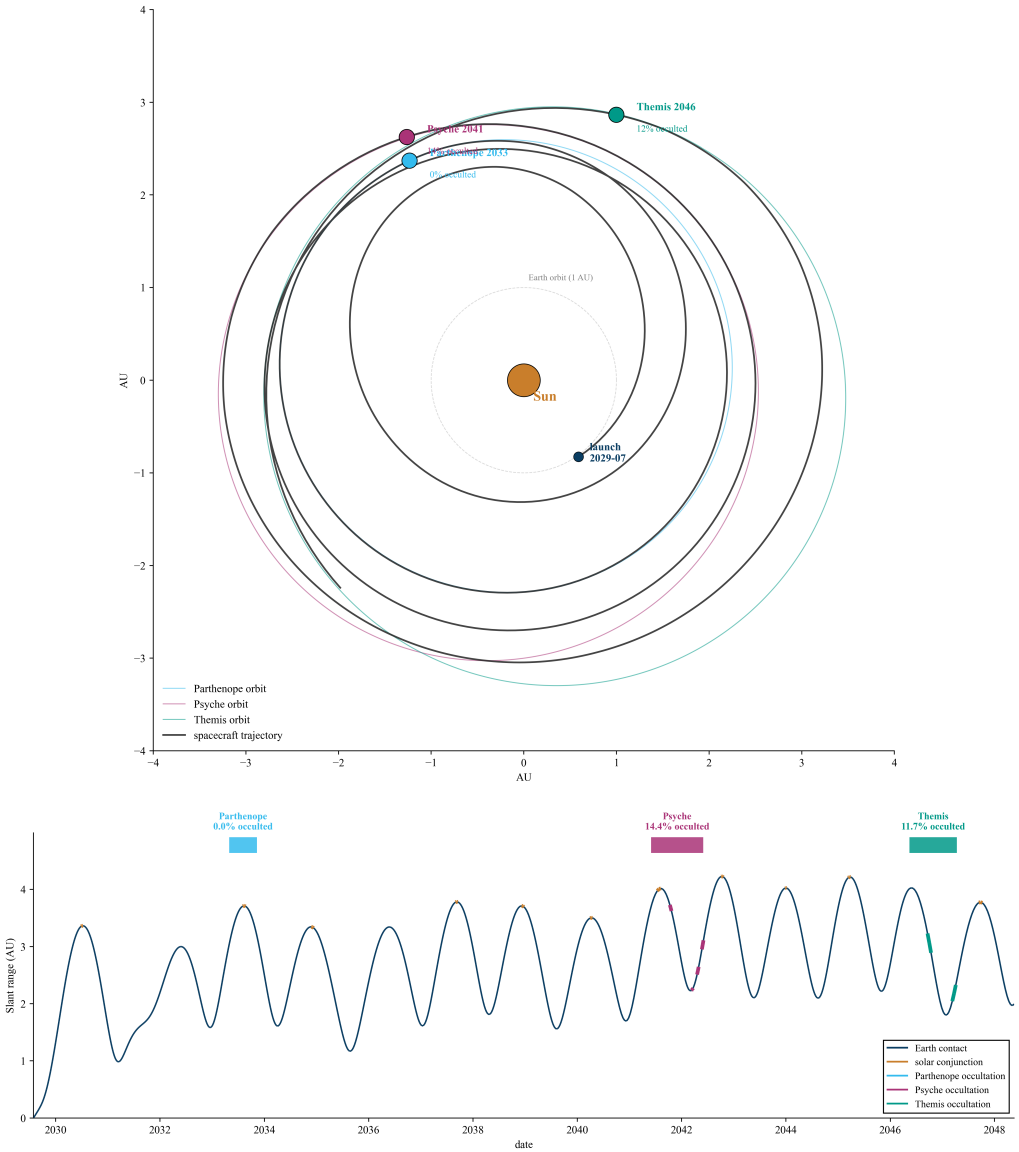


Figure 66: Mission geometry, with the heliocentric trajectory (top) and the Earth-spacecraft range with occultation and conjunction windows over the mission (bottom).

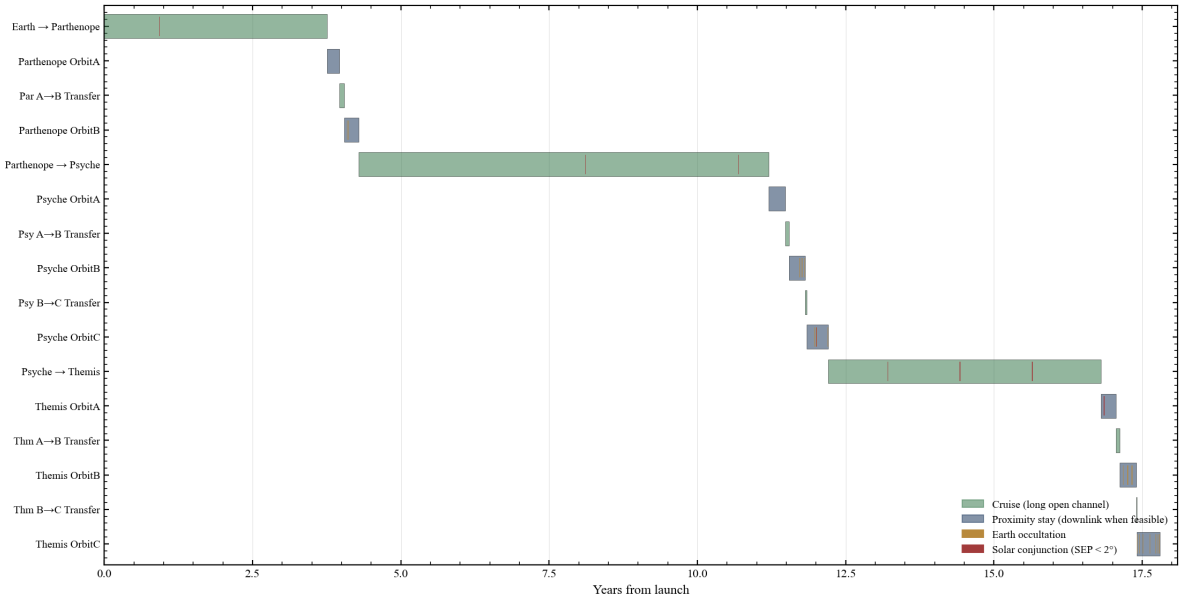


Figure 67: Mission-wide telecom timeline across the cruise and eight proximity stays, with the occultation and solar-conjunction blackouts marked.

3.9.5 Operations and Recorder Sizing

Because the dish is gimbaled, the spacecraft does not have to slew to point at Earth, so each proximity stay is split between communications days, when the instruments pause and the recorder drains, and science days, when the instruments collect. The per-stay split comes from the attitude control schedule data(Figure 68 and Table 50). Parthenope is the most attitude-limited at 14 to 16 communications days, while Psyche and Themis are nearly always available. The 256 GB recorder has the peak at 94 GB and never fills, with all data remaining during the proximity orbit being downlinked during cruise, so all stored science is eventually downlinked.

Stay	Altitude	Total (d)	Comm (d)	Science (d)	Sci. after 1st window (d)	Peak recorder (GB)
Parthenope A	300 km	77	14.0	62.9	27.9	51
Parthenope B	75 km	90	16.2	73.8	33.8	94
Psyche A	300 km	102	71.4	30.6	0.6	45
Psyche B	170 km	100	54.9	45.1	5.1	75
Psyche C	75 km	130	91.9	38.1	18.1	47
Themis A	170 km	90	35.0	55.0	0.0	83
Themis B	75 km	103	76.1	26.9	16.9	23
Themis C	25 km	140	87.0	53.0	18.0	65
Total		832	446.5	385.4	120.4	94

Table 50: Per-stay operations summary of communications days, science days, and peak recorder fill for each stay.

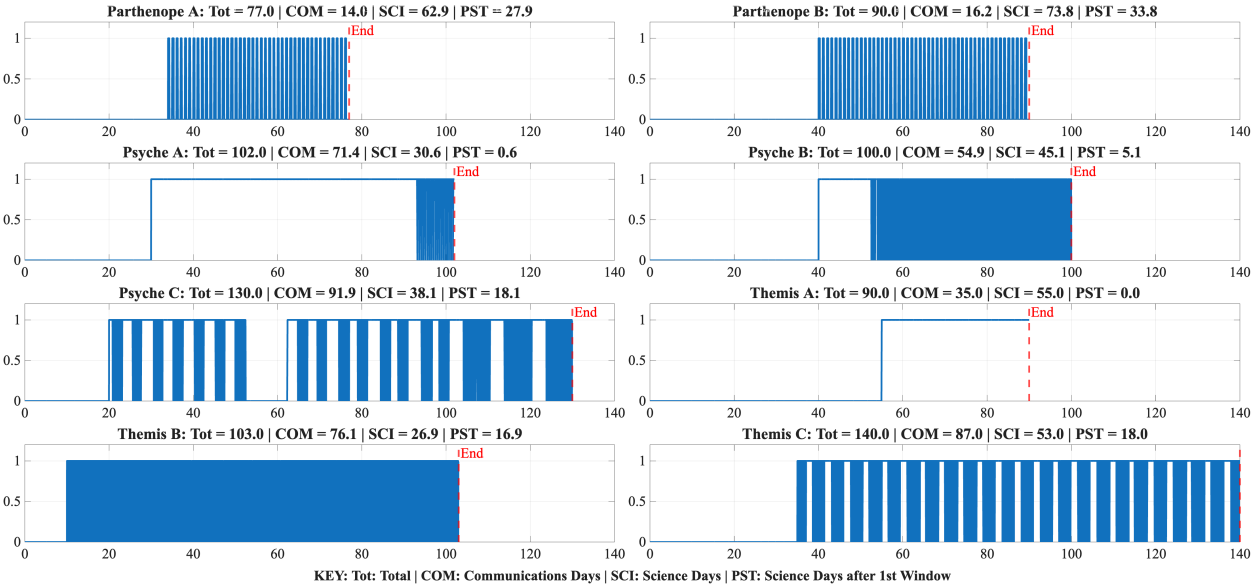


Figure 68: Per-stay downlink schedule, where the filled intervals mark communications (downlink) windows for each stay. (Figure Credit: Mohanad Ahmed)

4 Schedule Overview

4.1 Development Schedule (Hannah Ramsperger)

The development schedule follows the time line of the Psyche mission development schedule [68]. The mission proposal phase as well as the in-depth concept study phase, which last around 6 years before the preliminary design phase, are not included in this timeline. Given the launch date in July 28, 2029, the preliminary design phase is assumed to have started in the beginning of 2023. Therefore, high-level a development period of 6.5 years is shown in Figure 69.

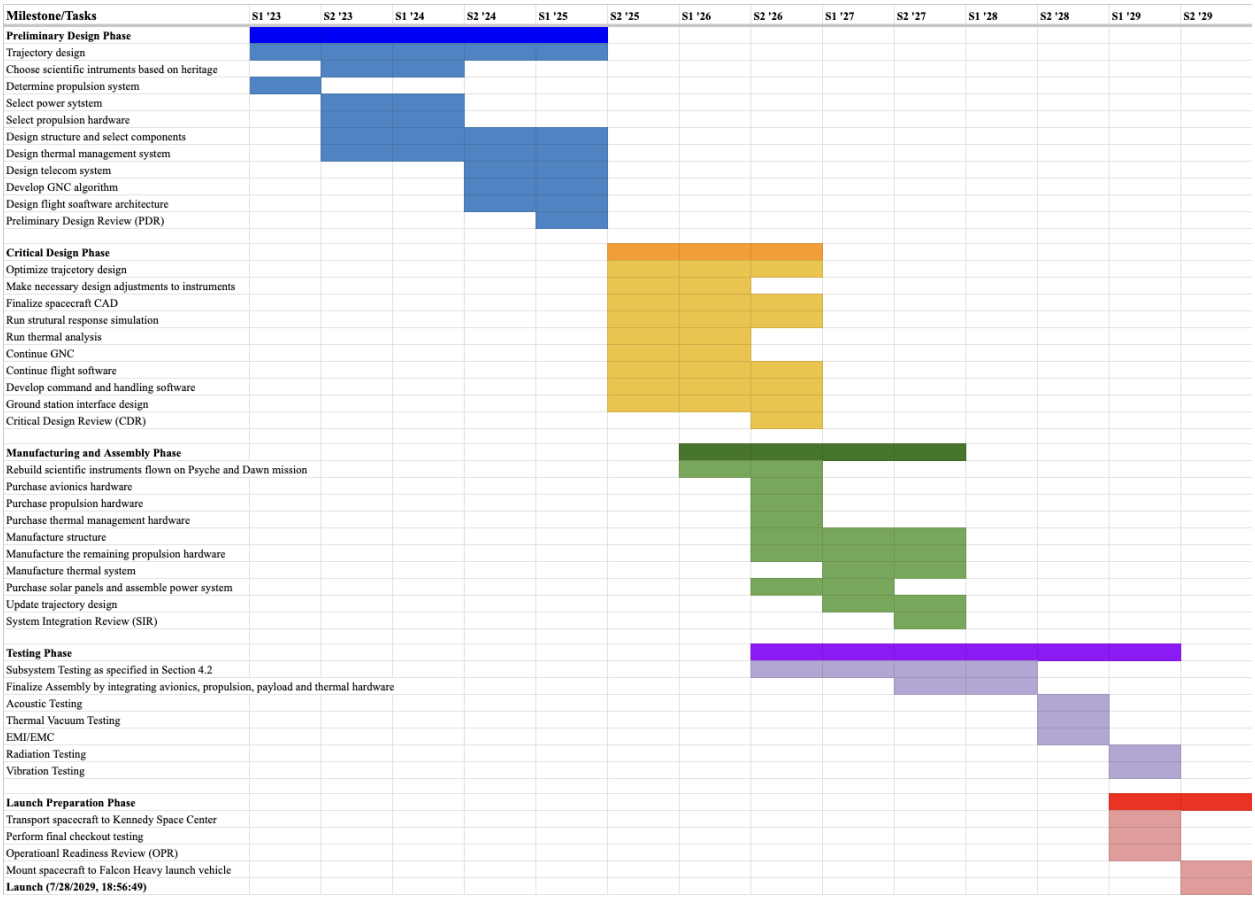


Figure 69: High-level development schedule of the AFONSO mission.

4.2 Testing Plan

4.2.1 Payload Testing Plan (Hannah Ramsperger)

Table 51: Payload testing plan.

Component	Test	Description
Spectrometer (VIR)	Software Testing	Verify onboard instrument control software including commanding, mode transitions, housekeeping telemetry.
VIR, Multispectral Imager	Calibration Testing	Characterize wavelength registration and radiometric response across the full visible and infrared spectral ranges at the full range of detectors operating temperatures. Quantify temperature-dependent wavelength shifts and responsivity changes and verify SNR expectations.
VIR	Cryocooler Performance Testing	Measure IR detector cooldown time to nominal operating setpoint and temperature hold stability.
Multispectral Imager	Optical Calibration	Verify geometric properties such as field of view and focal length of both camera heads independently.
Multispectral Imager	Filter Wheel Testing	Each of the 7 narrowband filters and the broadband clear filter positions was tested for proper actuation.
Multispectral Imager	Onboard Processing Tests	Test functions of DEA including image compression, image processing, and storage.
Multispectral Imager	Cover Mechanism Testing	Test the actuated cover to ensure Sun avoidance to confirm protection of the CCD from direct solar illumination.
Magnetometer	Ground Calibration	Characterize the instruments performance across its full allowable temperature range. The calibration process corrects for temperature-dependent effects.
Magnetometer	Magnetic Cleanliness Testing	Characterize magnetic fields generated by spacecraft assembly and verify that spacecraft-generated fields at the inboard sensor location meet cleanliness requirements.
Magnetometer	Software and Functional Testing	Verify onboard magnetometer commanding, sampling rate configuration, and housekeeping telemetry. Confirm cross-strapping between sensor and electronics units.
VIR, Imager, Magnetometer	Thermal Vacuum Testing	Subject instrument to representative thermal cycles in vacuum chamber for functional verification.
VIR, Imager, Magnetometer	Vibration Testing	Perform vibration and pyroshock testing to expected levels during the launch and spacecraft deployment and verify optical performance post-test.
VIR, Imager, Magnetometer	Electromagnetic Interference and Compatibility Testing	Ensure the spacecraft will operate correctly in the electrical and magnetic conditions of space, and that the electrical and magnetic components that make up the spacecraft are compatible and do not interfere with one another.
VIR, Imager, Magnetometer	Radiation Testing	Verify that detector electronics, and other sensitive components meet total ionizing dose.

4.2.2 Avionics Testing Plan (Alexis Herfurth)

The avionics testing plan is designed to ensure that all avionics meet their specified requirements and perform as intended at both the component and integrated subsystem levels.

Table 52: Avionics Testing Plan.

Component	**Test**	**Description**
RAD750 SBC, 256 GB NVM Card	Functional Testing	Verify component meets all operational requirements. Specific verifications include proper boot and startup sequence, autonomous and ground command execution, data routing, memory read/write functionality, interface communication, time synchronization, spacecraft operational mode transition, and swapping to redundant backup systems in the event of a failure
Northrop Grumman LN-200HPS IMU	Functional Testing	Verify component meets all operational requirements. Specific verifications include proper gyroscope and accelerometer performance within acceptable bias, noise, and random walk limits
Jena-Optronik ASTRO APS Star Trackers	Functional Testing	Verify component meets all operational requirements. Specific verifications include proper star field acquisition and attitude determination accuracy, as well as robustness to stray light
Redwire Digital Sun Sensors	Functional Testing	Verify component meets all operational requirements. Specific verifications include proper solar incidence angle measurement and field-of-view coverage, as well as robustness to stray light
Honeywell HR16-100 Reaction Wheels	Functional Testing	Verify component meets all operational requirements. Specific verifications include proper torque output, momentum storage, and saturation/desaturation performance
All Interfaces	Functional Testing	Verify each interface meets all operational requirements. Specific verifications include proper command and telemetry connectivity and adherence to interface protocols
RAD750 SBC, VxWorks RTOS, Flight Software	Software-in-the-Loop (SIL)	Connect software a real-time digital simulator that mimics on-board operations to test performance

Table 53: Avionics Testing Plan (Continued).

Component	**Test**	**Description**
All Avionics Components	Hardware-in-the-Loop (HIL)	Connect components a real-time digital simulator that mimics on-board operations to test performance and integration between interfaces
All Avionics Components	Electromagnetic Interference (EMI)	Confirm that unintentional signal emissions from each component, whether conducted through wiring or through electromagnetic radiation, are within allowable limits and do not cause interference with other avionics components or subsystems
All Avionics Components	Electromagnetic and Radio Frequency Compatibility	Verify that the each component can continue operating reliably when exposed to various amounts of electromagnetic radiation, ratio frequency voltages, or currents
All Avionics Components	Single Event Effects	Evaluates how each component reacts when struck by a single, highly energetic particle
All Avionics Components	Total Ionizing Dose	Evaluates how each component degrades when exposed to cumulative, long-term ionizing radiation
All Avionics Components	Thermal Cycling	Subject each component to various hot-cold cycles at ambient pressure to verify functionality
All Avionics Components	Thermal Balance/Vacuum	Subject each component to various hot-cold cycles in a vacuum environment to verify functionality
All Avionics Components	Burn-In	Ensure reliability of each component by operating the hardware and software for extended periods of time at elevated temperatures
All Avionics Components	Shock and Vibration Testing	Perform vibration and pyroshock testing at various levels, especially those during launch and spacecraft deployment, to ensure each component will continue to operate within requirements

4.2.3 Power Systems Testing Plan (Miina Anvelt)

Table 54: Power testing plan.

Component	Test	Description
Battery	Charge Testing	Verify ability to undergo charging and discharging when connected to power control unit.
Battery	Cyclic Testing	Verify the long-term ability of battery to charge and discharge over multiple cycles using extra test battery if available.
Power Control Unit, Solar Arrays	State of Charge Testing	Measure the state of charge of battery to verify that expected state of charge is being met and that expected solar cells output power is met.
Solar Arrays	Solar Cell Output Tests	Verify the solar cells output power and re-verify the efficiency of the solar cells used.
Power Control Unit	Functional Tests	Determine system ability to transfer power from solar arrays into voltage converters or into battery based on system requirements.
Power Control Unit	Voltage Converters Output Tests	Ensure that voltage converters are outputting expected voltages that meet subsystem component requirements.
Power Control Unit, Solar Arrays	Thermal Vacuum Testing	Verify functionality of power control unit and battery system when exposed to expected mission conditions.
Power Control Unit, Solar Arrays, Battery	Vibration Testing	Ensure that the systems are able to withstand launch vibrational modes.

4.2.4 Structure Testing Plan (Tamar Levin)

The structural verification plan is intended to confirm that the spacecraft meets the structural requirements.

Verification is performed first through finite element analysis, including static launch-load analysis, modal analysis, and dynamic steady-state response analysis. These analyses verify that the primary structure remains below allowable stress limits, and that the first lateral and axial modes exceed the launch vehicle requirements.

Hardware-level testing shall be performed after the design is complete to verify the FEA results.

- **Modal survey:** measuring the spacecraft natural frequencies and comparing them with the predicted modes.
- **Sine vibration test:** verifies the spacecraft response to launch vehicle excitation.

The spacecraft is mounted to a shaker table and excited with sinusoidal acceleration in the axial and lateral directions.

- **Random vibration test:** verifies survival under broadband launch excitation. The spacecraft is again excited using a shaker table, over a spectrum that corresponds to launch environment.
- **Shock test:** verifies survival of launch vehicle separation. A shock is applied and the response of sensitive components such as joints and interfaces is inspected.

A thermal vacuum test shall also be performed to verify that the spacecraft can survive vacuum and temperature conditions, as specified in thermal testing plan. Thermal cycles can cause deformation due to differences in coefficients of thermal expansion, which may lead to structural distortion or misalignment of sensors. Vacuum testing is especially important for the graphite-composite central cylinder, since composites are at risk of outgassing, which can cause contamination and possibly change mechanical properties.

Additional inspections include dimensional checks, mass measurement, visual inspection of joints, and alignment checks for the payload, antenna, solar arrays, and tank mounts. Deployment tests would be performed for the solar arrays. Verification steps are summarized in Table 55.

Table 55: Structural verification and testing plan.

Requirement	**Analysis Verification**	**Test Verification**
Remain withing launch envelope in stowed configuration	CAD model (in progress)	Dimensional inspection after integration
Minimum natural frequencies	Modal analysis (in progress)	Modal survey
Withstand launch loads	Static and dynamic steady-state analysis (in progress). Random vibration analysis (not yet completed)	sine vibration, and random vibration tests
Survive shock events	Shock analysis (not yet completed)	shock test
Deploy arrays, boom, and antenna	Deployment clearance checks in CAD (not yet completed)	Deployment test after integration.
Maintain mass budget	CAD-based mass estimated	Mass measurement before and after integration

4.2.5 Thermal Testing Plan (Michael Hwang)

The objective of the thermal testing plan is to verify that the thermal control system can maintain all temperature-sensitive components within their allowable operating ranges throughout the mission. In addition, these tests will serve to validate the multi-node thermal model.

Component-Level Testing

Patch heaters, temperature sensors, heat pipes, MLI blankets, and louvers will be individually tested before integration into the spacecraft.

Table 56: Component-level thermal testing plan.

Component	**Test Objective**	**Verification Method**
Patch heaters	Verify heater power, resistance, and redundancy	Apply required heater power, measure heat output, and confirm redundant heater operation
Temperature sensors / thermostats	Verify accurate temperature feedback	Calibrate sensors in a controlled thermal chamber over the expected operating range
Heat pipes	Verify heat transport capability	Apply required heat load at the evaporator and measure condenser temperature response and effective conductance
Louvers	Verify actuation temperature and open/closed thermal properties	Cycle louver temperature through actuation range and measure effective emissivity
MLI blankets	Verify MLI thermal properties and quality	Inspect seams, vent paths, grounding, attachment points, and coverage after installation, and test absorptivity/emissivity of random samples

Integrated Thermal Vacuum Testing

Following component-level testing, the thermal hardware will be integrated into the spacecraft. The entire spacecraft will then undergo thermal vacuum testing to validate the thermal design in flight conditions. Because most, if not all, hardware must be integrated for this test, it will occur near the end of the development timeline after most other subsystems have completed testing and integration. The spacecraft will be placed in a vacuum chamber, then subjected to several different thermal test conditions to verify the functionality of the louvers, heaters, and heat pipes working in tandem. For components such as the propulsion system that cannot be operated during a thermal vacuum test, heaters supplying the representative amount of power will be used instead.

Table 57: Integrated spacecraft thermal vacuum test cases.

Test Case	Purpose	Description
Cold case	Verify heater power is sufficient during operation	Operate spacecraft components at minimum expected power levels and verify heater-controlled nodes remain within temperature limits
Hot case	Verify components remain below maximum allowable temperatures	Simulate high solar input and maximum internal heat generation with louvers open
Eclipse transient case	Verify thermal response during asteroid eclipse periods	Simulate periodic changes in solar heat input and verify node temperature response
Heater fault case	Verify redundancy and fault tolerance	Disable selected heater circuits and verify backup heaters maintain acceptable temperatures

4.2.6 Propulsion Testing Plan (Mahak Mathur)

The propulsion testing plan involves the two major systems: the primary ion thruster and feed system; and the RCS thrusters and feed system.

1. The Next-C ion engine must be tuned with the PPU to achieve a peak thrust of 300 mN.

2. The nominal I_{sp} should be confirmed to be 3100 s.

3. The desired pulse modulation from the Aerojet thrusters must be verified, along with the true vector alignment of the final installations - modifications to the control system will be made accordingly.

4. The xenon, nitrogen, and hydrazine tanks must be checked for protection against leakages.

5. Through hardware-in-the-loop testing, the various components and software involved in the ion engines and RCS thruster systems must be verified.

6. The electromagnetic compatibility of the ion engines and RCS thrusters with the payload components must be verified.

7. The feed systems to the ion engines and RCS thrusters must be tested through electrical continuity and vacuum chamber testing.

8. In general, the strength and effectiveness of the various components must be validated under various pressure and flow conditions.

5 Risk Assessment

A failure modes and effects analysis (FMEA) was performed using the NASA risk matrix to assess risk based on likelihood (L) and consequence (C), shown in Figure 70.

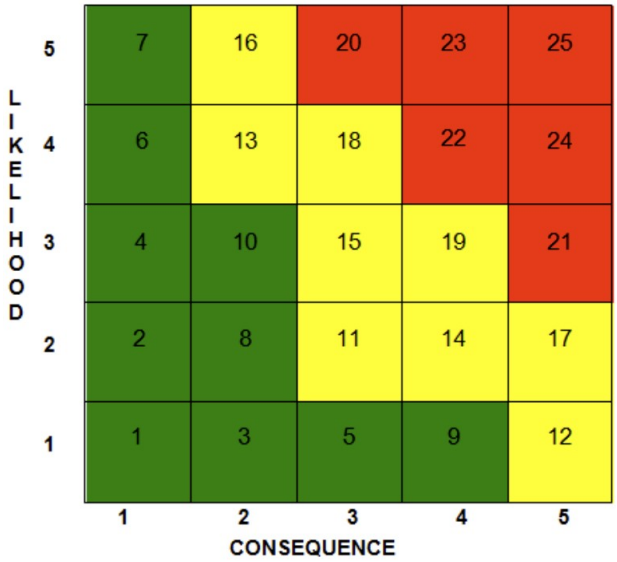


Figure 70: NASA risk assessment matrix.

The guidelines for estimating likelihood and consequence are outlined in Tables 58 and 59.

Likelihood	Sample Probabilities
1	Rare, $< 5\%$ probability
2	Unlikely, 5–25%
3	Possible, 25–50%
4	Likely, 50–75%
5	Certain, 75–100%

Table 58: Likelihood scale used for the risk matrix.

Severity	Classification	Effects on Mission
1	Negligible	Little to no effect on mission
2	Minor	Minor degradation of subsystem performance
3	Moderate	Measurable degradation of subsystem performance
4	Critical	Loss or significant degradation of subsystem, threatens mission objectives
5	Catastrophic	Loss of mission/spacecraft, irreversible mission-ending failure

Table 59: Severity scale used for the risk matrix.

5.1 Payload Risk Assessment (Hannah Ramsperger)

Table 60: Payload Risk Hazard Analysis.

Component	Failure Mode	Mitigation	L	C
Spectrometer	Degradation of optical elements from radiation exposure	In-flight heater annealing of focal plane detectors to reverse displacement damage and metallic cover mechanism [20]	3	3
Spectrometer	Detector readout failure	Safe mode commanding from ground	3	3
Spectrometer	Thermal cracking of instrument	Passive thermal control through MLI	2	3
Spectrometer	CCD temperature [69]	Use of cryocooler	4	2
Spectrometer	Cryocooler failure	IR channel is lost but visible channel is still operating, partial science return	2	4
Multispectral Imager	Cover mechanism failure	Redundant camera head, attitude control system avoids direct illumination with nadir pointing spacecraft	2	3
Multispectral Imager	DEA failure	DEA houses two independent electronics cards [16].	2	3
Multispectral Imager	Filter wheel mechanical jam	Redundant camera head, filter positions are redundant, and broadband clear filter still enables imaging if wheel is stuck.	2	2
Multispectral Imager	CCD radiation damage	Prelaunch radiation testing and heritage robustness.	2	3
Magnetometer	Sensor Unit Failure	Two sensors carried designed to meet scientific objectives in event of failure of one SU.	2	3
Magnetometer	Spacecraft magnetic contamination	Magnetic cleanliness program prelaunch and gradiometry through two SU.	2	3
Magnetometer	Electronics unit failure	Cross-strapping between sensor units and electronics units allows rerouting of signals.	2	2
Magnetometer	Asteroid magnetic field below instrument detection threshold	Progressively lower orbital altitudes increase field strength at sensor. Still informs about weakness of magnetic field.	4	3

5.2 Avionics Risk Assessment (Alexis Herfurth)

Table 61: Avionics Risk Hazard Analysis.

Component(s)	Failure Mode	Mitigation	L	C
RAD750 SBC	CPU failure	Complete functional testing for component before integration and utilize cold-spare block redundancy with cross-strapping. While swapping to backup RAD750, the spacecraft will automatically enter safe mode as flight software functionality will become temporarily unavailable.	1	5
RAD750 SBC	Delayed or faulty autonomous or ground command execution	Hardcode error monitoring and fault response logic within flight software to detect anomalous behavior and employ VxWorks RTOS.	2	4
RAD750 SBC, 256 GB NVM Card	Memory corruption	Complete functional testing for both components before integration and utilize cold-spare block redundancy with cross-strapping. Use RAD750's dedicated memory scrubbing process, where the SBC periodically scans all memory and detects and corrects errors before they accumulate.	2	4
All Interfaces	Interface or data bus failure	Complete functional testing for each interface, as well as hardware-in-the-loop to verify integration between interfaces. Utilize cold-spare block redundancy with cross-strapping.	2	5
All ACS Sensors (IMU, Star Trackers, Sun Sensors)	Failure due to sensor dropout, noise, or bias drift	Complete functional testing for each component before integration and utilize cold-spare block redundancy with cross-strapping. If both primary and backup star trackers fail, the spacecraft will automatically enter standby mode utilizing only IMU and sun sensors for attitude knowledge.	2	3

Table 62: Avionics Risk Hazard Analysis (Continued).

Component(s)	**Failure Mode**	**Mitigation**	**L**	**C**
Honeywell HR16-100 Reaction Wheels	Mechanical failure	Utilize redundant reaction wheel 4-wheel tetrahedral configuration. If more than two reaction wheels fail, the spacecraft will enter safe mode using RCS thrusters for attitude control.	3	4
All C&DH Components	Failure due to radiation	Only utilize radiation-hardened components, as well as perform single event effects and total ionizing does radiation testing.	1	5
All C&DH Components	Electromagnetic interference (EMI) between components and subsystems	Complete EMI testing, as well as electromagnetic and radio frequency compatibility verification, for each component before and after integration. Use proper shielding, enclosures, filtering, and ground connections.	1	5
All ACS Components	Failure due to radiation	Only utilize radiation-hardened components, as well as perform single event effects and total ionizing does radiation testing.	1	3
All ACS Components	Electromagnetic interference (EMI) between components and subsystems	Complete EMI testing, as well as electromagnetic and radio frequency compatibility verification, for each component before and after integration. Use proper shielding, enclosures, filtering, and ground connections.	1	3

5.3 Structural Risk Assessment (Brandon Lee)

Table 63: Thermal System Risk Hazard Analysis.

Component	**Failure Mode**	**Mitigation**	**L**	**C**
Honeycomb panels	Bottom or top honeycomb panels buckle during quasi-static loads	Add diagonal vertical panels at the bottom base of the spacecraft and structural beam support on the underside of the top panel of the spacecraft.	2	4
Primary structure	The structure vibrates at its resonant frequency	Preliminary modal analysis has been completed to verify the primary modes are above 25 Hz. A more detailed modal analysis will be utilized.	1	5
Tank brackets	Yielding of tank brackets	Static structural FEA analysis has been completed and it is determined that the von-Mises stress in the tank brackets is less than the yield stress by a factor of safety greater than 2.	1	4

5.4 Thermal Risk Assessment (Michael Hwang)

Table 64: Thermal System Risk Hazard Analysis.

Component	Failure Mode	Mitigation	L	C
Louvers	Mechanical louvers fail to actuate	Louvers are sourced from a vendor with extensive flight heritage and are unlikely to fail. Louvers will also be tested before integration.	1	5
Heat pipes	Wick capillary limit is reached	Heat pipes will be sized to carry the required heat load with 50% additional margin to prevent the wick from drying out.	1	4
Heat pipes	Leakage of working fluid	Heat pipes are implemented in parallel with single-fault tolerance so that loss of one heat pipe does not eliminate the heat rejection path.	1	4
Patch heaters	Loss of control over resistive heating element	Heating elements will be implemented with redundancy so failure of an isolated heating element will not affect spacecraft thermal control.	1	1
Temperature sensors / thermostats	Incorrect temperature feedback	Redundant sensors and thermostats will be used for heater control to reduce the likelihood of overheating or insufficient heating. Sensors will undergo testing before integration.	1	3
MLI	Degradation of optical properties over mission lifetime	Beginning-of-life and worst-case end-of-life MLI optical properties are included in the thermal analysis.	3	1

5.5 Power Risk Assessment (Miina Anvelt)

Table 65: Power System Risk Hazard Analysis.

Component	Failure Mode	Mitigation	L	C
Battery	Degradation of battery cells	Duty cycles of battery loading to maintain higher battery cell outputs and ensure within optimal thermal range[70].	1	3
Battery	State of charge readout failure	Determine other indicators of solar array input such as voltage monitoring	2	3
Battery	Battery cells failure	Ensure all cells are operational prior to launch and battery optimal conditions can be maintained throughout mission	2	5
Battery	Thermal runaway	Ensure limited thermal stresses on battery and ensure no mechanical stresses along the cells [71].	2	4
Solar Arrays	Deployment Partial failure	Ensure the full solar panel is able to functionally take in all possible power and re-calculate power budgets	1	4
Solar Arrays	Deployment Full failure	This would be a mission critical failure as solar panels support the power consumption of the electrical propulsion system	1	5
Battery	Degradation of solar cells	To be expected from mission and can be mitigated by planning for efficiency loss prior to launch [54].	4	2

5.6 Propulsion Risk Assessment (Jason Kamau)

Table 66: Propulsion System Risk Hazard Analysis.

Component	Failure Mode	Mitigation	L	C
NEXT-C ion thrusters	Thruster underperformance or degradation	Operate one thruster at a time and keep the second thruster as an operational spare. Throttle levels will be verified against the low-thrust trajectory simulations.	2	5
Xenon feed system	Regulator or flow-control failure	Use redundant regulation, pressure sensing, filters, check valves, and isolation valves so failed branches can be identified and isolated.	2	5

Table 66 continued.

Component	Failure Mode	Mitigation	L	C
Xenon tanks	Leak or loss of one tank branch	Xenon is distributed across six tanks with parallel feed paths so loss of one tank does not eliminate IPS propellant availability.	2	4
Hydrazine ACS feed system	Hydrazine depletion before end of mission	Hydrazine is sized from the ACS desaturation budget with margin. The nominal load is 80 kg, while the two-tank system can hold approximately 92 kg.	2	5
RCS thrusters	Individual thruster or valve failure	The ACS uses 12 MR-111G thrusters and dual manifolds so attitude control can continue after isolated thruster or branch failures.	2	4
Hydrazine tanks	Tank over-pressurization or thermal excursion	Tank pressure and temperature are monitored using pressure transducers and temperature sensors. Tanks are thermally controlled with insulation and heaters as needed.	1	5
Reaction wheel desaturation	Desaturation demand exceeds estimate	Maintain hydrazine margin and refine the ACS/RCS model with final reaction wheel geometry, thruster moment arms, and operations timeline.	2	4
Feed lines and valves	Propellant freezing, leakage, or blockage	Use filters, check valves, temperature sensors, heaters, and isolation valves to protect feed paths and limit fault propagation.	2	4
Thruster alignment	Thrust vector or RCS moment imbalance	Align thrusters with the spacecraft center of mass and verify pointing/moment balance using ACS simulations before final integration.	2	3

6 Major Topics and Contributors

Table 67: Major topics of this report and corresponding contributors.

Topic	**Contributors**
Mission Overview	Logan, Hannah, Riley, Michael, Miina, Mahak, Alexis, Edward, Tamar
Asteroid Scientific Weighting and Selection	Rajat, Logan, Marissa
Global Trajectory Optimization and Architecture	Donny, Logan, Marissa
Trajectory Refinement	Marissa
Science Orbit Design and Coverage Analysis	Rajat, Logan, Mohanad
Launch Window and Mission Robustness	Marissa, Donny
ConOps & Science Requirements	Hannah
Vehicle Overview	Hannah, Riley, Tamar, Michael, Miina, Citli, Sophia, Mahak, Jason, Alexis, Edward
Schedule Overview	Hannah, Alexis, Miina, Tamar, Riley, Mahak
Risk Assessment Overview	Hannah, Alexis, Brandon, Michael, Miina, Jason

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